

A-1. 부서별 사원 수

문제 번호(A-H)	A-1
분류	GROUP BY 종합실습
문제	[GROUP BY 1] 부서별 사원 수 HR 데이터를 활용하여 '부서별 사원 수' 결과를 조회하는 SQL 을 작성하십시오.
조건	GROUP BY 와 필요한 집계 함수를 사용하십시오. department_id 가 같은 사원을 그룹화하여 인원수를 계산합니다. 부서 미배정 사원은 '부서 미배정'으로 표시합니다.

실험결과

Filter connections by name

postgres

localhost:5432

Databases

postgres

skala_db

Schemas

app

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courses

enrollments

majors

students

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System info

postgres

day02_PostgresQL_HR_종합실습_문제.sql

postgres

day02_PostgresQL_HR_emp_dept_상표스키마.sql

postgres

day02_책리_임유리.sql

SELECT COALESCE(d.department_name, '부서 미배정') AS "부",
COUNT(*) AS "사원 수"
FROM employees e
LEFT JOIN departments d ON d.department_id = e.department_id
GROUP BY d.department_id, d.department_name
ORDER BY d.department_id NULLS LAST;

employees(*) 1 X

SELECT COALESCE(d.department_name, '부서 미배정') AS "부",
COUNT(*) AS "사원 수"
FROM employees e
LEFT JOIN departments d ON d.department_id = e.department_id
GROUP BY d.department_id, d.department_name
ORDER BY d.department_id NULLS LAST;

1 Administration 1
2 Marketing 2
3 Purchasing 6
4 Human Resources 3
5 Shipping 5
6 IT 5
7 Public Relations 1
8 Sales 11
9 Executive 3
10 Finance 6
11 Accounting 2
12 R&D Korea 3
13 부서 미배정 1

Value X

Text

Administration

Refresh

Save

Cancel

Export data

200

13

13 row(s) fetched - 0.006s, on 2026-07-28 at 21:40:09

KST en Writable

Smart Insert

28 : 27 : 1156

Set 0 | 0

A-2. 직무별 사원 수와 평균 급여

문제 번호(A-H)	A-2
분류	GROUP BY 종합실습
문제	[GROUP BY 2] 직무별 사원 수와 평균 급여 HR 데이터를 활용하여 '직무별 사원 수와 평균 급여' 결과를 조회하는 SQL 을 작성하십시오.
조건	GROUP BY 와 필요한 집계 함수를 사용하십시오. 직무별 인원수와 평균 급여를 동시에 집계합니다.

실험결과

DBBeaver 26.1.3 - <postgres> day02_쿼리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Filter connections by name

postgres localhost:5432

Databases

skala_db

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app

Tables

courses

enrollments

24K

majors

40K

students

56K

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day02

public

hr

Tables

countries

40K

departments

72K

employees

88K

job_history

56K

jobs

24K

locations

40K

regions

40K

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<postgres> Script-3 <postgres> day02_PostgreSQL_HR_종합실습_문제.sql <postgres> day02_PostgreSQL_HR_emp_dept_샘플_스키마.sql <postgres> day02_쿼리_임유리.sql

-- =====

-- A. GROUP BY 종합실습 (10문제)

-- =====

-- [GROUP BY 1] 부서별 사원 수

-- 문제: HR 데이터를 활용하여 '부서별 사원 수' 결과를 조회하는 SQL 을 작성하십시오.

-- 조건: GROUP BY와 필요한 집계 함수를 사용하십시오. department_id가 같은 사원을 그룹화하여 인원수를 계산합니다. 부서 이름은 '부서 이름'으로 표시합니다.

-- TODO: 아래의 SQL 을 작성하십시오.

SELECT

COALESCE(d.department_name, '부서 미배정') AS "부서명",

COUNT(*) AS "사원 수"

FROM employees e

LEFT JOIN departments d ON d.department_id = e.department_id

GROUP BY d.department_id, d.department_name

ORDER BY d.department_id NULLS LAST;

-- [GROUP BY 2] 직무별 사원 수와 평균 급여

-- 문제: HR 데이터를 활용하여 '직무별 사원 수와 평균 급여' 결과를 조회하는 SQL 을 작성하십시오.

-- 조건: GROUP BY와 필요한 집계 함수를 사용하십시오. 직무를 인원수와 평균 급여를 동시에 집계합니다.

-- TODO: 아래의 SQL 을 작성하십시오.

SELECT

COALESCE(j.job_title, '직무 미배정') AS "직무명",

COUNT(*) AS "사원 수",

ROUND(AVG(e.salary), 2) AS "평균 급여"

FROM employees e

LEFT JOIN jobs j ON j.job_id = e.job_id

GROUP BY j.job_id, j.job_title

ORDER BY j.job_id NULLS LAST;

-- [GROUP BY 3] 부서별 급여 합계·평균·최소값·최대값

-- 문제: HR 데이터를 활용하여 '부서별 급여 합계·평균·최소값·최대값' 결과를 조회하는 SQL 을 작성하십시오.

-- 조건: GROUP BY와 필요한 집계 함수를 사용하십시오. 하나의 그룹에 여러 집계 함수를 적용합니다.

-- TODO: 아래의 SQL 을 작성하십시오.

employees(*) 1 X

SELECT COALESCE(j.job_title, '직무 미배정') AS "직무명", COU

Enter a SQL expression to filter results (use Ctrl+Space)

Grid

1 Public Accountant 123 사원 수 1 123 평균 급여 8,300

2 Accounting Manager 1 12,000

3 Administration Assistant 1 4,400

4 President 1 24,000

5 Administration Vice President 2 17,000

6 Accountant 5 7,920

7 Finance Manager 1 12,000

8 Human Resources Representative 1 6,500

9 Programmer 8 6,950

10 Marketing Manager 1 13,000

11 Marketing Representative 2 6,100

12 Public Relations Representative 1 10,000

13 Purchasing Clerk 5 2,780

14 Purchasing Manager 1 11,000

15 Sales Manager 5 12,200

16 Sales Representative 6 8,500

17 Stock Manager 3 8,033.33

Record

Value X

Text

Public Accountant

Refresh Save Cancel Export data 200 17 17 row(s) fetched - 0.003s, on 2026-07-28 at 21:42:56

KST en Writable Smart Insert 56 : 1 : 2105 Sel: 0 | 0

A-3. 부서별 급여 합계·평균·최솟값·최댓값

문제 번호(A-H)	A-3
분류	GROUP BY 종합실습
문제	[GROUP BY 3] 부서별 급여 합계·평균·최솟값·최댓값 HR 데이터를 활용하여 '부서별 급여 합계·평균·최솟값·최댓값' 결과를 조회하는 SQL 을 작성하십시오.
조건	GROUP BY 와 필요한 집계 함수를 사용하십시오. 하나의 그룹에 여러 집계 함수를 적용합니다.

실험결과

DBaver 26.1.3 - <postgres> day02_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Commit Rollback Auto

postgres hr@skala_db

postgres day02_PostgreSQL_HR_종합실습_문제.sql postgres day02_PostgreSQL_HR_emp_dept_샘플_스키마.sql postgres day02_임유리.sql

Filter connections by name

postgres local host 5432

postgres

skala_db

app

Tables

courses

enrollments 24K

majors 40K

students 56K

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day02

public

hr

Tables

countries 40K

departments 72K

employees 88K

job_history 56K

jobs 24K

locations 40K

regions 40K

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```
COALESCE(j.job_title, '직무 미배정') AS "직무",
COUNT(*) AS "직원 수",
ROUND(AVG(e.salary), 2) AS "평균 급여"
FROM employees e
LEFT JOIN jobs j ON j.job_id = e.job_id
GROUP BY j.job_id, j.job_title
ORDER BY j.job_id NULLS LAST;
```

-- [GROUP BY 3] 부서별 급여 합계·평균·최솟값·최댓값
-- 문제: HR 데이터를 활용하여 '부서별 급여 합계·평균·최솟값·최댓값' 결과를 조회하는 SQL을 작성하십시오.
-- 조건: GROUP BY와 필요한 집계 함수를 사용하십시오. 하나의 그룹에 여러 집계 함수를 적용합니다.
-- TODO: 아래의 SQL을 작성하십시오.

```
SELECT
COALESCE(d.department_name, '부서 미배정') AS "부서",
SUM(e.salary) AS "급여 합계",
ROUND(AVG(e.salary), 2) AS "평균 급여",
MIN(e.salary) AS "최소 급여",
MAX(e.salary) AS "최대 급여"
FROM employees e
LEFT JOIN departments d ON d.department_id = e.department_id
GROUP BY d.department_id, d.department_name
ORDER BY d.department_id NULLS LAST;
```

-- [GROUP BY 4] 관리자별 부하 직원 수
-- 문제: HR 데이터를 활용하여 '관리자별 부하 직원 수' 결과를 조회하는 SQL을 작성하십시오.
-- 조건: GROUP BY와 필요한 집계 함수를 사용하십시오. 사원 테이블을 관리자 기준으로 그룹화합니다.
-- TODO: 아래의 SQL을 작성하십시오.

```
SELECT
COALESCE(e.manager_id::text, '관리자 없음') AS "관리자 ID",
COUNT(*) AS "부하 직원 수"
FROM employees e
GROUP BY e.manager_id
ORDER BY e.manager_id NULLS LAST;
```

-- [GROUP BY 5] 부서 연도별 입사자 수
-- 문제: HR 데이터를 활용하여 '부서 연도별 입사자 수' 결과를 조회하는 SQL을 작성하십시오.
-- 조건: GROUP BY와 필요한 집계 함수를 사용하십시오. EXTRACT로 날짜에서 연도만 꺼낸 뒤 그룹화합니다.

employees(+)

SELECT COALESCE(d.department_name, '부서 미배정') AS "부서", SUM(e.salary) AS "급여 합계", ROUND(AVG(e.salary), 2) AS "평균 급여", MIN(e.salary) AS "최소 급여", MAX(e.salary) AS "최대 급여" FROM employees e LEFT JOIN departments d ON d.department_id = e.department_id GROUP BY d.department_id, d.department_name ORDER BY d.department_id NULLS LAST;

1 Administration 4,400 4,400 4,400 4,400

2 Marketing 19,000 9,500 6,000 13,000

3 Purchasing 24,900 4,150 2,500 11,000

4 Human Resources 6,500 6,500 6,500 6,500

5 Shipping 24,100 8,033.33 7,900 8,200

6 IT 28,800 5,760 4,200 9,000

7 Public Relations 10,000 10,000 10,000 10,000

8 Sales 112,000 10,181.82 7,000 14,000

9 Executive 58,000 19,333.33 17,000 24,000

10 Finance 51,600 8,600 6,900 12,000

11 Accounting 20,300 10,150 8,300 12,000

12 R&D Korea 26,800 8,933.33 7,800 10,500

13 부서 미배정 6,200 6,200 6,200 6,200

Value X

Text

Administration

General

Files X

Bookmarks

Dashboards

Diagrams

Refresh Save Cancel

Export data

200

13

13 row(s) fetched - 0.001s, on 2026-07-28 at 21:43:14

KST en Writable Smart Insert 73 : 1 : 2644 Sel: 0 | 0

A-4. 관리자별 부하 직원 수

문제 번호(A-H)	A-4
분류	GROUP BY 종합실습
문제	[GROUP BY 4] 관리자별 부하 직원 수 HR 데이터를 활용하여 '관리자별 부하 직원 수' 결과를 조회하는 SQL 을 작성하십시오.
조건	GROUP BY 와 필요한 집계 함수를 사용하십시오. 사원 테이블을 관리자 기준으로 그룹화합니다.

실험결과

DBaver 26.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Connections x 프로젝트

Filter connections by name

DBaver Sample Database (SQLite)

postgres localhost-5432

Databases

postgres

skala_db

Schemas

app

Tables

courses

enrollments

24K

majors

40K

students

56K

Foreign Tables

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day02

public

hr

Tables

countries

40K

departments

72K

employees

88K

job_history

56K

jobs

24K

locations

40K

regions

40K

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Script-3

<postgres> day02_PostgresQL_HR_종합실습_문제.sql

<postgres> day02_PostgresQL_HR_emp_dept_상표_스키마.sql

<postgres> day02_책리_임유리.sql

-- TODO: 아래에 SQL 을 작성하십시오.

SELECT

COALESCE(d.department_name, '부서 미해당') AS "부서명",

SUM(e.salary) AS "관리자 급여",

ROUND(AVG(e.salary), 2) AS "평균 급여",

MIN(e.salary) AS "최소 급여",

MAX(e.salary) AS "최대 급여"

FROM employees e

LEFT JOIN departments d ON d.department_id = e.department_id

GROUP BY d.department_id, d.department_name

ORDER BY d.department_id NULLS LAST;

-- [GROUP BY 4] 관리자별 부하 직원 수

-- 문제: HR 데이터를 활용하여 '관리자별 부하 직원 수' 결과를 조회하는 SQL 을 작성하십시오.

-- 조건: GROUP BY와 필요한 집계 함수를 사용하십시오. 사원 테이블을 관리자 기준으로 그룹화합니다.

-- TODO: 아래에 SQL 을 작성하십시오.

SELECT

COALESCE(e.manager_id::text, '관리자 없음') AS "관리자 ID",

COUNT(*) AS "부하 직원 수"

FROM employees e

GROUP BY e.manager_id

ORDER BY e.manager_id NULLS LAST;

-- [GROUP BY 5] 일사 연도별 일사자 수

-- 문제: HR 데이터를 활용하여 '일사 연도별 일사자 수' 결과를 조회하는 SQL 을 작성하십시오.

-- 조건: GROUP BY와 필요한 집계 함수를 사용하십시오. EXTRACT로 날짜에서 연도만 꺼낸 뒤 그룹화합니다.

-- TODO: 아래에 SQL 을 작성하십시오.

SELECT

EXTRACT(YEAR FROM hire_date)::Integer AS "일사 연도",

COUNT(*) AS "일사자 수"

FROM employees

GROUP BY EXTRACT(YEAR FROM hire_date)

ORDER BY EXTRACT(YEAR FROM hire_date);

-- [GROUP BY 6] 커미션 수당 어분별 인원과 평균 급여

-- 문제: HR 데이터를 활용하여 '커미션 수당 어분별 인원과 평균 급여' 결과를 조회하는 SQL 을 작성하십시오.

-- 조건: GROUP BY와 필요한 집계 함수를 사용하십시오. CASE 표현식의 결과로 GROUP BY 기준으로 사용할 수 있습니다.

--

SELECT COALESCE(e.manager_id::text, '관리자 없음') AS "관리자 ID",

123 부하 직원 수

1 100 12

2 101 5

3 102 2

4 103 4

5 108 5

6 114 5

7 145 6

8 201 2

9 205 1

10 220 2

11 관리자 없음 1

Results 1 x

SELECT COALESCE(e.manager_id::text, '관리자 없음') AS "관리자 ID",

Enter a SQL expression to filter results (use Ctrl+Space)

Grid

Text

Value x

Text

100

Refresh Save Cancel

Export data

200

11

11 row(s) fetched - 0.001s, on 2026-07-28 at 21:43:26

KST en Writable

Smart Insert

84 : 1 : 2980

Set: 0 | 0

A-5. 입사 연도별 입사자 수

문제 번호(A-H)	A-5
분류	GROUP BY 종합실습
문제	[GROUP BY 5] 입사 연도별 입사자 수 HR 데이터를 활용하여 '입사 연도별 입사자 수' 결과를 조회하는 SQL 을 작성하십시오.
조건	GROUP BY 와 필요한 집계 함수를 사용하십시오. EXTRACT 로 날짜에서 연도만 꺼낸 뒤 그룹화합니다.

실험결과

DBBeaver 25.1.3 - <postgres> day02_쿼리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Auto Commit Rollback

postgres hr@skala_db

Connections 프로젝트

Filter connections by name

DBBeaver Sample Database (SQLite) postgres localhost:5432 Databases postgres schemas app Tables courses 24K enrollments 40K majors 56K students Foreign Tables Views Materialized Views Indexes Functions Sequences Data types Aggregate functions Event Triggers Extensions Storage System info Roles Administer System info

Script-3 postgres day02_PostgreSQL_HR_종합실습_문제.sql postgres day02_PostgreSQL_HR_emp_dept_샘플_스키마.sql postgres day02_쿼리_임유리.sql

MAX(e.salary) AS "최대 급여"
FROM employees e
LEFT JOIN departments d ON d.department_id = e.department_id
GROUP BY d.department_id, d.department_name
ORDER BY d.department_id NULLS LAST;

-- [GROUP BY 4] 관리자별 부하 직원 수
-- 문제: HR 데이터를 활용하여 '관리자를 부하 직원 수' 결과를 조회하는 SQL 을 작성하십시오.
-- 조건: GROUP BY와 필요한 집계 함수를 사용하십시오. 사용 데이터를 관리자 기준으로 그룹화합니다.
-- TODO: 아래의 SQL 을 작성하십시오.
SELECT
COLLECTSE(e.manager_id::text, '관리자 없음') AS "관리자 ID",
COUNT(*) AS "부하 직원 수"
FROM employees e
GROUP BY e.manager_id
ORDER BY e.manager_id NULLS LAST;

-- [GROUP BY 5] 입사 연도별 입사자 수
-- 문제: HR 데이터를 활용하여 '입사 연도별 입사자 수' 결과를 조회하는 SQL 을 작성하십시오.
-- 조건: GROUP BY와 필요한 집계 함수를 사용하십시오. EXTRACT로 날짜에서 연도만 꺼낸 뒤 그룹화합니다.
-- TODO: 아래의 SQL 을 작성하십시오.
SELECT
EXTRACT(YEAR FROM hire_date)::integer AS "입사 연도",
COUNT(*) AS "입사자 수"
FROM employees
GROUP BY EXTRACT(YEAR FROM hire_date)
ORDER BY EXTRACT(YEAR FROM hire_date);

-- [GROUP BY 6] 커미션 수월 여부별 인원과 평균 급여
-- 문제: HR 데이터를 활용하여 '커미션 수월 여부별 인원과 평균 급여' 결과를 조회하는 SQL 을 작성하십시오.
-- 조건: GROUP BY와 필요한 집계 함수를 사용하십시오. CASE 표현식의 결과를 GROUP BY 기준으로 사용할 수 있습니다.
-- TODO: 아래의 SQL 을 작성하십시오.
SELECT
CASE
WHEN commission_pct IS NOT NULL THEN '커미션 수월'
ELSE '커미션 미수월'
END AS "커미션 수월 여부",

Results 1 X

SELECT EXTRACT(YEAR FROM hire_date)::integer AS "입사 연도", COUNT(*) AS "입사자 수"

입사 연도	입사자 수
2001	1
2002	7
2003	4
2004	3
2005	13
2006	6
2007	6
2008	1
2019	1
2020	1
2021	1
2023	1

Value X
2001

12 row(s) fetched - 0.001s, on 2025-07-28 at 21:43:44

A-6. 커미션 수령 여부별 인원과 평균 급여

문제 번호(A-H)	A-6
분류	GROUP BY 종합실습
문제	[GROUP BY 6] 커미션 수령 여부별 인원과 평균 급여 HR 데이터를 활용하여 '커미션 수령 여부별 인원과 평균 급여' 결과를 조회하는 SQL 을 작성하십시오.
조건	GROUP BY 와 필요한 집계 함수를 사용하십시오. CASE 표현식의 결과도 GROUP BY 기준으로 사용할 수 있습니다.

실험결과

DBaver 16.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Commit Rollback Auto postgres hr@skala_db postgres day02_PostgreSQL_HR_종합실습_문제.sql postgres day02_PostgreSQL_HR_emp_dept_샘플_스키마.sql postgres day02_책리_임유리.sql

Filter connections by name

postgres localhost:5432

Databases

postgres

skala_db

Schemas

app

Tables

courses

enrollments

majors

students

Foreign Tables

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jobs

locations

regions

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System info

GROUP BY e.manager_id
ORDER BY e.manager_id NULLS LAST;

-- [GROUP BY 5] 인사 연도별 입사자 수
-- 출력: HR 데이터를 활용하여 '입사 연도별 입사자 수' 결과를 조회하는 SQL 을 작성하십시오.
-- 조건: GROUP BY와 필요한 집계 함수를 사용하십시오. EXTRACT로 날짜에서 연도만 꺼낸 뒤 그룹화합니다.
-- TODO: 아래의 SQL 을 작성하십시오.
SELECT
EXTRACT(YEAR FROM hire_date)::integer AS "입사 연도",
COUNT(*) AS "입사자 수"
FROM employees
GROUP BY EXTRACT(YEAR FROM hire_date)
ORDER BY EXTRACT(YEAR FROM hire_date);

-- [GROUP BY 6] 커미션 수령 여부별 인원과 평균 급여
-- 출력: HR 데이터를 활용하여 '커미션 수령 여부별 인원과 평균 급여' 결과를 조회하는 SQL 을 작성하십시오.
-- 조건: GROUP BY와 필요한 집계 함수를 사용하십시오. CASE 표현식의 결과도 GROUP BY 기준으로 사용할 수 있습니다.
-- TODO: 아래의 SQL 을 작성하십시오.
SELECT
CASE
WHEN commission_pct IS NOT NULL THEN '커미션 있음'
ELSE '커미션 없음'
END AS "커미션 수령 여부",
COUNT(*) AS "사람 수",
ROUND(AVG(salary), 2) AS "평균 급여"
FROM employees
GROUP BY
CASE
WHEN commission_pct IS NOT NULL THEN '커미션 있음'
ELSE '커미션 없음'
END
ORDER BY "커미션 수령 여부";

-- [GROUP BY 7] 국가별 근무 인원 수
-- 출력: HR 데이터를 활용하여 '국가별 근무 인원 수' 결과를 조회하는 SQL 을 작성하십시오.
-- 조건: GROUP BY와 필요한 집계 함수를 사용하십시오. employees테이블 countries까지 여러 테이블을 연결한 뒤 국가별로 집계합니다.
-- TODO: 아래의 SQL 을 작성하십시오.

Results 1 X

SELECT CASE WHEN commission_pct IS NOT NULL THEN '커미션 있음' ELSE '커미션 없음' END AS "커미션 수령 여부", COUNT(*) AS "사람 수", ROUND(AVG(salary), 2) AS "평균 급여" FROM employees GROUP BY "커미션 수령 여부" ORDER BY "커미션 수령 여부";

1 커미션 미수령 34 8,252.94
2 커미션 수령 11 10,181.82

Value X
Text
커미션 미수령

Refresh Save Cancel Export data 200 2 2 row(s) fetched - 0s, on 2026-07-28 at 21:43:59

General KST en Writable Smart Insert 117 : 1 : 3863 Sel: 0 | 0

A-7. 국가별 근무 인원 수

문제 번호(A-H)	A-7
분류	GROUP BY 종합실습
문제	[GROUP BY 7] 국가별 근무 인원 수 HR 데이터를 활용하여 '국가별 근무 인원 수' 결과를 조회하는 SQL 을 작성하십시오.
조건	GROUP BY 와 필요한 집계 함수를 사용하십시오. employees 부터 countries 까지 여러 테이블을 연결한 뒤 국가별로 집계합니다.

실험결과

DBBeaver 26.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Auto

postgres hr@skala_db

Connections postgres day02_PostgreSQL_HR_종합실습_문제.sql postgres day02_PostgreSQL_HR_emp_dept_상표_스키마.sql postgres day02_책리_임유리.sql

Filter connections by name

DBBeaver Sample Database (SQLite)

postgres localhost:5432

Databases

postgres

skala_db

Schemas

app

Tables

courses 24K

enrollments 24K

majors 40K

students 56K

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day02

public

hr

Tables

countries 40K

departments 72K

employees 88K

job_history 56K

jobs 24K

locations 40K

regions 40K

Foreign Tables

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Extensions

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Roles

Administer

System Info

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Name

Bookmarks

Dashboards

Diagrams

General

```
FROM employees e
GROUP BY
CASE
  WHEN commission_pct IS NOT NULL THEN '커미션 수월'
  ELSE '커미션 미수월'
END
ORDER BY "커미션 수월 여부";

-- [GROUP BY 7] 국가별 근무 인원 수
-- 문제: HR 데이터를 활용하여 '국가별 근무 인원 수' 결과를 조회하는 SQL 을 작성하십시오.
-- 조건: GROUP BY와 필요한 집계 함수를 사용하십시오. employees부터 countries까지 여러 테이블을 연결한 뒤 국가별로 집계합니다.
-- TODO: 아래의 SQL을 작성하십시오.
SELECT
  c.country_id AS "국가 코드",
  c.country_name AS "국가명",
  COUNT(e.employee_id) AS "인원 수"
FROM employees e
INNER JOIN departments d ON d.department_id = e.department_id
INNER JOIN locations l ON l.location_id = d.location_id
INNER JOIN countries c ON c.country_id = l.country_id
GROUP BY c.country_id, c.country_name
ORDER BY c.country_name;

-- [GROUP BY 8] 부서별 직무 종류 수
-- 문제: HR 데이터를 활용하여 '부서별 직무 종류 수' 결과를 조회하는 SQL을 작성하십시오.
-- 조건: GROUP BY와 필요한 집계 함수를 사용하십시오. COUNT(DISTINCT)을 사용하면 중복을 제외한 개수를 구할 수 있습니다.
-- TODO: 아래의 SQL을 작성하십시오.
SELECT
  COALESCE(d.department_name, '부서 미배정') AS "부서명",
  COUNT(DISTINCT e.job_id) AS "직무 종류 수"
FROM employees e
LEFT JOIN departments d ON d.department_id = e.department_id
GROUP BY d.department_id, d.department_name
ORDER BY d.department_id NULLS LAST;

-- [GROUP BY 9] 평균 급여가 8,000 이상인 부서
-- 문제: HR 데이터를 활용하여 '평균 급여가 8,000 이상인 부서' 결과를 조회하는 SQL을 작성하십시오.
-- TODO: 아래의 SQL을 작성하십시오.
```

countries 1

SELECT c.country_id AS "국가 코드", c.country_name AS "국가명", COUNT(e.employee_id) AS "인원 수" FROM employees e INNER JOIN departments d ON d.department_id = e.department_id INNER JOIN locations l ON l.location_id = d.location_id INNER JOIN countries c ON c.country_id = l.country_id GROUP BY c.country_id, c.country_name ORDER BY c.country_name;

Grid

1 CA Canada 2

2 DE Germany 1

3 KR South Korea 3

4 UK United Kingdom 12

5 US United States 26

Record

Refresh Save Cancel Export data 200 5 5 row(s) fetched - 0.004s, on 2026-07-28 at 21:44:18

KST en Writable Smart Insert 133 : 1 : 4413 Set: 0 | 0

A-8. 부서별 직무 종류 수

문제 번호(A-H)	A-8
분류	GROUP BY 종합실습
문제	[GROUP BY 8] 부서별 직무 종류 수 HR 데이터를 활용하여 '부서별 직무 종류 수' 결과를 조회하는 SQL 을 작성하십시오.
조건	GROUP BY 와 필요한 집계 함수를 사용하십시오. COUNT(DISTINCT 열)을 사용하면 중복을 제외한 개수를 구할 수 있습니다.

실험결과

DBEAVER 26.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Connections

Filter connections by name

DBeaver Sample Database (SQLite)

postgres local host 5432

Databases

postgres

skala_db

Schemas

app

Tables

courses

enrollments

majors

students

Foreign Tables

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hr

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countries

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employees

job_history

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locations

regions

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Administrator

System Info

*postgres> Script-3

<postgres> day02_PostgreSQL_HR_종합실습_문제.sql

<postgres> day02_PostgreSQL_HR_emp_dept_상표_스키마.sql

<postgres> day02_책리_임유리.sql

SQL

Commit

Rollback

Auto

postgres

hr@skala_db

postgres

day02_PostgreSQL_HR_종합실습_문제.sql

day02_PostgreSQL_HR_emp_dept_상표_스키마.sql

day02_책리_임유리.sql

SELECT

-- country_id AS "국가 코드",

-- c.country_name AS "국가명",

-- COUNT(e.employee_id) AS "직원 수"

FROM employees e

INNER JOIN departments d ON d.department_id = e.department_id

INNER JOIN locations l ON l.location_id = d.location_id

INNER JOIN countries c ON c.country_id = l.country_id

GROUP BY c.country_id, c.country_name

ORDER BY c.country_name;

-- [GROUP BY 8] 부서별 직무 종류 수

-- 문제: HR 데이터를 활용하여 '부서별 직무 종류 수' 결과를 조회하는 SQL 을 작성하십시오.

-- 조건: GROUP BY와 필요한 집계 함수를 사용하십시오. COUNT(DISTINCT 열)을 사용하면 중복을 제외한 개수를 구할 수 있습니다.

-- TODO: 아래에 SQL 을 작성하십시오.

SELECT

COALESCE(d.department_name, '부서 미배정') AS "부서명",

COUNT(DISTINCT e.job_id) AS "직무 종류 수"

FROM employees e

LEFT JOIN departments d ON d.department_id = e.department_id

GROUP BY d.department_id, d.department_name

ORDER BY d.department_id NULLS LAST;

-- [GROUP BY 9] 평균 급여가 8,000 이상인 부서

-- 문제: HR 데이터를 활용하여 '평균 급여가 8,000 이상인 부서' 결과를 조회하는 SQL 을 작성하십시오.

-- 조건: GROUP BY와 필요한 집계 함수를 사용하십시오. WHERE 는 그룹화 전 행을, HAVING 은 그룹화 후 결과를 필터링합니다.

-- TODO: 아래에 SQL 을 작성하십시오.

SELECT

COALESCE(d.department_name, '부서 미배정') AS "부서명",

ROUND(AVG(e.salary), 2) AS "평균 급여"

FROM employees e

LEFT JOIN departments d ON d.department_id = e.department_id

GROUP BY d.department_id, d.department_name

HAVING AVG(e.salary) >= 8000

ORDER BY AVG(e.salary) DESC;

employees(*) 1 X

SELECT COALESCE(d.department_name, '부서 미배정') AS "부서명",

COUNT(DISTINCT e.job_id) AS "직무 종류 수"

FROM employees e

LEFT JOIN departments d ON d.department_id = e.department_id

GROUP BY d.department_id, d.department_name

ORDER BY d.department_id NULLS LAST;

13 rows fetched - 0.001s, on 2026-07-28 at 21:44:34

Grid

1 Administration 1

2 Marketing 2

3 Purchasing 2

4 Human Resources 1

5 Shipping 1

6 IT 1

7 Public Relations 1

8 Sales 2

9 Executive 2

10 Finance 2

11 Accounting 2

12 R&D Korea 1

13 부서 미배정 1

Value X

Text

Administration

General

Name

Bookmarks

Dashboards

Diagrams

Refresh

Save

Cancel

Export data

200

13

13 rows fetched - 0.001s, on 2026-07-28 at 21:44:34

KST en Writable

Smart Insert

146 : 1 : 4868

Set 0 : 0

A-9. 평균 급여가 8,000 이상인 부서

문제 번호(A-H)	A-9
분류	GROUP BY 종합실습
문제	[GROUP BY 9] 평균 급여가 8,000 이상인 부서 HR 데이터를 활용하여 '평균 급여가 8,000 이상인 부서' 결과를 조회하는 SQL 을 작성하십시오.
조건	GROUP BY 와 필요한 집계 함수를 사용하십시오. WHERE 는 그룹화 전 행을, HAVING 은 그룹화 후 결과를 필터링합니다.

실험결과

[illegible]

A-10. 부서별 급여 구간 인원수

문제 번호(A-H)	A-10
분류	GROUP BY 종합실습
문제	[GROUP BY 10] 부서별 급여 구간 인원수 HR 데이터를 활용하여 '부서별 급여 구간 인원수' 결과를 조회하는 SQL 을 작성하십시오.
조건	GROUP BY 와 필요한 집계 함수를 사용하십시오. CASE 로 만든 급여 구간과 부서를 두 가지 기준으로 그룹화합니다.

실험결과

DBaver 26.1.3 - <postgres> day02_쿼리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Commit Rollback Auto postgres hr@skala_db postgres day02_PostgreSQL_HR_종합실습_문제.sql postgres day02_PostgreSQL_HR_emp_dept_샘플_스키마.sql postgres day02_쿼리_임유리.sql

Filter connections by name
Databases
postgres local:5432
Databases
postgres
skala_db
Schemas
app
Tables
courses
enrollments 24K
majors 40K
students 56K
Foreign Tables
Views
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Data types
Aggregate functions
day02
public
hr
Tables
countries 40K
departments 72K
employees 88K
job_history 56K
jobs 24K
locations 40K
regions 40K
Foreign Tables
Views
Materialized Views
Indexes
Functions
Sequences
Data types
Aggregate functions
Event Triggers
Extensions
Storage
System info
Roles
Administer
System info

```
SELECT
  COALESCE(d.department_name, '부서 미배정') AS "부서명",
  ROUND(AVG(e.salary), 2) AS "평균 급여"
FROM employees e
LEFT JOIN departments d ON d.department_id = e.department_id
GROUP BY d.department_id, d.department_name
HAVING AVG(e.salary) >= 8000
ORDER BY AVG(e.salary) DESC;
```

-- [GROUP BY 10] 부서별 급여 구간 인원수
-- 목적: HR 데이터를 활용하여 '부서별 급여 구간 인원수' 결과를 조회하는 SQL 을 작성하십시오.
-- 조건: GROUP BY와 필요한 집계 함수를 사용하십시오. CASE로 만든 급여 구간과 부서를 두 가지 기준으로 그룹화합니다.
-- TODO: 아래의 SQL 을 작성하십시오.

```
SELECT
  COALESCE(d.department_name, '부서 미배정') AS "부서명",
  CASE
    WHEN e.salary < 5000 THEN '5,000 미만'
    WHEN e.salary < 10000 THEN '5,000 이상 10,000 미만'
    ELSE '10,000 이상'
  END AS "급여 구간",
  COUNT(*) AS "사원 수"
FROM employees e
LEFT JOIN departments d ON d.department_id = e.department_id
GROUP BY
  d.department_id,
  d.department_name,
  CASE
    WHEN e.salary < 5000 THEN '5,000 미만'
    WHEN e.salary < 10000 THEN '5,000 이상 10,000 미만'
    ELSE '10,000 이상'
  END
ORDER BY d.department_id NULLS LAST, MIN(e.salary);
```

-- [서브쿼리 1] 전체 평균보다 급여가 높은 사원

employees(*) 1 X

SELECT COALESCE(d.department_name, '부서 미배정') AS "부서명", ROUND(AVG(e.salary), 2) AS "평균 급여", COUNT(*) AS "사원 수" FROM employees e LEFT JOIN departments d ON d.department_id = e.department_id GROUP BY d.department_id, d.department_name, CASE WHEN e.salary < 5000 THEN '5,000 미만' WHEN e.salary < 10000 THEN '5,000 이상 10,000 미만' ELSE '10,000 이상' END ORDER BY d.department_id NULLS LAST, MIN(e.salary);

부서명	급여 구간	123 사원 수
Administration	5,000 미만	1
Marketing	5,000 이상 10,000 미만	1
Marketing	10,000 이상	1
Purchasing	5,000 미만	5
Purchasing	10,000 이상	1
Human Resources	5,000 이상 10,000 미만	1
Shipping	5,000 이상 10,000 미만	3
IT	5,000 미만	3
IT	5,000 이상 10,000 미만	2
Public Relations	10,000 이상	1
Sales	5,000 이상 10,000 미만	5
Sales	10,000 이상	6
Executive	10,000 이상	3
Finance	5,000 이상 10,000 미만	5
Finance	10,000 이상	1
Accounting	5,000 이상 10,000 미만	1
Accounting	10,000 이상	1
R&D Korea	5,000 이상 10,000 미만	2
R&D Korea	10,000 이상	1
부서 미배정	5,000 이상 10,000 미만	1

Value X
Text
Administration

200 20 20 row(s) fetched - 0.001s, on 2025-07-28 at 21:44:59

General

B-1. 전체 평균보다 급여가 높은 사원

문제 번호(A-H)	B-1
분류	서브쿼리 종합실습
문제	[서브쿼리 1] 전체 평균보다 급여가 높은 사원 HR 데이터를 활용하여 '전체 평균보다 급여가 높은 사원' 결과를 조회하는 SQL 을 작성하십시오.
조건	서브쿼리를 사용해 조건을 해결하십시오. 단일 행 서브쿼리에서 계산한 전체 평균과 각 사원의 급여를 비교합니다.

실험결과

DBBeaver 26.1.3 - <postgres> day02_쿼리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Commit Rollback Auto

postgres hr@skala_db

Connections postgres local:localhost:5432

Filter connections by name

postgres local:localhost:5432

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enrollments

24K

majors

40K

students

56K

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88K

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56K

jobs

24K

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40K

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<postgres> Script-3

<postgres> day02_PostgreSQL_HR_종합실습_문제.sql

<postgres> day02_PostgreSQL_HR_emp_dept_샘플_스키마.sql

<postgres> day02_쿼리_임유리.sql

GROUP BY

d.department_id,

d.department_name,

CASE

WHEN e.salary < 5000 THEN '5,000 미만'

WHEN e.salary < 10000 THEN '5,000 이상 10,000 미만'

ELSE '10,000 이상'

END

ORDER BY d.department_id NULLS LAST, RIN(e.salary);

-- [서브쿼리 1] 전체 평균보다 급여가 높은 사원

-- 문제: HR 데이터를 활용하여 '전체 평균보다 급여가 높은 사원' 결과를 조회하는 SQL 을 작성하십시오.

-- 조건: 서브쿼리를 사용해 조건을 해결하십시오. 단일 행 서브쿼리에서 계산한 전체 평균과 각 사원의 급여를 비교합니다.

-- TODO: 아래에 SQL 을 작성하십시오.

SELECT

employee_id AS "사원 ID",

concat_ws(' ', first_name, last_name) AS "사원명",

salary AS "급여"

FROM employees

WHERE salary > (SELECT AVG(salary) FROM employees)

ORDER BY salary DESC, employee_id;

-- [서브쿼리 2] 소속 부서 평균보다 급여가 높은 사원

-- 문제: HR 데이터를 활용하여 '소속 부서 평균보다 급여가 높은 사원' 결과를 조회하는 SQL 을 작성하십시오.

-- 조건: 서브쿼리를 사용해 조건을 해결하십시오. 매달 행 서브쿼리의 department_id를 참조하는 상한 서브쿼리입니다.

-- TODO: 아래에 SQL 을 작성하십시오.

SELECT

e.employee_id AS "사원 ID",

concat_ws(' ', e.first_name, e.last_name) AS "사원명",

e.department_id AS "부서 ID",

e.salary AS "급여"

FROM employees e

WHERE e.salary > (

SELECT AVG(e2.salary)

employees 1 X

SELECT employee_id AS "사원 ID", concat_ws(' ', first_name, la

Enter a SQL expression to filter results (use Ctrl+Space)

Grid

1 100 Steven King 24,000

2 101 Neena Kochhar 17,000

3 102 Lex De Haan 17,000

4 145 John Russell 14,000

5 146 Karen Partners 13,500

6 201 Michael Hartstein 13,000

7 108 Nancy Greenberg 12,000

8 147 Alberto Errazuriz 12,000

9 205 Shelley Higgins 12,000

10 114 Den Raphaely 11,000

11 148 Gerald Cambrault 11,000

12 149 Elvi Zlotkey 10,500

13 220 Manjun Kim 10,500

14 150 Peter Tucker 10,000

15 204 Hermann Baer 10,000

16 151 David Bernstein 9,500

17 103 Alexander Hunold 9,000

18 109 Daniel Fiet 9,000

19 152 Peter Hall 9,000

Record

Refresh Save Cancel

Export data

200 19

19 row(s) fetched - 0.001s, on 2025-07-28 at 21:45:17

KST en Writable Smart Insert 201 : 1 : 6685 Sel: 0 | 0

Value X

100

B-2. 소속 부서 평균보다 급여가 높은 사원

문제 번호(A-H)	B-2
분류	서브쿼리 종합실습
문제	[서브쿼리 2] 소속 부서 평균보다 급여가 높은 사원 HR 데이터를 활용하여 '소속 부서 평균보다 급여가 높은 사원' 결과를 조회하는 SQL 을 작성하십시오.
조건	서브쿼리를 사용해 조건을 해결하십시오. 바깥 쿼리의 department_id 를 참조하는 상관 서브쿼리입니다.

실험결과

DBaver 26.1.3 - <postgres> day02_쿼리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Auto

postgres hr@skala_db

postgres day02_PostgreSQL_HR_종합실습_문제.sql postgres day02_PostgreSQL_HR_emp_dept_샘플_스키마.sql postgres day02_쿼리_임유리.sql

Filter connections by name

DBaver Sample Database (SQLite)

postgres local host 5432

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majors

40K

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56K

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88K

job_history

56K

jobs

24K

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40K

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1000: 아래의 SQL을 작성하십시오.

SELECT
employee_id AS "사원 ID",
concat_ws(' ', first_name, last_name) AS "사원명",
salary AS "급여"
FROM employees
WHERE salary > (SELECT AVG(salary) FROM employees)
ORDER BY salary DESC, employee_id;

1000: 아래의 SQL을 작성하십시오.

SELECT
employee_id AS "사원 ID",
concat_ws(' ', e.first_name, e.last_name) AS "사원명",
e.department_id AS "부서 ID",
e.salary AS "급여"
FROM employees e
WHERE e.salary > (
SELECT AVG(e2.salary)
FROM employees e2
WHERE e2.department_id = e.department_id
)
ORDER BY e.department_id NULLS LAST, e.salary DESC, e.employee_id;

1000: 아래의 SQL을 작성하십시오.

SELECT
d.department_id AS "부서 ID",
d.department_name AS "부서명"
FROM departments d
WHERE EXISTS (
SELECT 1
FROM employees e
WHERE e.department_id = d.department_id

employees 1 X

125 사원 ID 125 사원명 125 부서 ID 125 급여

1 201 Michael Hartstein 20 13,000

2 114 Den Raphaely 30 11,000

3 121 Adam Fripp 50 8,200

4 103 Alexander Hunold 60 9,000

5 104 Bruce Ernst 80 6,000

6 145 John Russell 80 14,000

7 146 Karen Partners 80 13,500

8 147 Alberto Errazuriz 80 12,000

9 148 Gerald Cambrault 80 11,000

10 149 Eleni Zlotkey 80 10,500

11 100 Steven King 90 24,000

12 108 Nancy Greenberg 100 12,000

13 109 Daniel Faviet 100 9,000

14 205 Shelley Higgins 110 12,000

15 220 Minjun Kim 120 10,500

Value X

201

Refresh Save Cancel

Export data

200 15

15 row(s) fetched - 0.001s, on 2026-07-28 at 21:45:31

KST en Writable Smart Insert 220 : 1 : 7224 Sel: 0 | 0

B-3. 사원이 한 명이라도 존재하는 부서

문제 번호(A-H)	B-3
분류	서브쿼리 종합실습
문제	[서브쿼리 3] 사원이 한 명이라도 존재하는 부서 HR 데이터를 활용하여 '사원이 한 명이라도 존재하는 부서' 결과를 조회하는 SQL 을 작성하십시오.
조건	서브쿼리를 사용해 조건을 해결하십시오. EXISTS 는 서브쿼리 결과 행의 존재 여부만 확인합니다.

실험결과

DBaver 26.1.3 - <postgres> day02_쿼리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Commit Rollback Auto

postgres hr@skala_db

Connections postgres Script-3 <postgres> day02_PostgreSQL_HR_종합실습_문제.sql <postgres> day02_PostgreSQL_HR_emp_dept_샘플_스키마.sql <postgres> day02_쿼리_임유리.sql

Filter connections by name

DBaver Sample Database (SQLite)

postgres localhost:5432

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courses

enrollments 24K

majors 40K

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jobs 24K

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regions 40K

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Roles

Administer

System info

SELECT AVG(e2.salary)
FROM employees e2
WHERE e2.department_id = e.department_id
ORDER BY e.department_id NULLS LAST, e.salary DESC, e.employee_id;

-- [서브쿼리 3] 사원이 한 명이라도 존재하는 부서
-- 목적: HR 데이터를 활용하여 '사원이 한 명이라도 존재하는 부서' 결과를 조회하는 SQL을 작성하십시오.
-- 조건: 서브쿼리를 사용해 조건을 해결하십시오. EXISTS는 서브쿼리 결과 행의 존재 여부만 확인합니다.
-- TODO: 아래의 SQL을 작성하십시오.
SELECT
d.department_id AS "부서 ID",
d.department_name AS "부서명"
FROM departments d
WHERE EXISTS (
SELECT 1
FROM employees e
WHERE e.department_id = d.department_id
)
ORDER BY d.department_id;

-- [서브쿼리 4] 가장 높은 평균 급여를 가진 부서
-- 목적: HR 데이터를 활용하여 '가장 높은 평균 급여를 가진 부서' 결과를 조회하는 SQL을 작성하십시오.
-- 조건: 서브쿼리를 사용해 조건을 해결하십시오. FROM 절 서브쿼리로 부서 평균을 만든 후 가장 큰 평균과 비교합니다.
-- TODO: 아래의 SQL을 작성하십시오.
SELECT
da.department_id AS "부서 ID",
da.department_name AS "부서명",
ROUND(da.avg_salary, 2) AS "평균 급여"
FROM (
SELECT department_id, AVG(salary) AS avg_salary
FROM employees
WHERE department_id IS NOT NULL
GROUP BY department_id
) da
INNER JOIN departments d ON d.department_id = da.department_id
SELECT MAX(x.avg_salary)

departments 1 X

SELECT d.department_id AS "부서 ID", d.department_name AS "부서명"

Enter a SQL expression to filter results (use Ctrl+Space)

Grid

1 10 Administration

2 20 Marketing

3 30 Purchasing

4 40 Human Resources

5 50 Shipping

6 60 IT

7 70 Public Relations

8 80 Sales

9 90 Executive

10 100 Finance

11 110 Accounting

12 120 R&D Korea

Record

Value X

10

Refresh Save Cancel

Export data

200 12

12 row(s) fetched - 0.001s, on 2026-07-28 at 21:45:43

KST en Writable Smart Insert 235 : 1 : 7617 Sel: 0 | 0

B-4. 가장 높은 평균 급여를 가진 부서

문제 번호(A-H)	B-4
분류	서브쿼리 종합실습
문제	[서브쿼리 4] 가장 높은 평균 급여를 가진 부서 HR 데이터를 활용하여 '가장 높은 평균 급여를 가진 부서' 결과를 조회하는 SQL 을 작성하십시오.
조건	서브쿼리를 사용해 조건을 해결하십시오. FROM 절 서브쿼리로 부서 평균을 만든 후 가장 큰 평균과 비교합니다.

실험결과

DBaver 26.1.3 - <postgres> day02_쿼리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Commit Rollback Auto postgres hr@skala_db postgres day02_PostgreSQL_HR_종합실습_문제.sql postgres day02_PostgreSQL_HR_emp_dept_샘플_스키마.sql postgres day02_쿼리_임유리.sql

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job_history 56K
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WHERE EXISTS (SELECT 1 FROM employees e WHERE e.department_id = d.department_id) ORDER BY d.department_id; -- [서브쿼리 4] 가장 높은 평균 급여를 가진 부서 -- 출처: HR 데이터를 활용하여 '가장 높은 평균 급여를 가진 부서' 결과를 조회하는 SQL을 작성하십시오. -- 조건: 서브쿼리를 사용해 조건을 해결하십시오, FROM 절 서브쿼리로 부서 평균을 만든 후 가장 큰 평균과 비교합니다. -- TODO: 아래의 SQL을 작성하십시오. SELECT da.department_id AS "부서 ID", d.department_name AS "부서명", ROUND(da.avg_salary, 2) AS "평균 급여" FROM (SELECT department_id, AVG(salary) AS avg_salary FROM employees WHERE department_id IS NOT NULL GROUP BY department_id) da INNER JOIN departments d ON d.department_id = da.department_id WHERE da.avg_salary = (SELECT MAX(x.avg_salary) FROM (SELECT AVG(salary) AS avg_salary FROM employees WHERE department_id IS NOT NULL GROUP BY department_id) x) ORDER BY da.department_id; -- [서브쿼리 5] 과거 직무 이력이 없는 사원 -- 출처: HR 데이터를 활용하여 '과거 직무 이력이 없는 사원' 결과를 조회하는 SQL을 작성하십시오. -- 조건: 서브쿼리를 사용해 조건을 해결하십시오, NOT EXISTS는 관련 행이 하나도 없는 대상을 찾을 때 사용합니다. -- TODO: 아래의 SQL을 작성하십시오. SELECT

employees(+) 1 X SELECT da.department_id AS "부서 ID", d.department_name A FROM employees e WHERE e.department_id = da.department_id

Grid 1 90 Executive 19,333.33

Record

Value X Edit value: 90 departments Type part of dictionary value to search Value Description 10 Administration 20 Marketing 30 Purchasing 40 Human Resources 50 Shipping 60 IT 70 Public Relations 80 Sales 90 Executive 100 Finance 110 Accounting 120 R&D Korea

Refresh Save Cancel Export data 200 1 1 row(s) fetched - 0.001s, on 2026-07-28 at 21:45:55

General KST en Writable Smart Insert 262 : 1 : 8365 Sel: 0 | 0

B-5. 과거 직무 이력이 없는 사원

문제 번호(A-H)	B-5
분류	서브쿼리 종합실습
문제	[서브쿼리 5] 과거 직무 이력이 없는 사원 HR 데이터를 활용하여 '과거 직무 이력이 없는 사원' 결과를 조회하는 SQL 을 작성하십시오.
조건	서브쿼리를 사용해 조건을 해결하십시오. NOT EXISTS 는 관련 행이 하나도 없는 대상을 찾을 때 안전합니다.

실험결과

DBBeaver 26.1.3 - <postgres> day02_쿼리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Auto

postgres hr@skala_db

Connections

프로젝트

<postgres> Script-3

<postgres> day02_PostgreSQL_HR_종합실습_문제.sql

<postgres> day02_PostgreSQL_HR_emp_dept_샘플_스키마.sql

<postgres> day02_쿼리_임유리.sql

Filter connections by name

DBBeaver Sample Database (SQLite)

postgres localhost:5432

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ORDER BY do.department_id;

-- [서브쿼리 5] 과거 직무 이력이 없는 사원
-- 문제: HR 데이터를 활용하여 '과거 직무 이력이 없는 사원' 결과를 조회하는 SQL 을 작성하십시오.
-- 조건: 서브쿼리를 사용해 조건을 해결하십시오. NOT EXISTS 는 관련 행이 하나도 없는 대상을 찾을 때 안전합니다.
-- TODO: 아래의 SQL 을 작성하십시오.

SELECT
e.employee_id AS "사원 ID",
concat_ws(' ', e.first_name, e.last_name) AS "사원명",
e.job_id AS "현재 직무 ID"
FROM employees e
WHERE NOT EXISTS (
SELECT 1
FROM job_history jh
WHERE jh.employee_id = e.employee_id
)
ORDER BY e.employee_id;

-- C. CTE 종합실습 (3문제)

-- [CTE 1] 부서별 평균 급여와 전체 평균 비교

employees 1 X

SELECT e.employee_id AS "사원 ID", concat_ws(' ', e.first_name, e.last_name) AS "사원명", e.job_id AS "현재 직무 ID"

1 100 Steven King AD_PRES
2 103 Alexander Hunold IT_PROG
3 104 Bruce Ernst IT_PROG
4 105 David Austin IT_PROG
5 106 Valli Pataballa IT_PROG
6 107 Diana Lorentz IT_PROG
7 108 Nancy Greenberg FL_MGR
8 109 Daniel Faviet FL_ACCOUNT
9 110 John Chen FL_ACCOUNT
10 111 Ismael Sciarra FL_ACCOUNT
11 112 Jose Manuel Urman FL_ACCOUNT
12 113 Luis Popp FL_ACCOUNT
13 115 Alexander Khoo PU_CLERK
14 116 Shell Baida PU_CLERK
15 117 Sigal Tobias PU_CLERK
16 118 Guy Himuro PU_CLERK
17 119 Karen Colmenares PU_CLERK
18 120 Matthew Weiss ST_MAN
19 121 Adam Fripp ST_MAN
20 145 John Russell SA_MAN
21 146 Karen Partners SA_MAN
22 147 Alberto Errazuriz SA_MAN
23 148 Gerald Cambrault SA_MAN
24 149 Eleni Zlotkey SA_MAN
25 150 Peter Tucker SA_REP
26 151 David Bernstein SA_REP
27 152 Peter Hall SA_REP
28 153 Christopher Olsen SA_REP
29 154 Nanette Cambrault SA_REP
30 155 Oliver Tuvault SA_REP
31 202 Pat Fay MK_REP
32 203 Susan Mavis HR_REP
33 204 Hermann Baer FR_REP
34 205 Shelley Higgins AC_MGR
35 206 William Gietz AC_ACCOUNT
36 221 Seoyeon Lee IT_PROG
37 222 Jihoon Park IT_PROG

Value X
151

Refresh Save Cancel Export data 200 37 37 row(s) fetched - 0.003s, on 2025-07-28 at 21:46:12

KST en Writable Smart Insert 279 : 1 : 881 Sel: 0 | 0

C-1. 부서별 평균 급여와 전체 평균 비교

문제 번호(A-H)	C-1
분류	CTE 종합실습
문제	[CTE 1] 부서별 평균 급여와 전체 평균 비교 HR 데이터를 활용하여 '부서별 평균 급여와 전체 평균 비교' 결과를 조회하는 SQL 을 작성하십시오.
조건	WITH 절(CTE)을 사용해 단계적으로 작성하십시오. WITH 절에서 부서 통계를 먼저 만든 뒤 메인 쿼리에서 사용합니다.

실험결과

Filter connections by name

postgres local:5432

postgres

skala.db

app

Tables

courses

enrollments

24K

majors

40K

students

56K

Foreign Tables

Views

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day02

public

hr

Tables

countries

40K

departments

72K

employees

88K

job_history

56K

jobs

24K

locations

40K

regions

40K

Foreign Tables

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Data types

Aggregate functions

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File Edit Navigate Search SQL Editor Database Window Help

Commit Rollback Auto

postgres hr@skala.db

postgres Script-3 postgres day02_PostgreSQL_HR_종합실습_문제.sql postgres day02_PostgreSQL_HR_emp_dept_샘플_스키마.sql postgres day02_쿼리_임유리.sql

Filter connections by name

postgres local:5432

postgres

skala.db

app

Tables

courses

enrollments

24K

majors

40K

students

56K

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

day02

public

hr

Tables

countries

40K

departments

72K

employees

88K

job_history

56K

jobs

24K

locations

40K

regions

40K

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Data types

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Administer

System Info

-- C. CTE 종합실습 (3문제)

--

-- [CTE 1] 부서별 평균 급여와 전체 평균 비교

-- 급여: HR 데이터를 활용하여 '부서별 평균 급여와 전체 평균 비교' 결과를 조회하는 SQL 을 작성하십시오.

-- 조건: WITH 절(CTE)을 사용해 단계적으로 작성하십시오. WITH 절에서 부서 통계를 먼저 만든 뒤 메인 쿼리에서 사용합니다.

-- TODO: 아래에 SQL 을 작성하십시오.

WITH department_stats AS (
SELECT department_id, AVG(salary) AS department_avg_salary
FROM employees
GROUP BY department_id
)
,
overall_stats AS (
SELECT AVG(salary) AS overall_avg_salary
FROM employees
)
SELECT
COLLECT(d.department_name, '부서 이름') AS "부서명",
ROUND(ds.department_avg_salary, 2) AS "부서 평균 급여",
ROUND(os.overall_avg_salary, 2) AS "전체 평균 급여",
ROUND(ds.department_avg_salary - os.overall_avg_salary, 2) AS "평균 차이",
CASE
WHEN ds.department_avg_salary >= os.overall_avg_salary THEN '전체 평균 이상'
ELSE '전체 평균 미만'
END AS "비교 결과"
FROM department_stats ds
LEFT JOIN overall_stats os ON d.department_id = ds.department_id
CROSS JOIN overall_stats os
ORDER BY ds.department_avg_salary DESC;

-- [CTE 2] 두 메인 CTE로 부서별 최고 급여자 찾기

4

Grid

1 2 3 4 5 6 7 8 9 10 11 12 13

1 Executive 19,333.33 8,724.44 10,608.89 전체 평균 이상

2 Sales 10,181.82 8,724.44 1,457.37 전체 평균 이상

3 Accounting 10,150 8,724.44 1,425.56 전체 평균 이상

4 Public Relations 10,000 8,724.44 1,275.56 전체 평균 이상

5 Marketing 9,500 8,724.44 775.56 전체 평균 이상

6 R&D Korea 8,933.33 8,724.44 208.89 전체 평균 이상

7 Finance 8,600 8,724.44 -124.44 전체 평균 미만

8 Shipping 8,033.33 8,724.44 -691.11 전체 평균 미만

9 Human Resources 6,500 8,724.44 -2,224.44 전체 평균 미만

10 부서 미배정 6,200 8,724.44 -2,524.44 전체 평균 미만

11 IT 5,760 8,724.44 -2,964.44 전체 평균 미만

12 Administration 4,400 8,724.44 -4,324.44 전체 평균 미만

13 Purchasing 4,150 8,724.44 -4,574.44 전체 평균 미만

Value X

Text

Executive

Refresh Save Cancel

Export data

200

13

13 row(s) fetched - 0.001s, on 2025-07-28 at 21:47:04

KST en Writable Smart Insert 311 : 1 : 9942 Sel: 0 | 0

C-2. 두 단계 CTE 로 부서별 최고 급여자 찾기

문제 번호(A-H)	C-2
분류	CTE 종합실습
문제	[CTE 2] 두 단계 CTE 로 부서별 최고 급여자 찾기 HR 데이터를 활용하여 '두 단계 CTE 로 부서별 최고 급여자 찾기' 결과를 조회하는 SQL 을 작성하십시오.
조건	WITH 절(CTE)을 사용해 단계적으로 작성하십시오. 첫 번째 CTE 의 결과를 두 번째 CTE 가 다시 참조합니다.

실험결과

DBBeaver 26.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Commit Rollback Auto postgres hr@skala_db postgres day02_PostgreSQL_HR_종합실습_문제.sql postgres day02_PostgreSQL_HR_emp_dept_샘플_스키마.sql postgres day02_책리_임유리.sql

Filter connections by name

postgres local:localhost-5432

postgres

skala_db

Schemas

app

Tables

courses

enrollments

24K

majors

40K

students

56K

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day02

public

hr

Tables

countries

40K

departments

72K

employees

88K

job_history

56K

jobs

24K

locations

40K

regions

40K

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WITH department_max_salary AS (SELECT department_id, MAX(salary) AS max_salary FROM employees GROUP BY department_id), department_top_earners AS (SELECT e.employee_id, e.first_name, e.last_name, e.department_id, e.salary FROM employees e INNER JOIN department_max_salary dm ON dm.department_id IS NOT DISTINCT FROM e.department_id AND dm.max_salary = e.salary) SELECT COALESCE(d.department_name, '부서 미배정') AS "부서명", dte.employee_id AS "사원 ID", concat_ws(' ', dte.first_name, dte.last_name) AS "최고 급여자", dte.salary AS "최고 급여" FROM department_top_earners dte LEFT JOIN departments d ON d.department_id = dte.department_id ORDER BY dte.department_id NULLS LAST, dte.employee_id;

employees 1 X

WITH department_max_salary AS (SELECT department_id, MA

Grid

123 부서명

123 사원 ID

AZ 최고 급여자

123 최고 급여

1 Administration 200 Jennifer Whalen 4,400

2 Marketing 201 Michael Hartstein 13,000

3 Purchasing 114 Den Raphaely 11,000

4 Human Resources 203 Susan Mavris 6,500

5 Shipping 121 Adam Fripp 8,200

6 IT 103 Alexander Hunold 9,000

7 Public Relations 204 Hermann Baer 10,000

8 Sales 145 John Russell 14,000

9 Executive 100 Steven King 24,000

10 Finance 108 Nancy Greenberg 12,000

11 Accounting 205 Shelley Higgins 12,000

12 R&D Korea 220 Minjux Kim 10,500

13 부서 미배정 223 Yuna Choi 6,200

Value X

Text

Administration

Refresh Save Cancel Export data 200 13 13 row(s) fetched - 0.001s, on 2025-07-28 at 21:47:21

General

C-3. 재귀 CTE 로 사원 조직도 조회

문제 번호(A-H)	C-3
분류	CTE 종합실습
문제	[CTE 3] 재귀 CTE 로 사원 조직도 조회 HR 데이터를 활용하여 '재귀 CTE 로 사원 조직도 조회' 결과를 조회하는 SQL 을 작성하십시오.
조건	WITH 절(CTE)을 사용해 단계적으로 작성하십시오. 최고경영자를 시작점으로 manager_id 관계를 반복하여 내려갑니다.

실험결과

DBaveer 26.1.3 - <postgres> day02_쿼리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Connections postgres localhost:5432

Filter connections by name

postgres localhost:5432

Databases

postgres

skala_db

Schemas

app

Tables

courses

enrollments

24K

majors

40K

students

56K

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day02

public

hr

Tables

countries

40K

departments

72K

employees

88K

job_history

56K

jobs

24K

locations

40K

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40K

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Roles

Administer

System info

SQL Editor

<postgres> Script-3 <postgres> day02_PostgreSQL_HR_종합실습_문제.sql <postgres> day02_PostgreSQL_HR_emp_dept_샘플_스키마.sql <postgres> day02_쿼리_임유리.sql

-- [CTE 3] 재귀 CTE로 사원 조직도 조회
-- 목적: HR 데이터를 활용하여 '재귀 CTE로 사원 조직도 조회' 결과를 조회하는 SQL을 작성하십시오.
-- 조건: WITH 절(CTE)을 사용해 단계적으로 작성하십시오. 최고경영자를 시작점으로 manager_id 관계를 반복하여 내려갑니다.
-- TODO: 아래에 SQL을 작성하십시오.
WITH RECURSIVE organization AS (
SELECT
e.employee_id, e.first_name, e.last_name, e.manager_id,
e.department_id, 1 AS level,
e.employee_id AS organization_path
FROM employees e
WHERE e.employee_id = 100

UNION ALL

SELECT
e.employee_id, e.first_name, e.last_name, e.manager_id,
e.department_id, o.level + 1,
o.organization_path || e.employee_id
FROM employees e
INNER JOIN organization o ON e.manager_id = o.employee_id
WHERE NOT e.employee_id = ANY(o.organization_path)
)
SELECT
level AS "레벨",
employee_id AS "사원 ID",
repeat(' ', level - 1) || concat_ws(' ', first_name, last_name) AS "조직도",
manager_id AS "관리자 ID",
department_id AS "부서 ID"
FROM organization
ORDER BY organization_path;

organization 1 x

WITH RECURSIVE organization AS (SELECT e.employee_id, e.first_name, e.last_name, e.manager_id, e.department_id, 1 AS level, e.employee_id AS organization_path FROM employees e WHERE e.employee_id = 100 UNION ALL SELECT e.employee_id, e.first_name, e.last_name, e.manager_id, e.department_id, o.level + 1, o.organization_path || e.employee_id FROM employees e INNER JOIN organization o ON e.manager_id = o.employee_id WHERE NOT e.employee_id = ANY(o.organization_path)) SELECT level AS "레벨", employee_id AS "사원 ID", repeat(' ', level - 1) || concat_ws(' ', first_name, last_name) AS "조직도", manager_id AS "관리자 ID", department_id AS "부서 ID" FROM organization ORDER BY organization_path;

1 1 100 Steven King [NULL] 90
2 2 101 Neena Kochhar 100 90
3 3 108 Nancy Greenberg 101 100
4 4 109 Daniel Faviet 108 100
5 4 110 John Chen 108 100
6 4 111 Ismael Sierra 108 100
7 4 112 Jose Manuel Urman 108 100
8 4 113 Luis Popp 108 100
9 3 200 Jennifer Whalen 101 10
10 3 203 Susan Mavris 101 40
11 3 204 Hermann Baer 101 70
12 3 205 Shelley Higgins 101 110
13 4 206 William Gietz 205 110
14 2 102 Lex De Haan 100 90
15 3 103 Alexander Hunold 102 60
16 4 104 Bruce Ernst 103 60
17 4 105 David Austin 103 60
18 4 106 Vali Pataballa 103 60
19 4 107 Diana Lorentz 103 60
20 3 220 Minjun Kim 102 120
21 4 221 Seoyeon Lee 220 120
22 4 222 Jihoon Park 220 120
23 2 114 Den Raphaely 100 30
24 3 115 Alexander Khoo 114 30
25 3 116 Shelli Baida 114 30
26 3 117 Sigal Tobias 114 30
27 3 118 Guy Himuro 114 30
28 3 119 Karen Colmenares 114 30
29 2 120 Matthew Weiss 100 50
30 2 121 Adam Fripp 100 50
31 2 122 Payam Kauffling 100 50
32 2 145 John Russell 100 80
33 3 150 Peter Tucker 145 80
34 3 151 David Bernstein 145 80

Refresh Save Cancel Export data 200 45 ... 45 row(s) fetched - 0.002s, on 2026-07-28 at 21:47:42

General KST en Writable Smart Insert 370 : 1 : 11884 Sel: 0 | 0

DBeaver 25.1.3 - <postgres> day02_관리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Commit Rollback Auto postgres hr@skala_db hr@skala_db <postgres> day02_관리_임유리.sql

Connections 프로젝트

Filter connections by name

DBeaver Sample Database (SQLite)

postgres localhost:5432

Databases

postgres

skala_db

Schemas

app

Tables

courses 24K

enrollments 24K

majors 40K

students 56K

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day02

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departments 72K

employees 88K

job_history 56K

jobs 24K

locations 40K

regions 40K

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Script-3

<postgres> day02_PostgresQL_HR_종합실습_문제.sql

<postgres> day02_PostgresQL_HR_emp_dept_샘플_스키마.sql

<postgres> day02_관리_임유리.sql

```
-- [CTE 3] 관리 CTE로 사용 조직도 조회
-- 출처: HR 데이터베이스를 활용하여 '관리 CTE로 사용 조직도 조회' 결과를 조회하는 SQL을 작성하십시오.
-- 조건: WITH 절(CTE)을 사용해 단계적으로 작성하십시오. 최종결과를 시작점으로 manager_id 관계를 반복하여 내려갑니다.
-- TODO: 아래와 SQL을 작성하십시오.

WITH RECURSIVE organization AS (
    SELECT
        e.employee_id, e.first_name, e.last_name, e.manager_id,
        e.department_id, 1 AS level,
        HRNVL(e.employee_id) AS organization_path
    FROM employees e
    WHERE e.employee_id = 100

    UNION ALL

    SELECT
        e.employee_id, e.first_name, e.last_name, e.manager_id,
        e.department_id, o.level + 1,
        o.organization_path || e.employee_id
    FROM employees e
    INNER JOIN organization o ON e.manager_id = o.employee_id
    WHERE NOT e.employee_id = ANY(o.organization_path)
)

SELECT
    level AS "레벨",
    employee_id AS "사원 ID",
    repeat('-', level - 1) || concat_ws(' ', first_name, last_name) AS "조직도",
    department_id AS "관리자 ID",
    FROM organization
ORDER BY organization_path;
```

organization 1 X

WITH RECURSIVE organization AS (SELECT e.employee_id, e.first_name, e.last_name, e.manager_id, e.department_id, 1 AS level, HRNVL(e.employee_id) AS organization_path FROM employees e WHERE e.employee_id = 100 UNION ALL SELECT e.employee_id, e.first_name, e.last_name, e.manager_id, e.department_id, o.level + 1, o.organization_path || e.employee_id FROM employees e INNER JOIN organization o ON e.manager_id = o.employee_id WHERE NOT e.employee_id = ANY(o.organization_path))

레벨	123 사원 ID	A7 조직도	123 관리자 ID	123 부서 ID
13	4	William Gietz	205	110
14	2	Lex De Haan	100	90
15	3	Alexander Hunold	102	60
16	4	Bruce Ernst	103	60
17	4	David Austin	103	60
18	4	Valli Pataballa	103	60
19	4	Diana Lorentz	103	60
20	3	Minjun Kim	102	120
21	4	Seoyeon Lee	220	120
22	4	Jihoon Park	220	120
23	2	Den Raphaely	100	30
24	3	Alexander Khoo	114	30
25	3	Shelli Baida	114	30
26	3	Sigal Tobias	114	30
27	3	Guy Himuro	114	30
28	3	Karen Colmenares	114	30
29	2	Matthew Weiss	100	50
30	2	Adam Frippe	100	50
31	2	Payam Kaufling	100	50
32	2	John Russell	100	80
33	3	Peter Tucker	145	80
34	3	David Bernstein	145	80
35	3	Peter Hall	145	80
36	3	Christopher Olsen	145	80
37	3	Nanette Cambrault	145	80
38	3	Oliver Tuvault	145	80
39	2	Karen Partners	100	80
40	2	Alberto Errazuriz	100	80
41	2	Gerald Cambrault	100	80
42	2	Eleni Zlotkey	100	80
43	2	Michael Hartstein	100	20
44	3	Pat Fay	201	20
45	3	Yuna Choi	201	[NULL]

Record

Refresh Save Cancel

Export data 200 45 45 row(s) fetched - 0.002s, on 2025-07-28 at 21:47:42

KST en Writable Smart Insert 370 : 1 : 11884 Sel: 0 | 0

D-1. 사원과 부서

문제 번호(A-H)	D-1
분류	JOIN 종합실습
문제	[JOIN 1: INNER JOIN] 사원과 부서 HR 데이터를 활용하여 '사원과 부서' 결과를 조회하는 SQL 을 작성하십시오.
조건	INNER JOIN 방식을 활용하십시오. 부서가 배정된 사원만 출력합니다.

실험결과

DBEaver 26.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Rollback Commit SQL

Connections

Filter connections by name

DBEaver Sample Database (SQLite)

postgres localhost-5432

id postgres

skala_db

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<postgres> Script-3

<postgres> day02_PostgresSQL_HR_종합실습_문제.sql

<postgres> day02_PostgresSQL_HR_emp_dept_샘플_스키마.sql

<postgres> day02_책리_임유리.sql

-- D. JOIN 종합실습 (19문제)

-- [JOIN 1: INNER JOIN] 사원과 부서

-- 문제: HR 데이터를 활용하여 '사원과 부서' 결과를 조회하는 SQL 을 작성하십시오.

-- 조건: INNER JOIN 방식을 활용하십시오. 부서가 배정된 사원만 출력합니다.

-- TODO: 아래에 SQL 을 작성하십시오.

SELECT

e.employee_id AS "사원 ID",

concat_ws(' ', e.first_name, e.last_name) AS "사원명",

d.department_id AS "부서 ID",

d.department_name AS "부서명"

FROM employees e

INNER JOIN departments d ON d.department_id = e.department_id

ORDER BY e.employee_id;

employees(*) 1 X

SELECT e.employee_id AS "사원 ID", concat_ws(' ', e.first_name, e.last_name) AS "사원명", d.department_id AS "부서 ID", d.department_name AS "부서명"

1 100 Steven King 90 Executive

2 101 Neena Kochhar 90 Executive

3 102 Lee De Haan 90 Executive

4 103 Alexander Humold 60 IT

5 104 Bruce Ernst 60 IT

6 105 David Austin 60 IT

7 106 Valli Pataballa 60 IT

8 107 Diana Lorentz 60 IT

9 108 Nancy Greenberg 100 Finance

10 109 Daniel Faviet 100 Finance

11 110 John Chen 100 Finance

12 111 Ismael Sciarra 100 Finance

13 112 Jose Manuel Uman 100 Finance

14 113 Luis Popp 100 Finance

15 114 Den Raphaely 30 Purchasing

16 115 Alexander Khoo 30 Purchasing

17 116 Shelli Baida 30 Purchasing

18 117 Sigal Tobias 30 Purchasing

19 118 Guy Himuro 30 Purchasing

20 119 Karen Colmenares 30 Purchasing

21 120 Matthew Weiss 50 Shipping

22 121 Adam Fripp 50 Shipping

23 122 Payam Kaufling 50 Shipping

24 145 John Russell 80 Sales

25 146 Karen Partners 80 Sales

26 147 Alberto Errazuriz 80 Sales

27 148 Gerald Cambrault 80 Sales

28 149 Eleni Zlotkey 80 Sales

29 150 Peter Tucker 80 Sales

30 151 David Bernstein 80 Sales

31 152 Peter Hall 80 Sales

32 153 Christopher Olsen 80 Sales

33 154 Nanette Cambrault 80 Sales

34 155 Oliver Tuvault 80 Sales

35 200 Jennifer Whalen 10 Administration

36 201 Michael Hartstein 20 Marketing

37 202 Pat Fay 20 Marketing

38 203 Susan Mavis 40 Human Resources

39 204 Hermann Baer 70 Public Relations

40 205 Shelley Higgins 110 Accounting

41 206 William Gietz 110 Accounting

42 220 Minjun Kim 120 R&D Korea

43 221 Seoyeon Lee 120 R&D Korea

44 222 Jihoon Park 120 R&D Korea

Value X

100

Refresh Save Cancel

Export data

200

44

44 row(s) fetched - 0.001s, on 2026-07-28 at 21:48:19

KST en Writable Smart Insert 388 : 1 : 12487 Sel: 0 | 0

D-2. 사원과 직무

문제 번호(A-H)	D-2
분류	JOIN 종합실습
문제	[JOIN 2: INNER JOIN] 사원과 직무 HR 데이터를 활용하여 '사원과 직무' 결과를 조회하는 SQL 을 작성하십시오.
조건	INNER JOIN 방식을 활용하십시오. job_id 가 일치하는 직무명을 사원 정보에 결합합니다.

실험결과

DBeaver 21.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Auto Rollback

Connections X 프로젝트

Filter connections by name

- DBeaver Sample Database (SQLite)
 - postgres localhost:5432
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 - postgres
 - skala_db
 - Schemas
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 - enrollments 40K
 - enrollments 56K
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 - countries 40K
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 - job_history 56K
 - jobs 24K
 - locations 40K
 - regions 40K
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 - Administrator
 - System info

SQL Editor

```
-- [JOIN 2: INNER JOIN] 사람과 직무
-- 문제: HR 데이터를 활용하여 '사람과 직무' 결과를 조회하는 SQL을 작성하십시오.
-- 조건: INNER JOIN 방식을 활용하십시오. job_id가 일치하는 직무를 사용 정보에 결합합니다.
-- TODO: 아래의 SQL을 작성하십시오.

SELECT
  e.employee_id AS "사람 ID",
  concat_ws(' ', e.first_name, e.last_name) AS "이름",
  j.job_id AS "직무 ID",
  j.job_title AS "직무명",
  e.salary AS "급여"
FROM employees e
INNER JOIN jobs j ON j.job_id = e.job_id
ORDER BY e.employee_id;
```

employees(+) 1 X

SQL Editor

```
-- SELECT e.employee_id AS "사람 ID", concat_ws(' ', e.first_name, e.last_name) AS "이름",
-- j.job_id AS "직무 ID", j.job_title AS "직무명", e.salary AS "급여"
-- FROM employees e
-- INNER JOIN jobs j ON j.job_id = e.job_id
-- ORDER BY e.employee_id;
```

Grid

| | 103 사람 ID | AZ 사람명 | AZ 직무 ID | AZ 직무명 | 123 급여 |
|----|-----------|-------------------|------------|---------------------------------|--------|
| 1 | 100 | Steven King | AD_PRES | President | 24,000 |
| 2 | 101 | Neena Kochhar | AD_VP | Administration Vice President | 17,000 |
| 3 | 102 | Lee De Haan | AD_VP | Administration Vice President | 17,000 |
| 4 | 103 | Alexander Hunold | IT_PROG | Programmer | 9,000 |
| 5 | 104 | Bruce Ernst | IT_PROG | Programmer | 6,000 |
| 6 | 105 | David Austin | IT_PROG | Programmer | 4,800 |
| 7 | 106 | Valli Pataballa | IT_PROG | Programmer | 4,800 |
| 8 | 107 | Diana Lorentz | IT_PROG | Programmer | 4,200 |
| 9 | 108 | Nancy Greenberg | FI_MGR | Finance Manager | 12,000 |
| 10 | 109 | Daniel Faviet | FI_ACCOUNT | Accountant | 9,000 |
| 11 | 110 | John Chen | FI_ACCOUNT | Accountant | 8,200 |
| 12 | 111 | Ismael Scarra | FI_ACCOUNT | Accountant | 7,700 |
| 13 | 112 | Jose Manuel Urman | FI_ACCOUNT | Accountant | 7,800 |
| 14 | 113 | Luis Popp | FI_ACCOUNT | Accountant | 6,900 |
| 15 | 114 | Den Raphaely | PU_MAN | Purchasing Manager | 11,000 |
| 16 | 115 | Alexander Khoo | PU_CLERK | Purchasing Clerk | 3,100 |
| 17 | 116 | Shelli Baida | PU_CLERK | Purchasing Clerk | 2,900 |
| 18 | 117 | Sigal Tobias | PU_CLERK | Purchasing Clerk | 2,800 |
| 19 | 118 | Guy Himuro | PU_CLERK | Purchasing Clerk | 2,600 |
| 20 | 119 | Karen Colmenares | PU_CLERK | Purchasing Clerk | 2,500 |
| 21 | 120 | Matthew Weiss | ST_MAN | Stock Manager | 8,000 |
| 22 | 121 | Adam Fripp | ST_MAN | Stock Manager | 8,200 |
| 23 | 122 | Payam Kaufling | ST_MAN | Stock Manager | 7,900 |
| 24 | 145 | John Russell | SA_MAN | Sales Manager | 14,000 |
| 25 | 146 | Karen Partners | SA_MAN | Sales Manager | 13,500 |
| 26 | 147 | Alberto Errazuriz | SA_MAN | Sales Manager | 12,000 |
| 27 | 148 | Gerald Cambrault | SA_MAN | Sales Manager | 11,000 |
| 28 | 149 | Eleni Zlotkey | SA_MAN | Sales Manager | 10,500 |
| 29 | 150 | Peter Tucker | SA_REP | Sales Representative | 10,000 |
| 30 | 151 | David Bernstein | SA_REP | Sales Representative | 9,500 |
| 31 | 152 | Peter Hall | SA_REP | Sales Representative | 9,000 |
| 32 | 153 | Christopher Olsen | SA_REP | Sales Representative | 8,000 |
| 33 | 154 | Nanette Cambrault | SA_REP | Sales Representative | 7,500 |
| 34 | 155 | Oliver Tuvault | SA_REP | Sales Representative | 7,000 |
| 35 | 200 | Jennifer Whalen | AD_ASST | Administration Assistant | 4,400 |
| 36 | 201 | Michael Hartstein | MK_MAN | Marketing Manager | 13,000 |
| 37 | 202 | Pat Fay | MK_REP | Marketing Representative | 6,000 |
| 38 | 203 | Susan Mavris | HR_REP | Human Resources Representative | 6,500 |
| 39 | 204 | Hermann Baer | PR_REP | Public Relations Representative | 10,000 |
| 40 | 205 | Shelley Higgins | AC_MGR | Accounting Manager | 12,000 |
| 41 | 206 | William Gietz | AC_ACCOUNT | Public Accountant | 8,300 |
| 42 | 220 | Marjun Kim | IT_PROG | Programmer | 10,500 |
| 43 | 221 | Seoyeon Lee | IT_PROG | Programmer | 8,500 |
| 44 | 222 | Jihoon Park | IT_PROG | Programmer | 7,800 |
| 45 | 223 | Yuna Choi | MK_REP | Marketing Representative | 6,200 |

Value X

100

Refresh Save Cancel Export data 200 45 45 row(s) fetched - 0s, on 2026-07-28 at 21:48:45

KST en Writable Smart Insert 403 : 1 : 12907 Sel: 0 | 0

D-3. 사원의 부서·도시·국가

| | |
|------------|---|
| 문제 번호(A-H) | D-3 |
| 분류 | JOIN 종합실습 |
| 문제 | [JOIN 3: 4 개 테이블 INNER JOIN] 사원의 부서·도시·국가
HR 데이터를 활용하여 '사원의 부서·도시·국가' 결과를 조회하는 SQL 을 작성하십시오. |
| 조건 | 4 개 테이블 INNER JOIN 방식을 활용하십시오. 사원에서 국가까지 FK 관계를 차례로 연결합니다. |

실험결과

DBEaver 26.1.3 - <postgres> day02_쿼리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Filter connections by name

postgres localhost:5432

postgres

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Script-3

<postgres> day02_PostgreSQL_HR_종합실습_문제.sql

<postgres> day02_PostgreSQL_HR_emp_dept_샘플_스키마.sql

<postgres> day02_쿼리_임유리.sql

SELECT

e.employee_id AS "사원 ID",

concat_ws(' ', e.first_name, e.last_name) AS "사원명",

d.department_name AS "부서명",

l.city AS "도시",

c.country_name AS "국가명"

FROM employees e

INNER JOIN departments d ON d.department_id = e.department_id

INNER JOIN locations l ON l.location_id = d.location_id

INNER JOIN countries c ON c.country_id = l.country_id

employees(*) 1 X

SELECT e.employee_id AS "사원 ID", concat_ws(' ', e.first_name, e.last_name) AS "사원명", d.department_name AS "부서명", l.city AS "도시", c.country_name AS "국가명"

1 100 Steven King Executive Seattle United States

2 101 Neena Kochhar Executive Seattle United States

3 102 Lex De Haan Executive Seattle United States

4 103 Alexander Hunsold IT Southlake United States

5 104 Bruce Ernst IT Southlake United States

6 105 David Austin IT Southlake United States

7 106 Valli Pataballa IT Southlake United States

8 107 Diana Lorentz IT Southlake United States

9 108 Nancy Greenberg Finance Seattle United States

10 109 Daniel Faviet Finance Seattle United States

11 110 John Chen Finance Seattle United States

12 111 Ismael Scarra Finance Seattle United States

13 112 Jose Manuel Urman Finance Seattle United States

14 113 Luis Popp Finance Seattle United States

15 114 Den Raphaely Purchasing Seattle United States

16 115 Alexander Khoo Purchasing Seattle United States

17 116 Shelli Baida Purchasing Seattle United States

18 117 Sigal Tobias Purchasing Seattle United States

19 118 Guy Himuro Purchasing Seattle United States

20 119 Karen Colmenares Purchasing Seattle United States

21 120 Matthew Weiss Shipping South San Francisco United States

22 121 Adam Fripp Shipping South San Francisco United States

23 122 Payam Kaufing Shipping South San Francisco United States

24 145 John Russell Sales Oxford United Kingdom

25 146 Karen Partners Sales Oxford United Kingdom

26 147 Alberto Errazuriz Sales Oxford United Kingdom

27 148 Gerald Cambrault Sales Oxford United Kingdom

28 149 Eleni Zlotkey Sales Oxford United Kingdom

29 150 Peter Tucker Sales Oxford United Kingdom

30 151 David Bernstein Sales Oxford United Kingdom

31 152 Peter Hall Sales Oxford United Kingdom

32 153 Christopher Olsen Sales Oxford United Kingdom

33 154 Nanette Cambrault Sales Oxford United Kingdom

34 155 Oliver Tuvault Sales Oxford United Kingdom

35 200 Jennifer Whalen Administration Seattle United States

36 201 Michael Hartstein Marketing Toronto Canada

37 202 Pat Fay Marketing Toronto Canada

38 203 Susan Mavris Human Resources London United Kingdom

39 204 Hermann Baer Public Relations Munich Germany

40 205 Shelley Higgins Accounting Seattle United States

41 206 William Gietz Accounting Seattle United States

42 220 Minjun Kim R&D Korea Seoul South Korea

43 221 Seoyeon Lee R&D Korea Seoul South Korea

44 222 Jihoon Park R&D Korea Seoul South Korea

Value X

100

Refresh

Save

Cancel

Export data

200

44

44 row(s) fetched - 0.001s, on 2025-07-28 at 21:49:01

KST en Writable Smart Insert 418 : 54 : 13463 Sel: 0 | 0

D-4. 사원과 직속 관리자

| | |
|------------|--|
| 문제 번호(A-H) | D-4 |
| 분류 | JOIN 종합실습 |
| 문제 | [JOIN 4: SELF JOIN] 사원과 직속 관리자
HR 데이터를 활용하여 '사원과 직속 관리자' 결과를 조회하는 SQL 을 작성하십시오. |
| 조건 | SELF JOIN 방식을 활용하십시오. 동일한 employees 테이블을 사원용 e 와 관리자용 m 으로 두 번 사용합니다. |

실험결과

The screenshot shows the DBeaver 21.0.6 interface with a SQL query and its results. The query is a self-join on the employees table to find managers. The results table shows columns for employee ID, name, manager ID, and manager name.

SQL Query:

```
-- [JOIN 4: SELF JOIN] 사원과 직속 관리자
-- 목적: HR 데이터를 활용하여 '사원과 직속 관리자' 결과를 조회하는 SQL을 작성하십시오.
-- 조건: SELF JOIN 방식을 활용하십시오. 동일한 employees 테이블을 사원과 e 관리자 테이블을 m으로 두 번 사용합니다.
-- TODO: 아래와 SQL을 작성하십시오.

SELECT
    e.employee_id AS "사원 ID",
    concat_ws(' ', e.first_name, e.last_name) AS "사원명",
    m.employee_id AS "관리자 ID",
    concat_ws(' ', m.first_name, m.last_name) AS "직속 관리자명"
FROM employees e
INNER JOIN employees m ON m.employee_id = e.manager_id
ORDER BY e.employee_id;
```

Query Results:

| 사원 ID | 사원명 | 관리자 ID | 직속 관리자명 |
|-------|-------------------|--------|-------------------|
| 101 | Neena Kochhar | 100 | Steven King |
| 102 | Lex De Haan | 100 | Steven King |
| 103 | Alexander Humold | 102 | Lex De Haan |
| 104 | Bruce Ernst | 103 | Alexander Humold |
| 105 | David Austin | 103 | Alexander Humold |
| 106 | Valli Pataballa | 103 | Alexander Humold |
| 107 | Diana Lorentz | 103 | Alexander Humold |
| 108 | Nancy Greenberg | 101 | Neena Kochhar |
| 109 | Daniel Faviet | 108 | Nancy Greenberg |
| 110 | John Chen | 108 | Nancy Greenberg |
| 111 | Ismael Scarra | 108 | Nancy Greenberg |
| 112 | Jose Manuel Urman | 108 | Nancy Greenberg |
| 113 | Luis Popp | 108 | Nancy Greenberg |
| 114 | Den Raphaely | 100 | Steven King |
| 115 | Alexander Khoo | 114 | Den Raphaely |
| 116 | Shelli Baida | 114 | Den Raphaely |
| 117 | Sigal Tobias | 114 | Den Raphaely |
| 118 | Guy Himuro | 114 | Den Raphaely |
| 119 | Karen Colmenares | 114 | Den Raphaely |
| 120 | Matthew Weiss | 100 | Steven King |
| 121 | Adam Fripp | 100 | Steven King |
| 122 | Payam Kauffing | 100 | Steven King |
| 145 | John Russell | 100 | Steven King |
| 146 | Karen Partners | 100 | Steven King |
| 147 | Alberto Errazuriz | 100 | Steven King |
| 148 | Gerald Cambrault | 100 | Steven King |
| 149 | Eleni Zlotkey | 100 | Steven King |
| 150 | Peter Tucker | 145 | John Russell |
| 151 | David Bernstein | 145 | John Russell |
| 152 | Peter Hall | 145 | John Russell |
| 153 | Christopher Olsen | 145 | John Russell |
| 154 | Nanette Cambrault | 145 | John Russell |
| 155 | Oliver Tuvault | 145 | John Russell |
| 200 | Jennifer Whalen | 101 | Neena Kochhar |
| 201 | Michael Hartstein | 100 | Steven King |
| 202 | Pat Fay | 201 | Michael Hartstein |
| 203 | Susan Mavris | 101 | Neena Kochhar |
| 204 | Hermann Baer | 101 | Neena Kochhar |
| 205 | Shelley Higgins | 101 | Neena Kochhar |
| 206 | William Gietz | 205 | Shelley Higgins |
| 220 | Minjun Kim | 102 | Lex De Haan |
| 221 | Seoyeon Lee | 220 | Minjun Kim |
| 222 | Jihoon Park | 220 | Minjun Kim |
| 223 | Yuna Choi | 201 | Michael Hartstein |

D-5. 부서 미배정 직원 포함

| | |
|------------|--|
| 문제 번호(A-H) | D-5 |
| 분류 | JOIN 종합실습 |
| 문제 | [JOIN 5: LEFT OUTER JOIN] 부서 미배정 직원 포함
HR 데이터를 활용하여 '부서 미배정 직원 포함' 결과를 조회하는 SQL 을 작성하십시오. |
| 조건 | LEFT OUTER JOIN 방식을 활용하십시오. 왼쪽 employees 의 모든 행을 보존하므로
부서가 없는 직원도 나옵니다. |

실험결과

DBEAVER 26.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Connections x 프로젝트

Filter connections by name

postgres localhost:5432

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SQL Script-3

<postgres> day02_PostgreSQL_HR_종합실습_문제.sql

<postgres> day02_PostgreSQL_HR_emp_dept_상행_스키마.sql

<postgres> day02_책리_임유리.sql

ORDER BY e.employee_id;

-- [JOIN 5: LEFT OUTER JOIN] 부서 미배정 직원 포함
-- 목적: HR 데이터를 활용하여 '부서 미배정 직원 포함' 결과를 조회하는 SQL을 작성하십시오.
-- 조건: LEFT OUTER JOIN 방식을 활용하십시오. 왼쪽 employees의 모든 행을 보존하므로 부서가 없는 직원도 나옵니다.
-- TODO: 아래에 SQL을 작성하십시오.

SELECT
 e.employee_id AS "직원 ID",
 concat_ws(' ', e.first_name, e.last_name) AS "이름",
 COALESCE(d.department_name, '부서 미배정') AS "부서명"
FROM employees e
LEFT OUTER JOIN departments d ON d.department_id = e.department_id
ORDER BY e.employee_id;

employees 1 x

SELECT e.employee_id AS "직원 ID", concat_ws(' ', e.first_name, e.last_name) AS "이름", COALESCE(d.department_name, '부서 미배정') AS "부서명"

1 100 Steven King Executive

2 101 Neena Kochhar Executive

3 102 Lex De Haan Executive

4 103 Alexander Hunsold IT

5 104 Bruce Ernst IT

6 105 David Austin IT

7 106 Vali Pataballa IT

8 107 Diana Lorentz IT

9 108 Nancy Greenberg Finance

10 109 Daniel Faviet Finance

11 110 John Chen Finance

12 111 Ismael Sciarra Finance

13 112 Jose Manuel Urman Finance

14 113 Luis Popp Finance

15 114 Den Raphaely Purchasing

16 115 Alexander Khoo Purchasing

17 116 Shelli Baida Purchasing

18 117 Sigal Tobias Purchasing

19 118 Guy Himuro Purchasing

20 119 Karen Colmenares Purchasing

21 120 Matthew Weiss Shipping

22 121 Adam Fripp Shipping

23 122 Payam Kauffing Shipping

24 145 John Russell Sales

25 146 Karen Partners Sales

26 147 Alberto Errazuriz Sales

27 148 Gerald Cambrault Sales

28 149 Eleni Zlotkey Sales

29 150 Peter Tucker Sales

30 151 David Bernstein Sales

31 152 Peter Hall Sales

32 153 Christopher Olsen Sales

33 154 Nanette Cambrault Sales

34 155 Oliver Tuvault Sales

35 200 Jennifer Whalen Administration

36 201 Michael Hartstein Marketing

37 202 Pat Fay Marketing

38 203 Susan Mavis Human Resources

39 204 Hermann Baer Public Relations

40 205 Shelley Higgins Accounting

41 206 William Gietz Accounting

42 220 Minjun Kim R&D Korea

43 221 Seoyeon Lee R&D Korea

44 222 Jihoon Park R&D Korea

45 223 Yuna Choi 부서 미배정

Value x

100

Refresh Save Cancel

Export data

200

45

45 row(s) fetched - 0.001s, on 2026-07-28 at 21:49:32

KST en Writable

Smart Insert

447 : 1 : 14418

Set: 0 : 0

D-6. 사원이 없는 부서 포함

| | |
|------------|--|
| 문제 번호(A-H) | D-6 |
| 분류 | JOIN 종합실습 |
| 문제 | [JOIN 6: LEFT OUTER JOIN] 사원이 없는 부서 포함
HR 데이터를 활용하여 '사원이 없는 부서 포함' 결과를 조회하는 SQL 을 작성하십시오. |
| 조건 | LEFT OUTER JOIN 방식을 활용하십시오. 모든 부서를 보존하고 소속 사원이 없으면
사원 열을 NULL 로 표시합니다. |

실험결과

DBEAVER 26.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Filter connections by name

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Script-3

<postgres> day02_PostgreSQL_HR_종합실습_문제.sql

<postgres> day02_PostgreSQL_HR_emp_dept_상표_스키마.sql

<postgres> day02_책리_임유리.sql

SELECT

d.department_id AS "부서 ID",

d.department_name AS "부서명",

e.employee_id AS "사원 ID",

concat_ws(' ', e.first_name, e.last_name) AS "사원명"

FROM departments d

LEFT OUTER JOIN employees e ON e.department_id = d.department_id

ORDER BY d.department_id, e.employee_id NULLS LAST;

departments(+) 1 X

SELECT d.department_id AS "부서 ID", d.department_name AS "부서명"

Enter a SQL expression to filter results (use Ctrl+Space)

1 10 Administration 200 Jennifer Whalen

2 20 Marketing 201 Michael Hartstein

3 20 Marketing 202 Pat Fay

4 30 Purchasing 114 Den Raphaely

5 30 Purchasing 115 Alexander Khoo

6 30 Purchasing 116 Shelli Baida

7 30 Purchasing 117 Sigal Tobias

8 30 Purchasing 118 Guy Himuro

9 30 Purchasing 119 Karen Colmenares

10 40 Human Resources 203 Susan Mavris

11 50 Shipping 120 Matthew Weiss

12 50 Shipping 121 Adam Fripp

13 50 Shipping 122 Payam Kaufing

14 60 IT 103 Alexander Hunold

15 60 IT 104 Bruce Ernst

16 60 IT 105 David Austin

17 60 IT 106 Valli Pataballa

18 60 IT 107 Diana Lorentz

19 70 Public Relations 204 Hermann Baer

20 80 Sales 145 John Russell

21 80 Sales 146 Karen Partners

22 80 Sales 147 Alberto Erasmuz

23 80 Sales 148 Gerald Cambrault

24 80 Sales 149 Eleni Zlotkey

25 80 Sales 150 Peter Tucker

26 80 Sales 151 David Bernstein

27 80 Sales 152 Peter Hall

28 80 Sales 153 Christopher Olsen

29 80 Sales 154 Nanette Cambrault

30 80 Sales 155 Oliver Tuvault

31 90 Executive 100 Steven King

32 90 Executive 101 Neena Kochhar

33 90 Executive 102 Lex De Haan

34 100 Finance 108 Nancy Greenberg

35 100 Finance 109 Daniel Faviat

36 100 Finance 110 John Chen

37 100 Finance 111 Ismael Sciarra

38 100 Finance 112 Jose Manuel Urman

39 100 Finance 113 Luis Popp

40 110 Accounting 205 Shelley Higgins

41 110 Accounting 206 William Gietz

42 120 R&D Korea 220 Minjun Kim

43 120 R&D Korea 221 Seoyeon Lee

44 120 R&D Korea 222 Jihoon Park

Value X

10

Refresh

Save

Cancel

Export data

200

44

44 row(s) fetched - 0.001s, on 2026-07-28 at 21:49:53

KST en Writable

Smart Insert

462 : 1 : 1:49:16

Set: 0 | 0

D-7. 모든 직무 포함

| | |
|------------|---|
| 문제 번호(A-H) | D-7 |
| 분류 | JOIN 종합실습 |
| 문제 | [JOIN 7: RIGHT OUTER JOIN] 모든 직무 포함
HR 데이터를 활용하여 '모든 직무 포함' 결과를 조회하는 SQL 을 작성하십시오. |
| 조건 | RIGHT OUTER JOIN 방식을 활용하십시오. 오른쪽 jobs 의 모든 행을 보존하여 현재 담당
사원이 없는 직무도 보여줍니다. |

실험결과

The screenshot displays the DBeaver 26.1.3 interface with a PostgreSQL database connection. The left sidebar shows the database structure, including the 'public' schema and the 'hr' database. The main window shows a SQL query editor with a query that joins the 'employees' and 'jobs' tables. The results grid shows a list of employees and their job details.

SQL Query:

```
-- [JOIN 7: RIGHT OUTER JOIN] 로한 직무 포함
-- job_id: 직위와 동일하며 '로한' 직무 포함 결과로 조회하는 SQL을 작성하십시오.
-- 조건: RIGHT OUTER JOIN 방식을 활용하십시오. 오른쪽 jobs의 로한 항목 보존하여 일치 항목이 없는 직위로 보여줍니다.
-- TODO: 아래의 SQL을 작성하십시오.

SELECT
    j.job_id AS "직위 ID",
    j.job_title AS "직무명",
    e.employee_id AS "사원 ID",
    concat_ws(" ", e.first_name, e.last_name) AS "사원명"
FROM employees e
RIGHT OUTER JOIN jobs j ON j.job_id = e.job_id
ORDER BY j.job_id, e.employee_id NULLS LAST;
```

Query Results:

| 1 | A2 - 직위 ID | A2 직무명 | I23 - 사원 ID | A2 사원명 |
|----|------------|---------------------------------|-------------|-------------------|
| 1 | AC_ACCOUNT | Public Accountant | 206 | William Gietz |
| 2 | AC_MGR | Accounting Manager | 205 | Shelley Higgins |
| 3 | AD_ASST | Administration Assistant | 200 | Jennifer Whalen |
| 4 | AD_PRES | President | 100 | Steven King |
| 5 | AD_VP | Administration Vice President | 101 | Neena Kochhar |
| 6 | AD_VP | Administration Vice President | 102 | Lex De Haan |
| 7 | FI_ACCOUNT | Accountant | 109 | Daniel Faviet |
| 8 | FI_ACCOUNT | Accountant | 110 | John Chen |
| 9 | FI_ACCOUNT | Accountant | 111 | Ismael Sciarra |
| 10 | FI_ACCOUNT | Accountant | 112 | Jose Manuel Urman |
| 11 | FI_ACCOUNT | Accountant | 113 | Luis Popp |
| 12 | FI_MGR | Finance Manager | 108 | Nancy Greenberg |
| 13 | HR_REP | Human Resources Representative | 203 | Susan Mavris |
| 14 | IT_PROG | Programmer | 103 | Alexander Hunold |
| 15 | IT_PROG | Programmer | 104 | Bruce Ernst |
| 16 | IT_PROG | Programmer | 105 | David Austin |
| 17 | IT_PROG | Programmer | 106 | Vali Pataballa |
| 18 | IT_PROG | Programmer | 107 | Diana Lorentz |
| 19 | IT_PROG | Programmer | 220 | Minjun Kim |
| 20 | IT_PROG | Programmer | 221 | Seoyeon Lee |
| 21 | IT_PROG | Programmer | 222 | Jihoon Park |
| 22 | MK_MAN | Marketing Manager | 201 | Michael Hartstein |
| 23 | MK_REP | Marketing Representative | 202 | Pat Fay |
| 24 | MK_REP | Marketing Representative | 223 | Yuna Choi |
| 25 | PR_REP | Public Relations Representative | 204 | Hermann Baer |
| 26 | PU_CLERK | Purchasing Clerk | 115 | Alexander Khoo |
| 27 | PU_CLERK | Purchasing Clerk | 116 | Shelli Baida |
| 28 | PU_CLERK | Purchasing Clerk | 117 | Sigal Tobias |
| 29 | PU_CLERK | Purchasing Clerk | 118 | Guy Himuro |
| 30 | PU_CLERK | Purchasing Clerk | 119 | Karen Colmenares |
| 31 | PU_MAN | Purchasing Manager | 114 | Den Raphaels |
| 32 | SA_MAN | Sales Manager | 145 | John Russell |
| 33 | SA_MAN | Sales Manager | 146 | Karen Partners |
| 34 | SA_MAN | Sales Manager | 147 | Alberto Erazuiz |
| 35 | SA_MAN | Sales Manager | 148 | Gerald Cambrault |
| 36 | SA_MAN | Sales Manager | 149 | Eleni Zlotkey |
| 37 | SA_REP | Sales Representative | 150 | Peter Tucker |
| 38 | SA_REP | Sales Representative | 151 | David Bernstein |
| 39 | SA_REP | Sales Representative | 152 | Peter Hall |
| 40 | SA_REP | Sales Representative | 153 | Christopher Olsen |
| 41 | SA_REP | Sales Representative | 154 | Nanette Cambrault |
| 42 | SA_REP | Sales Representative | 155 | Oliver Tuvault |
| 43 | SH_CLERK | Shipping Clerk | [NULL] | |
| 44 | ST_CLERK | Stock Clerk | [NULL] | |
| 45 | ST_MAN | Stock Manager | 120 | Matthew Weiss |
| 46 | ST_MAN | Stock Manager | 121 | Adam Fripp |
| 47 | ST_MAN | Stock Manager | 122 | Payam Kaufling |

D-8. 모든 부서와 모든 사원

| | |
|------------|--|
| 문제 번호(A-H) | D-8 |
| 분류 | JOIN 종합실습 |
| 문제 | [JOIN 8: FULL OUTER JOIN] 모든 부서와 모든 사원
HR 데이터를 활용하여 '모든 부서와 모든 사원' 결과를 조회하는 SQL 을 작성하십시오. |
| 조건 | FULL OUTER JOIN 방식을 활용하십시오. 어느 쪽에도 일치 상대가 없는 행까지 모두 보존합니다. |

실험결과

DBaver 26.13 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Auto

postgres hr@skala_db

postgres day02_PostgresQL_HR_종합실습_문제.sql postgres day02_PostgresQL_HR_emp_dept_상표_스키마.sql postgres day02_책리_임유리.sql

Filter connections by name

DBaver Sample Database (SQLite)

postgres localhost:5432

Databases

postgres

skala_db

Schemas

app

Tables

courses

enrollments

majors

students

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

day02

public

hr

Tables

countries

departments

employees

job_history

jobs

locations

regions

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

Event Triggers

Extensions

Storage

System info

Roles

Administer

System info

-- [JOIN 8: FULL OUTER JOIN] 모든 부서와 모든 사원
-- 목적: HR 데이터를 활용하여 '모든 부서와 모든 사원' 결과를 조회하는 SQL 을 작성하십시오.
-- 조건: FULL OUTER JOIN 방식을 활용하십시오. 어느 쪽에도 일치 상대가 없는 행까지 모두 보존합니다.
-- TODO: 아래에 SQL 을 작성하십시오.

SELECT
d.department_id AS "부서 ID",
COALESCE(d.department_name, '부서 미배정') AS "부서명",
e.employee_id AS "사원 ID",
concat_ws(' ', e.first_name, e.last_name) AS "사원명"
FROM departments d
FULL OUTER JOIN employees e ON e.department_id = d.department_id
ORDER BY d.department_id NULLS LAST, e.employee_id NULLS LAST;

departments(+) 1 X

SELECT d.department_id AS "부서 ID", COALESCE(d.departme

Enter a SQL expression to filter results (use Ctrl+Space)

1 10 Administration 200 Jennifer Whalen

2 20 Marketing 201 Michael Hartstein

3 20 Marketing 202 Pat Fay

4 30 Purchasing 114 Den Raphaely

5 30 Purchasing 115 Alexander Khoo

6 30 Purchasing 116 Shelli Baida

7 30 Purchasing 117 Sigal Tobias

8 30 Purchasing 118 Guy Himuro

9 30 Purchasing 119 Karen Colmenares

10 40 Human Resources 203 Susan Mavris

11 50 Shipping 120 Matthew Weiss

12 50 Shipping 121 Adam Fripp

13 50 Shipping 122 Payam Kaufing

14 60 IT 103 Alexander Hunold

15 60 IT 104 Bruce Ernst

16 60 IT 105 David Austin

17 60 IT 106 Valli Pataballa

18 60 IT 107 Diana Lorentz

19 70 Public Relations 204 Hermann Baer

20 80 Sales 145 John Russell

21 80 Sales 146 Karen Partners

22 80 Sales 147 Alberto Erasmuiz

23 80 Sales 148 Gerald Cambrault

24 80 Sales 149 Eleni Zlotkey

25 80 Sales 150 Peter Tucker

26 80 Sales 151 David Bernstein

27 80 Sales 152 Peter Hall

28 80 Sales 153 Christopher Olsen

29 80 Sales 154 Nanette Cambrault

30 80 Sales 155 Oliver Tuvault

31 90 Executive 100 Steven King

32 90 Executive 101 Neena Kochhar

33 90 Executive 102 Lex De Haan

34 100 Finance 108 Nancy Greenberg

35 100 Finance 109 Daniel Faviet

36 100 Finance 110 John Chen

37 100 Finance 111 Ismael Sciarra

38 100 Finance 112 Jose Manuel Urman

39 100 Finance 113 Luis Popp

40 110 Accounting 205 Shelley Higgins

41 110 Accounting 206 William Gietz

42 120 R&D Korea 220 Minjun Kim

43 120 R&D Korea 221 Seoyeon Lee

44 120 R&D Korea 222 Jihoon Park

45 [NULL] 부서 미배정 223 Yuna Choi

Value X

10

Refresh Save Cancel

Export data

200

45

45 row(s) fetched - 0s on 2026-07-28 at 21:50:26

KST en Writable

Smart Insert

488 : 63 : 15885

Set: 0 | 0

D-9. 사원의 과거 직무 이력

| | |
|------------|---|
| 문제 번호(A-H) | D-9 |
| 분류 | JOIN 종합실습 |
| 문제 | [JOIN 9: INNER JOIN] 사원의 과거 직무 이력
HR 데이터를 활용하여 '사원의 과거 직무 이력' 결과를 조회하는 SQL 을 작성하십시오. |
| 조건 | INNER JOIN 방식을 활용하십시오. job_history 를 employees, jobs, departments 와 결합합니다. |

실험결과

DBaver 26.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Filter connections by name

postgres localhost-5432

postgres

skala_db

app

Tables

courses

enrollments

majors

students

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

Event Triggers

Extensions

Storage

System info

Roles

Administer

System info

Script-3

<postgres> day02_PostgresQL_HR_종합실습_문제.sql

<postgres> day02_PostgresQL_HR_emp_dept_상표_스키마.sql

<postgres> day02_책리_임유리.sql

SELECT

-- [JOIN 9: INNER JOIN] 사원의 과거 직무 이력

-- 문제: HR 데이터를 활용하여 '사원의 과거 직무 이력' 결과를 조회하는 SQL 을 작성하십시오.

-- 조건: INNER JOIN 방식을 활용하십시오. job_history를 employees, jobs, departments 와 결합합니다.

-- TDDO: 이 문제 SQL을 작성하십시오.

SELECT

e.employee_id AS "사원 ID",

concat_ws(' ', e.first_name, e.last_name) AS "사원명",

jh.start_date AS "시작일",

jh.end_date AS "종료일",

j.job_title AS "과거 직무명",

d.department_name AS "과거 부서명"

FROM job_history jh

INNER JOIN employees e ON e.employee_id = jh.employee_id

INNER JOIN jobs j ON j.job_id = jh.job_id

INNER JOIN departments d ON d.department_id = jh.department_id

ORDER BY e.employee_id, jh.start_date;

employees(+)

1 101 Neena Kochhar 1997-09-21 2001-10-27 Public Accountant Accounting

2 101 Neena Kochhar 2001-10-28 2005-03-15 Accounting Manager Accounting

3 102 Lex De Haan 1993-01-13 1998-07-24 Programmer IT

4 114 Den Raphaely 1998-03-24 1999-12-31 Stock Clerk Shipping

5 122 Payam Kaufing 1999-01-01 1999-12-31 Stock Clerk Shipping

6 200 Jennifer Whalen 1995-09-17 2001-06-17 Administration Assistant Executive

7 201 Michael Hartstein 1996-02-17 1999-12-19 Marketing Representative Marketing

8 220 Minjun Kim 2015-01-05 2019-03-03 Programmer IT

9 223 Yuna Choi 2021-01-04 2023-02-19 Administration Assistant Administration

Value

101

Refresh

Save

Cancel

Export data

200

9

9 row(s) fetched - 0.002s, on 2026-07-28 at 21:50:44

KST en Writable

Smart Insert

506 : 39 : 16519

Set: 0 | 0

D-10. 사원의 급여가 직무 급여 범위에 맞는지 확인

| | |
|------------|---|
| 문제 번호(A-H) | D-10 |
| 분류 | JOIN 종합실습 |
| 문제 | [JOIN 10: 비등가 JOIN] 사원의 급여가 직무 급여 범위에 맞는지 확인
HR 데이터를 활용하여 '사원의 급여가 직무 급여 범위에 맞는지 확인' 결과를
조회하는 SQL 을 작성하십시오. |
| 조건 | 비등가 JOIN 방식을 활용하십시오. 등호가 아닌 BETWEEN 조건으로 급여와 직무
범위를 결합합니다. |

실험결과

DBeaver 26.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Auto Rollback postgres hr@skala_db

Connections X 프로젝트

Filter connections by name

- > DBeaver Sample Database (SQLite)
 - > postgres localhost:5432
 - > Databases
 - > postgres
 - > skala_db
 - > Schemas
 - > app
 - > Tables
 - > courses
 - > enrollments 24K
 - > majors 40K
 - > students 56K
 - > Foreign Tables
 - > Views
 - > Materialized Views
 - > Indexes
 - > Functions
 - > Sequences
 - > Data types
 - > Aggregate functions
 - > day02
 - > public
 - > hr
 - > Tables
 - > countries 40K
 - > departments 72K
 - > employees 88K
 - > job_history 56K
 - > jobs 24K
 - > locations 40K
 - > regions 40K
 - > Foreign Tables
 - > Views
 - > Materialized Views
 - > Indexes
 - > Functions
 - > Sequences
 - > Data types
 - > Aggregate functions
 - > Event Triggers
 - > Extensions
 - > Storage
 - > System info
 - > Roles
 - > Administrator
 - > System info

Files X

Name

 - > Bookmarks
 - > Dashboards
 - > Diagrams

General

```
-- [JOIN 10: 비둘기 JOIN] 사원의 급여가 직무 급여 범위 안에 있는지 확인
-- 출처: HR 데이터에 들어있는 "사원" 테이블과 "직무" 테이블의 관계를 확인하고 결과를 조회하는 SQL을 작성하십시오.
-- 목적: 비둘기 JOIN 방식을 활용하십시오. 동료가 아닌 BETWEEN 조건으로 급여와 직무 범위를 결합합니다.
-- TODO: 아래의 SQL을 작성하십시오.
```

```
SELECT
    e.employee_id AS "사원 ID",
    concat_ws(' ', e.first_name, e.last_name) AS "서명",
    e.salary AS "사원 급여",
    j.job_id AS "급여 범위 ID",
    j.job_title AS "급여 범위 직위명",
    j.min_salary AS "최소 급여",
    j.max_salary AS "최대 급여"
FROM employees e
INNER JOIN jobs j
ON e.salary BETWEEN j.min_salary AND j.max_salary
ORDER BY e.employee_id, j.min_salary, j.job_id;
```

employees(*) 1 X

Enter a SQL expression to filter results (use Ctrl+Space)

| | AZ *사원 ID | AZ 서명 | 123 사원 급여 | AZ - 급여 범위 ID | AZ 급여 범위 직위명 | 123 최소 급여 | 123 최대 |
|----|-----------|------------------|-----------|---------------|---------------------------------|-----------|--------|
| 1 | 100 | Steven King | 24,000 | AD_VP | Administration Vice President | 15,000 | |
| 2 | 100 | Steven King | 24,000 | AD_PRES | President | 20,000 | |
| 3 | 101 | Neena Kochhar | 17,000 | SA_MAN | Sales Manager | 10,000 | |
| 4 | 101 | Neena Kochhar | 17,000 | AD_VP | Administration Vice President | 15,000 | |
| 5 | 102 | Lex De Haan | 17,000 | SA_MAN | Sales Manager | 10,000 | |
| 6 | 102 | Lex De Haan | 17,000 | AD_VP | Administration Vice President | 15,000 | |
| 7 | 103 | Alexander Hunold | 9,000 | HR_REP | Human Resources Representative | 4,000 | |
| 8 | 103 | Alexander Hunold | 9,000 | IT_PROG | Programmer | 4,000 | |
| 9 | 103 | Alexander Hunold | 9,000 | MKT_REP | Marketing Representative | 4,000 | |
| 10 | 103 | Alexander Hunold | 9,000 | AC_ACCOUNT | Public Accountant | 4,200 | |
| 11 | 103 | Alexander Hunold | 9,000 | FI_ACCOUNT | Accountant | 4,200 | |
| 12 | 103 | Alexander Hunold | 9,000 | PR_REP | Public Relations Representative | 4,500 | |
| 13 | 103 | Alexander Hunold | 9,000 | SA_REP | Sales Representative | 6,000 | |
| 14 | 103 | Alexander Hunold | 9,000 | PJ_MAN | Purchasing Manager | 8,000 | |
| 15 | 103 | Alexander Hunold | 9,000 | AC_MGR | Accounting Manager | 8,200 | |
| 16 | 103 | Alexander Hunold | 9,000 | FI_MGR | Finance Manager | 8,200 | |
| 17 | 103 | Alexander Hunold | 9,000 | MKT_MAN | Marketing Manager | 9,000 | |
| 18 | 104 | Bruce Ernst | 6,000 | AD_ASST | Administration Assistant | 3,000 | |
| 19 | 104 | Bruce Ernst | 6,000 | HR_REP | Human Resources Representative | 4,000 | |
| 20 | 104 | Bruce Ernst | 6,000 | IT_PROG | Programmer | 4,000 | |
| 21 | 104 | Bruce Ernst | 6,000 | MKT_REP | Marketing Representative | 4,000 | |
| 22 | 104 | Bruce Ernst | 6,000 | AC_ACCOUNT | Public Accountant | 4,200 | |
| 23 | 104 | Bruce Ernst | 6,000 | FI_ACCOUNT | Accountant | 4,200 | |
| 24 | 104 | Bruce Ernst | 6,000 | PR_REP | Public Relations Representative | 4,500 | |
| 25 | 104 | Bruce Ernst | 6,000 | ST_MAN | Stock Manager | 5,500 | |
| 26 | 104 | Bruce Ernst | 6,000 | SA_REP | Sales Representative | 6,000 | |
| 27 | 105 | David Austin | 4,800 | ST_CLERK | Stock Clerk | 2,000 | |
| 28 | 105 | David Austin | 4,800 | PJ_CLERK | Purchasing Clerk | 2,500 | |
| 29 | 105 | David Austin | 4,800 | SH_CLERK | Shipping Clerk | 2,500 | |
| 30 | 105 | David Austin | 4,800 | AD_ASST | Administration Assistant | 3,000 | |
| 31 | 105 | David Austin | 4,800 | HR_REP | Human Resources Representative | 4,000 | |
| 32 | 105 | David Austin | 4,800 | IT_PROG | Programmer | 4,000 | |
| 33 | 105 | David Austin | 4,800 | MKT_REP | Marketing Representative | 4,000 | |
| 34 | 105 | David Austin | 4,800 | AC_ACCOUNT | Public Accountant | 4,200 | |
| 35 | 105 | David Austin | 4,800 | FI_ACCOUNT | Accountant | 4,200 | |
| 36 | 105 | David Austin | 4,800 | PR_REP | Public Relations Representative | 4,500 | |
| 37 | 106 | Valli Pataballa | 4,800 | ST_CLERK | Stock Clerk | 2,000 | |
| 38 | 106 | Valli Pataballa | 4,800 | PJ_CLERK | Purchasing Clerk | 2,500 | |
| 39 | 106 | Valli Pataballa | 4,800 | SH_CLERK | Shipping Clerk | 2,500 | |
| 40 | 106 | Valli Pataballa | 4,800 | AD_ASST | Administration Assistant | 3,000 | |
| 41 | 106 | Valli Pataballa | 4,800 | HR_REP | Human Resources Representative | 4,000 | |
| 42 | 106 | Valli Pataballa | 4,800 | IT_PROG | Programmer | 4,000 | |
| 43 | 106 | Valli Pataballa | 4,800 | MKT_REP | Marketing Representative | 4,000 | |
| 44 | 106 | Valli Pataballa | 4,800 | AC_ACCOUNT | Public Accountant | 4,200 | |

DBeaver 25.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Commit Rollback Auto

Connections 프로젝트

Filter connections by name

DBeaver Sample Database (SQLite)

postgres localhost:5432

Databases

- postgres
 - skala_db
 - Schemas
 - app
 - Tables
- day02
 - public
 - Tables

Tables

- courses
- enrollments
- majors
- students

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General

Script-3

postgres day02_PostgreSQL_HR_종합실습_문제.sql

postgres day02_PostgreSQL_HR_emp_dept_샘플_스키마.sql

postgres day02_책리_임유리.sql

```
-- [JOIN 18: 비종가 JOIN] 사용의 급여가 직무 급여 범위와 일치지 확인
-- 문제: HR 테이블을 활용하여 '사원'의 급여가 직무 급여 범위에 있는지 확인 결과를 조회하는 SQL을 작성하십시오.
-- 조건: 비종가 JOIN 방식을 활용하십시오. 결과가 아닌 BETWEEN 조건으로 급여와 직무 범위를 결합합니다.
-- TODO: 아래의 SQL을 작성하십시오.

SELECT
    e.employee_id AS "사원 ID",
    concat_ws(' ', e.first_name, e.last_name) AS "사원명",
    e.salary AS "사원 급여",
    j.job_id AS "급여 범위 직무 ID",
    j.job_title AS "급여 범위 직무명",
    j.min_salary AS "최소 급여",
    j.max_salary AS "최대 급여"
FROM employees e
INNER JOIN jobs j
    ON e.salary BETWEEN j.min_salary AND j.max_salary
ORDER BY e.employee_id, j.min_salary, j.job_id;
```

employees(*) 1 X

SELECT e.employee_id AS "사원 ID", concat_ws(' ', e.first_name, e.last_name) AS "사원명", e.salary AS "사원 급여", j.job_id AS "급여 범위 직무 ID", j.job_title AS "급여 범위 직무명", j.min_salary AS "최소 급여", j.max_salary AS "최대 급여"

| 사원 ID | 사원명 | 사원 급여 | 급여 범위 직무 ID | 급여 범위 직무명 | 최소 급여 | 최대 급여 |
|-------|-----|-----------------|-------------|------------|---------------------------------|--------|
| 43 | 106 | Valli Pataballa | 4,800 | MK_REP | Marketing Representative | 4,000 |
| 44 | 106 | Valli Pataballa | 4,800 | AC_ACCOUNT | Public Accountant | 4,200 |
| 45 | 106 | Valli Pataballa | 4,800 | FI_ACCOUNT | Accountant | 4,200 |
| 46 | 106 | Valli Pataballa | 4,800 | PR_REP | Public Relations Representative | 4,500 |
| 47 | 107 | Diana Lorentz | 4,200 | ST_CLERK | Stock Clerk | 2,000 |
| 48 | 107 | Diana Lorentz | 4,200 | PU_CLERK | Purchasing Clerk | 2,500 |
| 49 | 107 | Diana Lorentz | 4,200 | SH_CLERK | Shipping Clerk | 2,500 |
| 50 | 107 | Diana Lorentz | 4,200 | AD_ASST | Administration Assistant | 3,000 |
| 51 | 107 | Diana Lorentz | 4,200 | HR_REP | Human Resources Representative | 4,000 |
| 52 | 107 | Diana Lorentz | 4,200 | IT_PROG | Programmer | 4,000 |
| 53 | 107 | Diana Lorentz | 4,200 | MK_REP | Marketing Representative | 4,000 |
| 54 | 107 | Diana Lorentz | 4,200 | AC_ACCOUNT | Public Accountant | 4,200 |
| 55 | 107 | Diana Lorentz | 4,200 | FI_ACCOUNT | Accountant | 4,200 |
| 56 | 108 | Nancy Greenberg | 12,000 | SA_REP | Sales Representative | 6,000 |
| 57 | 108 | Nancy Greenberg | 12,000 | PU_MAN | Purchasing Manager | 8,000 |
| 58 | 108 | Nancy Greenberg | 12,000 | AC_MGR | Accounting Manager | 8,200 |
| 59 | 108 | Nancy Greenberg | 12,000 | FI_MGR | Finance Manager | 8,200 |
| 60 | 108 | Nancy Greenberg | 12,000 | MK_MAN | Marketing Manager | 9,000 |
| 61 | 108 | Nancy Greenberg | 12,000 | SA_MAN | Sales Manager | 10,000 |
| 62 | 109 | Daniel Faviet | 9,000 | HR_REP | Human Resources Representative | 4,000 |
| 63 | 109 | Daniel Faviet | 9,000 | IT_PROG | Programmer | 4,000 |
| 64 | 109 | Daniel Faviet | 9,000 | MK_REP | Marketing Representative | 4,000 |
| 65 | 109 | Daniel Faviet | 9,000 | AC_ACCOUNT | Public Accountant | 4,200 |
| 66 | 109 | Daniel Faviet | 9,000 | FI_ACCOUNT | Accountant | 4,200 |
| 67 | 109 | Daniel Faviet | 9,000 | PR_REP | Public Relations Representative | 4,500 |
| 68 | 109 | Daniel Faviet | 9,000 | SA_REP | Sales Representative | 6,000 |
| 69 | 109 | Daniel Faviet | 9,000 | PU_MAN | Purchasing Manager | 8,000 |
| 70 | 109 | Daniel Faviet | 9,000 | AC_MGR | Accounting Manager | 8,200 |
| 71 | 109 | Daniel Faviet | 9,000 | FI_MGR | Finance Manager | 8,200 |
| 72 | 109 | Daniel Faviet | 9,000 | MK_MAN | Marketing Manager | 9,000 |
| 73 | 110 | John Chen | 8,200 | HR_REP | Human Resources Representative | 4,000 |
| 74 | 110 | John Chen | 8,200 | IT_PROG | Programmer | 4,000 |
| 75 | 110 | John Chen | 8,200 | MK_REP | Marketing Representative | 4,000 |
| 76 | 110 | John Chen | 8,200 | AC_ACCOUNT | Public Accountant | 4,200 |
| 77 | 110 | John Chen | 8,200 | FI_ACCOUNT | Accountant | 4,200 |
| 78 | 110 | John Chen | 8,200 | PR_REP | Public Relations Representative | 4,500 |
| 79 | 110 | John Chen | 8,200 | ST_MAN | Stock Manager | 5,500 |
| 80 | 110 | John Chen | 8,200 | SA_REP | Sales Representative | 6,000 |
| 81 | 110 | John Chen | 8,200 | PU_MAN | Purchasing Manager | 8,000 |
| 82 | 110 | John Chen | 8,200 | AC_MGR | Accounting Manager | 8,200 |
| 83 | 110 | John Chen | 8,200 | FI_MGR | Finance Manager | 8,200 |
| 84 | 111 | Ismael Sciarra | 7,700 | HR_REP | Human Resources Representative | 4,000 |
| 85 | 111 | Ismael Sciarra | 7,700 | IT_PROG | Programmer | 4,000 |
| 86 | 111 | Ismael Sciarra | 7,700 | MK_REP | Marketing Representative | 4,000 |
| 87 | 111 | Ismael Sciarra | 7,700 | AC_ACCOUNT | Public Accountant | 4,200 |

Refresh Save Cancel Export data 200 200+ 200 row(s) fetched - 0.003s (0.002s fetch), on 2026-07-28 at 21:51:01

KST en Writeable Smart Insert 524 : 48 : 17106 Sel: 0 | 0

The screenshot displays the DBeaver 26.1.3 interface. The top menu bar includes File, Edit, Navigate, Search, SQL Editor, Database, Window, and Help. The toolbar contains icons for various database actions. The left sidebar shows the 'Connections' pane with 'DBeaver Sample Database (SQLite)' selected, and a tree view of the database schema including 'Databases', 'Schemas', 'Tables', 'Views', 'Materialized Views', 'Indexes', 'Functions', 'Sequences', 'Data types', 'Aggregate functions', 'Event Triggers', 'Extensions', 'Storage', 'System info', 'Roles', 'Administrator', and 'System info'.

The main SQL Editor window shows a query for the 'employees' table, filtered by '111'. The query is as follows:

```
SELECT e.employee_id AS "사원 ID",
       concat_ws(' ', e.first_name, e.last_name) AS "사원명",
       e.salary AS "사원 급여",
       j.job_id AS "급여 범위 직무 ID",
       j.job_title AS "급여 범위 직무명",
       j.min_salary AS "최소 급여",
       j.max_salary AS "최대 급여"
FROM employees e
INNER JOIN jobs j
ON e.salary BETWEEN j.min_salary AND j.max_salary
ORDER BY e.employee_id, j.min_salary, j.job_id;
```

The results pane shows a table with 7 columns: 사원 ID, 사원명, 급여, 급여 범위 직무 ID, 급여 범위 직무명, 최소 급여, and 최대 급여. The table contains 111 rows of data, with the first row highlighted in blue. The status bar at the bottom indicates that 200 rows were fetched (0.002s) on 2026-07-28 at 21:51:01.

DBeaver 25.1.3 - <postgres> day02_쿼리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Commit Rollback Auto

Connections 프로젝트

Filter connections by name

DBeaver Sample Database (SQLite)

postgres localhost:5432

Databases

- postgres
 - skala_db
 - Schemas
 - app
 - Tables

Tables

- courses
- enrollments
- majors
- students

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General

Script-3

postgres day02_PostgreSQL_HR_종합실습_문제.sql

postgres day02_PostgreSQL_HR_emp_dept_샘플_스키마.sql

postgres day02_쿼리_임유리.sql

```
-- [JOIN 18: 비종가 JOIN] 사원의 급여가 직무 급여 범위에 있는지 확인
-- 논리: 매 직위를 통해 해당 사원의 급여가 직무 급여 범위에 있는지 확인 결과를 조회하는 SQL을 작성하십시오.
-- 조건: 비종가 JOIN 방식을 활용하십시오. 결과가 아닌 BETWEEN 조건으로 급여와 직무 범위를 결합합니다.
-- TODO: 아래의 SQL을 작성하십시오.

SELECT
    e.employee_id AS "사원 ID",
    concat_ws(' ', e.first_name, e.last_name) AS "사원명",
    e.salary AS "사원 급여",
    j.job_id AS "급여 범위 직무 ID",
    j.job_title AS "급여 범위 직무명",
    j.min_salary AS "최소 급여",
    j.max_salary AS "최대 급여"
FROM employees e
INNER JOIN jobs j
    ON e.salary BETWEEN j.min_salary AND j.max_salary
ORDER BY e.employee_id, j.min_salary, j.job_id;
```

employees(*) 1 X

SELECT e.employee_id AS "사원 ID", concat_ws(' ', e.first_name, e.last_name) AS "사원명", e.salary AS "사원 급여", j.job_id AS "급여 범위 직무 ID", j.job_title AS "급여 범위 직무명", j.min_salary AS "최소 급여", j.max_salary AS "최대 급여"

| 사원 ID | 사원명 | 사원 급여 | 급여 범위 직무 ID | 급여 범위 직무명 | 최소 급여 | 최대 급여 |
|-------|-------------------|--------|-------------|---------------------------------|--------|--------|
| 169 | Alberto Errazuriz | 12,000 | PU_MAN | Purchasing Manager | 8,000 | 8,000 |
| 170 | Alberto Errazuriz | 12,000 | FI_MGR | Finance Manager | 8,200 | 8,200 |
| 171 | Alberto Errazuriz | 12,000 | SA_MAN | Sales Manager | 9,000 | 9,000 |
| 172 | Alberto Errazuriz | 12,000 | SA_REP | Sales Representative | 6,000 | 6,000 |
| 173 | Gerald Cambrault | 11,000 | PU_MAN | Purchasing Manager | 8,000 | 8,000 |
| 174 | Gerald Cambrault | 11,000 | AC_MGR | Accounting Manager | 8,200 | 8,200 |
| 175 | Gerald Cambrault | 11,000 | FI_MGR | Finance Manager | 8,200 | 8,200 |
| 176 | Gerald Cambrault | 11,000 | MK_MAN | Marketing Manager | 9,000 | 9,000 |
| 177 | Gerald Cambrault | 11,000 | SA_MAN | Sales Manager | 10,000 | 10,000 |
| 178 | Gerald Cambrault | 11,000 | SA_REP | Sales Representative | 6,000 | 6,000 |
| 179 | Eleni Zlotkey | 10,500 | PR_REP | Public Relations Representative | 4,500 | 4,500 |
| 180 | Eleni Zlotkey | 10,500 | SA_REP | Sales Representative | 6,000 | 6,000 |
| 181 | Eleni Zlotkey | 10,500 | PU_MAN | Purchasing Manager | 8,000 | 8,000 |
| 182 | Eleni Zlotkey | 10,500 | AC_MGR | Accounting Manager | 8,200 | 8,200 |
| 183 | Eleni Zlotkey | 10,500 | FI_MGR | Finance Manager | 8,200 | 8,200 |
| 184 | Eleni Zlotkey | 10,500 | MK_MAN | Marketing Manager | 9,000 | 9,000 |
| 185 | Eleni Zlotkey | 10,500 | SA_MAN | Sales Manager | 10,000 | 10,000 |
| 186 | Peter Tucker | 10,000 | IT_PROG | Programmer | 4,000 | 4,000 |
| 187 | Peter Tucker | 10,000 | PR_REP | Public Relations Representative | 4,500 | 4,500 |
| 188 | Peter Tucker | 10,000 | SA_REP | Sales Representative | 6,000 | 6,000 |
| 189 | Peter Tucker | 10,000 | PU_MAN | Purchasing Manager | 8,000 | 8,000 |
| 190 | Peter Tucker | 10,000 | AC_MGR | Accounting Manager | 8,200 | 8,200 |
| 191 | Peter Tucker | 10,000 | FI_MGR | Finance Manager | 8,200 | 8,200 |
| 192 | Peter Tucker | 10,000 | MK_MAN | Marketing Manager | 9,000 | 9,000 |
| 193 | Peter Tucker | 10,000 | SA_MAN | Sales Manager | 10,000 | 10,000 |
| 194 | David Bernstein | 9,500 | IT_PROG | Programmer | 4,000 | 4,000 |
| 195 | David Bernstein | 9,500 | PR_REP | Public Relations Representative | 4,500 | 4,500 |
| 196 | David Bernstein | 9,500 | SA_REP | Sales Representative | 6,000 | 6,000 |
| 197 | David Bernstein | 9,500 | PU_MAN | Purchasing Manager | 8,000 | 8,000 |
| 198 | David Bernstein | 9,500 | AC_MGR | Accounting Manager | 8,200 | 8,200 |
| 199 | David Bernstein | 9,500 | FI_MGR | Finance Manager | 8,200 | 8,200 |
| 200 | David Bernstein | 9,500 | MK_MAN | Marketing Manager | 9,000 | 9,000 |
| 201 | Peter Hall | 9,000 | HR_REP | Human Resources Representative | 4,000 | 4,000 |
| 202 | Peter Hall | 9,000 | IT_PROG | Programmer | 4,000 | 4,000 |
| 203 | Peter Hall | 9,000 | MK_REP | Marketing Representative | 4,000 | 4,000 |
| 204 | Peter Hall | 9,000 | AC_ACCOUNT | Public Accountant | 4,200 | 4,200 |
| 205 | Peter Hall | 9,000 | FI_ACCOUNT | Accountant | 4,200 | 4,200 |
| 206 | Peter Hall | 9,000 | PR_REP | Public Relations Representative | 4,500 | 4,500 |
| 207 | Peter Hall | 9,000 | SA_REP | Sales Representative | 6,000 | 6,000 |
| 208 | Peter Hall | 9,000 | PU_MAN | Purchasing Manager | 8,000 | 8,000 |
| 209 | Peter Hall | 9,000 | AC_MGR | Accounting Manager | 8,200 | 8,200 |
| 210 | Peter Hall | 9,000 | FI_MGR | Finance Manager | 8,200 | 8,200 |
| 211 | Peter Hall | 9,000 | MK_MAN | Marketing Manager | 9,000 | 9,000 |
| 212 | Christopher Olsen | 8,000 | HR_REP | Human Resources Representative | 4,000 | 4,000 |

Refresh Save Cancel Export data 200 327 327 row(s) fetched - 0.001s, on 2026-07-28 at 21:51:57

KST en Writable Smart Insert 524 : 48 : 17106 Sel: 0 | 0

DBeaver 25.1.3 - <postgres> day02_쿼리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Commit Rollback Auto postgres hr@skala_db hr@skala_db postgres day02_PostgresQL_HR_종합실습_문제.sql postgres day02_PostgresQL_HR_emp_dept_샘플_스키마.sql postgres day02_쿼리_임유리.sql

Filter connections by name

DBeaver Sample Database (SQLite)

postgres localhost:5432

Databases

- postgres
 - skala_db
 - Schemas
 - app
 - Tables

Tables

- courses
- enrollments
- majors
- students

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- Bookmarks
- Dashboards
- Diagrams

General

```
-- [JOIN 18: 비종가 JOIN] 사용의 급여가 직무 급여 범위에 있는지 확인
-- 논리: HR 테이블을 활용하여 '사용의 급여가 직무 급여 범위에 있는지 확인' 결과를 조회하는 SQL을 작성하십시오.
-- 조건: 비종가 JOIN 방식을 활용하십시오. 결과가 아닌 BETWEEN 조건으로 급여와 직무 범위를 결합합니다.
-- TODO: 아래의 SQL을 작성하십시오.

SELECT
    e.employee_id AS "사원 ID",
    concat_ws(' ', e.first_name, e.last_name) AS "사원명",
    e.salary AS "사원 급여",
    j.job_id AS "급여 범위 직무 ID",
    j.job_title AS "급여 범위 직무명",
    j.min_salary AS "최소 급여",
    j.max_salary AS "최대 급여"
FROM employees e
INNER JOIN jobs j
    ON e.salary BETWEEN j.min_salary AND j.max_salary
ORDER BY e.employee_id, j.min_salary, j.job_id;
```

employees(*) 1 X

SELECT e.employee_id AS "사원 ID", concat_ws(' ', e.first_name, e.last_name) AS "사원명", e.salary AS "사원 급여", j.job_id AS "급여 범위 직무 ID", j.job_title AS "급여 범위 직무명", j.min_salary AS "최소 급여", j.max_salary AS "최대 급여"

| 사원 ID | 사원명 | 사원 급여 | 급여 범위 직무 ID | 급여 범위 직무명 | 최소 급여 | 최대 급여 |
|-------|-------------------|--------|-------------|---------------------------------|--------|--------|
| 211 | Peter Hall | 9,000 | FI_MGR | Finance Manager | 8,200 | 9,000 |
| 212 | Peter Hall | 9,000 | MK_MAN | Marketing Manager | 9,000 | 9,000 |
| 213 | Christopher Olsen | 8,000 | HR_REP | Human Resources Representative | 4,000 | 4,000 |
| 214 | Christopher Olsen | 8,000 | IT_PROG | Programmer | 4,000 | 4,000 |
| 215 | Christopher Olsen | 8,000 | MK_REP | Marketing Representative | 4,000 | 4,000 |
| 216 | Christopher Olsen | 8,000 | AC_ACCOUNT | Public Accountant | 4,200 | 4,200 |
| 217 | Christopher Olsen | 8,000 | FI_ACCOUNT | Accountant | 4,200 | 4,200 |
| 218 | Christopher Olsen | 8,000 | PR_REP | Public Relations Representative | 4,500 | 4,500 |
| 219 | Christopher Olsen | 8,000 | ST_MAN | Stock Manager | 5,500 | 5,500 |
| 220 | Christopher Olsen | 8,000 | SA_REP | Sales Representative | 6,000 | 6,000 |
| 221 | Christopher Olsen | 8,000 | PU_MAN | Purchasing Manager | 8,000 | 8,000 |
| 222 | Nanette Cambrault | 7,500 | HR_REP | Human Resources Representative | 4,000 | 4,000 |
| 223 | Nanette Cambrault | 7,500 | IT_PROG | Programmer | 4,000 | 4,000 |
| 224 | Nanette Cambrault | 7,500 | MK_REP | Marketing Representative | 4,000 | 4,000 |
| 225 | Nanette Cambrault | 7,500 | AC_ACCOUNT | Public Accountant | 4,200 | 4,200 |
| 226 | Nanette Cambrault | 7,500 | FI_ACCOUNT | Accountant | 4,200 | 4,200 |
| 227 | Nanette Cambrault | 7,500 | PR_REP | Public Relations Representative | 4,500 | 4,500 |
| 228 | Nanette Cambrault | 7,500 | ST_MAN | Stock Manager | 5,500 | 5,500 |
| 229 | Nanette Cambrault | 7,500 | SA_REP | Sales Representative | 6,000 | 6,000 |
| 230 | Oliver Tuvault | 7,000 | HR_REP | Human Resources Representative | 4,000 | 4,000 |
| 231 | Oliver Tuvault | 7,000 | IT_PROG | Programmer | 4,000 | 4,000 |
| 232 | Oliver Tuvault | 7,000 | MK_REP | Marketing Representative | 4,000 | 4,000 |
| 233 | Oliver Tuvault | 7,000 | AC_ACCOUNT | Public Accountant | 4,200 | 4,200 |
| 234 | Oliver Tuvault | 7,000 | FI_ACCOUNT | Accountant | 4,200 | 4,200 |
| 235 | Oliver Tuvault | 7,000 | PR_REP | Public Relations Representative | 4,500 | 4,500 |
| 236 | Oliver Tuvault | 7,000 | ST_MAN | Stock Manager | 5,500 | 5,500 |
| 237 | Oliver Tuvault | 7,000 | SA_REP | Sales Representative | 6,000 | 6,000 |
| 238 | Jennifer Whalen | 4,400 | ST_CLERK | Stock Clerk | 2,000 | 2,000 |
| 239 | Jennifer Whalen | 4,400 | PU_CLERK | Purchasing Clerk | 2,500 | 2,500 |
| 240 | Jennifer Whalen | 4,400 | SH_CLERK | Shipping Clerk | 2,500 | 2,500 |
| 241 | Jennifer Whalen | 4,400 | AD_ASST | Administration Assistant | 3,000 | 3,000 |
| 242 | Jennifer Whalen | 4,400 | HR_REP | Human Resources Representative | 4,000 | 4,000 |
| 243 | Jennifer Whalen | 4,400 | IT_PROG | Programmer | 4,000 | 4,000 |
| 244 | Jennifer Whalen | 4,400 | MK_REP | Marketing Representative | 4,000 | 4,000 |
| 245 | Jennifer Whalen | 4,400 | AC_ACCOUNT | Public Accountant | 4,200 | 4,200 |
| 246 | Jennifer Whalen | 4,400 | FI_ACCOUNT | Accountant | 4,200 | 4,200 |
| 247 | Michael Hartstein | 13,000 | PU_MAN | Purchasing Manager | 8,000 | 8,000 |
| 248 | Michael Hartstein | 13,000 | AC_MGR | Accounting Manager | 8,200 | 8,200 |
| 249 | Michael Hartstein | 13,000 | FI_MGR | Finance Manager | 8,200 | 8,200 |
| 250 | Michael Hartstein | 13,000 | MK_MAN | Marketing Manager | 9,000 | 9,000 |
| 251 | Michael Hartstein | 13,000 | SA_MAN | Sales Manager | 10,000 | 10,000 |
| 252 | Pat Fay | 6,000 | AD_ASST | Administration Assistant | 3,000 | 3,000 |
| 253 | Pat Fay | 6,000 | HR_REP | Human Resources Representative | 4,000 | 4,000 |
| 254 | Pat Fay | 6,000 | IT_PROG | Programmer | 4,000 | 4,000 |
| 255 | Pat Fay | 6,000 | MK_REP | Marketing Representative | 4,000 | 4,000 |

Refresh Save Cancel Export data 200 327 327 row(s) fetched - 0.001s, on 2026-07-28 at 21:51:57

KST en Writable Smart Insert 524 : 48 : 17106 Sel: 0 | 0

DBeaver 25.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Commit Rollback Auto postgres hr@skala_db day02_PostgresQL_HR_종합실습_문제.sql day02_PostgresQL_HR_emp_dept_샘플_스키마.sql day02_책리_임유리.sql

Filter connections by name

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 - postgres
 - skala_db
 - Schemas
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 - Views
 - Materialized Views
 - Indexes
 - Functions
 - Sequences
 - Data types
 - Aggregate functions
 - Event Triggers
 - Extensions
 - Storage
 - System info
 - Roles
 - Administer
 - System info

Tables

- courses
- enrollments
- majors
- students

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- Name
- Bookmarks
- Dashboards
- Diagrams

General

```
-- [JOIN 18: 비종가 JOIN] 사원의 급여가 직무 급여 범위에 있는지 확인
-- 논리: 매 직위를 통해 해당 사원의 급여가 직무 급여 범위에 있는지 확인 결과를 조회하는 SQL을 작성하십시오.
-- 조건: 비종가 JOIN 방식을 활용하십시오. 결과가 아닌 BETWEEN 조건으로 급여와 직무 범위를 결합합니다.
-- TODO: 아래의 SQL을 작성하십시오.

SELECT
    e.employee_id AS "사원 ID",
    concat_ws(' ', e.first_name, e.last_name) AS "사원명",
    e.salary AS "사원 급여",
    j.job_id AS "급여 범위 직무 ID",
    j.job_title AS "급여 범위 직무명",
    j.min_salary AS "최소 급여",
    j.max_salary AS "최대 급여"
FROM employees e
INNER JOIN jobs j
    ON e.salary BETWEEN j.min_salary AND j.max_salary
ORDER BY e.employee_id, j.job_id;
```

employees(*) 1 X

SELECT e.employee_id AS "사원 ID", concat_ws(' ', e.first_name, e.last_name) AS "사원명", e.salary AS "사원 급여", j.job_id AS "급여 범위 직무 ID", j.job_title AS "급여 범위 직무명", j.min_salary AS "최소 급여", j.max_salary AS "최대 급여"

| 사원 ID | 사원명 | 사원 급여 | 급여 범위 직무 ID | 급여 범위 직무명 | 최소 급여 | 최대 급여 |
|-------|-----------------|--------|-------------|---------------------------------|--------|--------|
| 253 | Pat Fay | 6,000 | HR_REP | Human Resources Representative | 4,000 | 8,000 |
| 254 | Pat Fay | 6,000 | IT_PROG | Programmer | 4,000 | 8,000 |
| 255 | Pat Fay | 6,000 | MK_REP | Marketing Representative | 4,000 | 8,000 |
| 256 | Pat Fay | 6,000 | AC_ACCOUNT | Public Accountant | 4,200 | 8,200 |
| 257 | Pat Fay | 6,000 | FL_ACCOUNT | Accountant | 4,200 | 8,200 |
| 258 | Pat Fay | 6,000 | PR_REP | Public Relations Representative | 4,500 | 8,500 |
| 259 | Pat Fay | 6,000 | ST_MAN | Stock Manager | 5,500 | 9,500 |
| 260 | Pat Fay | 6,000 | SA_REP | Sales Representative | 6,000 | 10,000 |
| 261 | Susan Mavis | 6,500 | HR_REP | Human Resources Representative | 4,000 | 8,000 |
| 262 | Susan Mavis | 6,500 | IT_PROG | Programmer | 4,000 | 8,000 |
| 263 | Susan Mavis | 6,500 | MK_REP | Marketing Representative | 4,000 | 8,000 |
| 264 | Susan Mavis | 6,500 | AC_ACCOUNT | Public Accountant | 4,200 | 8,200 |
| 265 | Susan Mavis | 6,500 | FL_ACCOUNT | Accountant | 4,200 | 8,200 |
| 266 | Susan Mavis | 6,500 | PR_REP | Public Relations Representative | 4,500 | 8,500 |
| 267 | Susan Mavis | 6,500 | ST_MAN | Stock Manager | 5,500 | 9,500 |
| 268 | Susan Mavis | 6,500 | SA_REP | Sales Representative | 6,000 | 10,000 |
| 269 | Hermann Baer | 10,000 | IT_PROG | Programmer | 4,000 | 8,000 |
| 270 | Hermann Baer | 10,000 | PR_REP | Public Relations Representative | 4,500 | 8,500 |
| 271 | Hermann Baer | 10,000 | SA_REP | Sales Representative | 6,000 | 10,000 |
| 272 | Hermann Baer | 10,000 | PU_MAN | Purchasing Manager | 8,000 | 12,000 |
| 273 | Hermann Baer | 10,000 | AC_MGR | Accounting Manager | 8,200 | 12,200 |
| 274 | Hermann Baer | 10,000 | FI_MGR | Finance Manager | 8,200 | 12,200 |
| 275 | Hermann Baer | 10,000 | MK_MAN | Marketing Manager | 9,000 | 13,000 |
| 276 | Hermann Baer | 10,000 | SA_MAN | Sales Manager | 10,000 | 14,000 |
| 277 | Shelley Higgins | 12,000 | SA_REP | Sales Representative | 6,000 | 10,000 |
| 278 | Shelley Higgins | 12,000 | PU_MAN | Purchasing Manager | 8,000 | 12,000 |
| 279 | Shelley Higgins | 12,000 | AC_MGR | Accounting Manager | 8,200 | 12,200 |
| 280 | Shelley Higgins | 12,000 | FI_MGR | Finance Manager | 8,200 | 12,200 |
| 281 | Shelley Higgins | 12,000 | MK_MAN | Marketing Manager | 9,000 | 13,000 |
| 282 | Shelley Higgins | 12,000 | SA_MAN | Sales Manager | 10,000 | 14,000 |
| 283 | William Gietz | 8,300 | HR_REP | Human Resources Representative | 4,000 | 8,000 |
| 284 | William Gietz | 8,300 | IT_PROG | Programmer | 4,000 | 8,000 |
| 285 | William Gietz | 8,300 | MK_REP | Marketing Representative | 4,000 | 8,000 |
| 286 | William Gietz | 8,300 | AC_ACCOUNT | Public Accountant | 4,200 | 8,200 |
| 287 | William Gietz | 8,300 | FL_ACCOUNT | Accountant | 4,200 | 8,200 |
| 288 | William Gietz | 8,300 | PR_REP | Public Relations Representative | 4,500 | 8,500 |
| 289 | William Gietz | 8,300 | ST_MAN | Stock Manager | 5,500 | 9,500 |
| 290 | William Gietz | 8,300 | SA_REP | Sales Representative | 6,000 | 10,000 |
| 291 | William Gietz | 8,300 | PU_MAN | Purchasing Manager | 8,000 | 12,000 |
| 292 | William Gietz | 8,300 | AC_MGR | Accounting Manager | 8,200 | 12,200 |
| 293 | William Gietz | 8,300 | FI_MGR | Finance Manager | 8,200 | 12,200 |
| 294 | Minjun Kim | 10,500 | PR_REP | Public Relations Representative | 4,500 | 8,500 |
| 295 | Minjun Kim | 10,500 | SA_REP | Sales Representative | 6,000 | 10,000 |
| 296 | Minjun Kim | 10,500 | PU_MAN | Purchasing Manager | 8,000 | 12,000 |
| 297 | Minjun Kim | 10,500 | AC_MGR | Accounting Manager | 8,200 | 12,200 |

Record

Refresh Save Cancel Export data 200 327 327 row(s) fetched - 0.001s, on 2026-07-28 at 21:51:57

KST en Writable Smart Insert 524 : 48 : 17106 Sel: 0 | 0

DBeaver 25.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Commit Rollback Auto postgres hr@skala_db hr@skala_db day02_PostgreSQL_HR_종합실습_문책.sql day02_PostgreSQL_HR_emp_dept_샘플_스키마.sql day02_책리_임유리.sql

Filter connections by name

- DBEaver Sample Database (SQLite)
- postgres localhost:5432
 - Databases
 - postgres
 - skala_db
 - Schemas
 - app
 - Tables

employees(*) 1 X

```
SELECT e.employee_id AS "사원 ID",
       concat_ws(' ', e.first_name, e.last_name) AS "사원명",
       e.salary AS "사원 급여",
       j.job_id AS "급여 범위 직무 ID",
       j.job_title AS "급여 범위 직무명",
       j.min_salary AS "최소 급여",
       j.max_salary AS "최대 급여"
FROM employees e
INNER JOIN jobs j
ON e.salary BETWEEN j.min_salary AND j.max_salary
ORDER BY e.employee_id, j.min_salary, j.job_id;
```

| 123 => 사원 ID | AZ 사원명 | 123 사원 급여 | AZ => 급여 범위 직무 ID | AZ 급여 범위 직무명 | 123 최소 급여 | 123 최대 |
|--------------|--------|---------------|-------------------|--------------|---------------------------------|--------|
| 285 | 206 | William Gietz | 8,300 | MK_REP | Marketing Representative | 4,000 |
| 286 | 206 | William Gietz | 8,300 | AC_ACCOUNT | Public Accountant | 4,200 |
| 287 | 206 | William Gietz | 8,300 | FI_ACCOUNT | Accountant | 4,200 |
| 288 | 206 | William Gietz | 8,300 | PR_REP | Public Relations Representative | 4,500 |
| 289 | 206 | William Gietz | 8,300 | ST_MAN | Stock Manager | 5,500 |
| 290 | 206 | William Gietz | 8,300 | SA_REP | Sales Representative | 6,000 |
| 291 | 206 | William Gietz | 8,300 | PU_MAN | Purchasing Manager | 8,000 |
| 292 | 206 | William Gietz | 8,300 | AC_MGR | Accounting Manager | 8,200 |
| 293 | 206 | William Gietz | 8,300 | FI_MGR | Finance Manager | 8,200 |
| 294 | 220 | Minjun Kim | 10,500 | PR_REP | Public Relations Representative | 4,500 |
| 295 | 220 | Minjun Kim | 10,500 | SA_REP | Sales Representative | 6,000 |
| 296 | 220 | Minjun Kim | 10,500 | PU_MAN | Purchasing Manager | 8,000 |
| 297 | 220 | Minjun Kim | 10,500 | AC_MGR | Accounting Manager | 8,200 |
| 298 | 220 | Minjun Kim | 10,500 | FI_MGR | Finance Manager | 8,200 |
| 299 | 220 | Minjun Kim | 10,500 | MK_MAN | Marketing Manager | 9,000 |
| 300 | 220 | Minjun Kim | 10,500 | SA_MAN | Sales Manager | 10,000 |
| 301 | 221 | Seoyeon Lee | 8,500 | HR_REP | Human Resources Representative | 4,000 |
| 302 | 221 | Seoyeon Lee | 8,500 | IT_PROG | Programmer | 4,000 |
| 303 | 221 | Seoyeon Lee | 8,500 | MK_REP | Marketing Representative | 4,000 |
| 304 | 221 | Seoyeon Lee | 8,500 | AC_ACCOUNT | Public Accountant | 4,200 |
| 305 | 221 | Seoyeon Lee | 8,500 | FI_ACCOUNT | Accountant | 4,200 |
| 306 | 221 | Seoyeon Lee | 8,500 | PR_REP | Public Relations Representative | 4,500 |
| 307 | 221 | Seoyeon Lee | 8,500 | ST_MAN | Stock Manager | 5,500 |
| 308 | 221 | Seoyeon Lee | 8,500 | SA_REP | Sales Representative | 6,000 |
| 309 | 221 | Seoyeon Lee | 8,500 | PU_MAN | Purchasing Manager | 8,000 |
| 310 | 221 | Seoyeon Lee | 8,500 | AC_MGR | Accounting Manager | 8,200 |
| 311 | 221 | Seoyeon Lee | 8,500 | FI_MGR | Finance Manager | 8,200 |
| 312 | 222 | Jihoon Park | 7,800 | HR_REP | Human Resources Representative | 4,000 |
| 313 | 222 | Jihoon Park | 7,800 | IT_PROG | Programmer | 4,000 |
| 314 | 222 | Jihoon Park | 7,800 | MK_REP | Marketing Representative | 4,000 |
| 315 | 222 | Jihoon Park | 7,800 | AC_ACCOUNT | Public Accountant | 4,200 |
| 316 | 222 | Jihoon Park | 7,800 | FI_ACCOUNT | Accountant | 4,200 |
| 317 | 222 | Jihoon Park | 7,800 | PR_REP | Public Relations Representative | 4,500 |
| 318 | 222 | Jihoon Park | 7,800 | ST_MAN | Stock Manager | 5,500 |
| 319 | 222 | Jihoon Park | 7,800 | SA_REP | Sales Representative | 6,000 |
| 320 | 223 | Yuna Choi | 6,200 | HR_REP | Human Resources Representative | 4,000 |
| 321 | 223 | Yuna Choi | 6,200 | IT_PROG | Programmer | 4,000 |
| 322 | 223 | Yuna Choi | 6,200 | MK_REP | Marketing Representative | 4,000 |
| 323 | 223 | Yuna Choi | 6,200 | AC_ACCOUNT | Public Accountant | 4,200 |
| 324 | 223 | Yuna Choi | 6,200 | FI_ACCOUNT | Accountant | 4,200 |
| 325 | 223 | Yuna Choi | 6,200 | PR_REP | Public Relations Representative | 4,500 |
| 326 | 223 | Yuna Choi | 6,200 | ST_MAN | Stock Manager | 5,500 |
| 327 | 223 | Yuna Choi | 6,200 | SA_REP | Sales Representative | 6,000 |

Refresh Save Cancel Export data 200 327 327 row(s) fetched - 0.001s, on 2026-07-28 at 21:51:57

KST en Writable Smart Insert 524 : 48 : 17106 Sel: 0 | 0

E-1. 사원 기본 정보 View

| | |
|------------|---|
| 문제 번호(A-H) | E-1 |
| 분류 | VIEW 종합실습 |
| 문제 | [VIEW 1] 사원 기본 정보 View

HR 데이터를 활용하여 '사원 기본 정보 View'를 제공하는 일반 View 를 생성하고 조회하십시오. |
| 조건 | 지정된 이름의 일반 View 를 생성한 뒤 결과를 조회하십시오. |

실험결과

DBaver 26.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Auto postgres hr@skala_db

Connections postgres Sample Database (SQLite) postgres localhost:5432 schemas app tables countries enrollments majors students Foreign Tables Views Materialized Views Indices Functions Sequences Data types Aggregate functions day02 public tables countries departments employees job_history jobs locations regions Foreign Tables Views Materialized Views Indices Functions Sequences Data types Aggregate functions Event Triggers Extensions Storage System Info Roles Administrator System Info

<postgres> Script-3 <postgres> day02_PostgresQL_HR_종합실습_문제.sql <postgres> day02_PostgresQL_HR_emp_dept_삼출스키마.sql <postgres> day02_책리_임유리.sql

-- [VIEW 1] 사원 기본 정보 View
-- 목적: HR 데이터를 활용하여 '사원 기본 정보 View'를 제공하는 일반 View를 생성하고 조회하십시오.
-- 조건: 지정된 이름의 일반 View를 생성한 뒤 결과를 조회하십시오.
-- TDD: 아래의 SQL을 작성하십시오.
DROP VIEW IF EXISTS v_employee_basic CASCADE;
CREATE VIEW v_employee_basic AS
SELECT
employee_id,
concat_ws(' ', first_name, last_name) AS employee_name,
email,
phone_number,
hire_date,
job_id,
department_id
FROM employees;
SELECT *
FROM v_employee_basic
ORDER BY employee_id;

v_employee_basic 1 Statistics 1
DROP VIEW IF EXISTS v_employee_basic CASCADE; CREATE VII
Enter a SQL expression to filter results (use Ctrl+Space)
Grid
1 100 Steven King SKING 515.123.4567 2003-06-17 AD_PRES 24,000
2 101 Neena Kochhar NKOCHHAR 515.123.4568 2005-09-21 AD_VP 17,000
3 102 Lex De Haan LDEHAAN 515.123.4569 2001-01-13 AD_VP 17,000
4 103 Alexander Humold AHUMOLD 590.423.4567 2006-01-03 IT_PROG 9,000
5 104 Bruce Ernst BERNST 590.423.4568 2007-05-21 IT_PROG 6,000
6 105 David Austin DAUSTIN 590.423.4569 2005-06-25 IT_PROG 4,800
7 106 Valli Pataballa VPATABAL 590.423.4560 2006-02-05 IT_PROG 4,800
8 107 Diana Lorentz DLORENTZ 590.423.5567 2007-02-07 IT_PROG 4,200
9 108 Nancy Greenberg NGREENBE 515.124.4569 2002-08-17 FL_MGR 12,000
10 109 Daniel Faviat DFAVIET 515.124.4169 2002-08-16 FL_ACCOUNT 9,000
11 110 John Chen JCHEN 515.124.4269 2005-09-28 FL_ACCOUNT 8,200
12 111 Ismael Sciarra ISCIARRA 515.124.4369 2005-09-30 FL_ACCOUNT 7,700
13 112 Jose Manuel Urman JMURMAN 515.124.4469 2006-03-07 FL_ACCOUNT 7,800
14 113 Luis Popp LPOPP 515.124.4567 2007-12-07 FL_ACCOUNT 6,900
15 114 Den Raphaely DRAPHEAL 515.127.4561 2002-12-07 PU_MAN 11,000
16 115 Alexander Khoo AKHOO 515.127.4562 2003-05-18 PU_CLERK 3,100
17 116 Shelli Baida SBAIDA 515.127.4563 2005-12-24 PU_CLERK 2,900
18 117 Sigal Tobias STOBias 515.127.4564 2005-07-24 PU_CLERK 2,800
19 118 Guy Himuro GHIMURO 515.127.4565 2006-11-15 PU_CLERK 2,600
20 119 Karen Colmenares KCOLMENA 515.127.4566 2007-08-10 PU_CLERK 2,500
21 120 Matthew Weiss MWEISS 650.123.1234 2004-07-18 ST_MAN 8,000
22 121 Adam Fripp AFRIPP 650.123.2234 2005-04-10 ST_MAN 8,200
23 122 Payam Kaufing PKAUFUN 650.123.3234 2003-05-01 ST_MAN 7,900
24 145 John Russell JRUSSEL [NULL] 2004-10-01 SA_MAN 14,000
25 146 Karen Partners KPARTNER [NULL] 2005-01-05 SA_MAN 13,500
26 147 Alberto Erzuriz AERRAZUR [NULL] 2005-03-10 SA_MAN 12,000
27 148 Gerald Cambrault GCAMBRAU [NULL] 2007-10-15 SA_MAN 11,000
28 149 Eleni Zlotkey EZLOTKEY [NULL] 2008-01-29 SA_MAN 10,500
29 150 Peter Tucker PTUCKER [NULL] 2005-01-30 SA_REP 10,000
30 151 David Bernstein DBERNSTE [NULL] 2005-03-24 SA_REP 9,500
31 152 Peter Hall PHALL [NULL] 2005-08-20 SA_REP 9,000
32 153 Christopher Olsen COLSEN [NULL] 2006-03-30 SA_REP 8,000
33 154 Nanette Cambrault NCAMBRAU [NULL] 2006-12-09 SA_REP 7,500
34 155 Oliver Tuvault OTUVAULT [NULL] 2007-11-23 SA_REP 7,000
35 200 Jennifer Whalen JWHALEN 515.123.4444 2003-09-17 AD_ASST 4,400
36 201 Michael Hartstein MHARTSTE 515.123.5555 2004-02-17 MK_MAN 13,000
37 202 Pat Fay PFAY 603.123.6666 2005-08-17 MK_REP 6,000
38 203 Susan Mavris SMAVRIS 515.123.7777 2002-06-07 HR_REP 6,500
39 204 Hermann Baer HBAER 515.123.8888 2002-06-07 PR_REP 10,000
40 205 Shelley Higgins SHIGGINS 515.123.8080 2002-06-07 AC_MGR 12,000
41 206 William Gietz WGIEZT 515.123.8181 2002-06-07 AC_ACCOUNT 8,300
Refresh Save Cancel Export data 200 45 45 row(s) fetched - 0.002s, on 2026-07-28 at 21:52:49
KST en Writable Smart insert 551 : 1 : 17768 Sel: 0 | 0

[illegible]

E-2. 사원·부서·직무 통합 View

| | |
|------------|---|
| 문제 번호(A-H) | E-2 |
| 분류 | VIEW 종합실습 |
| 문제 | [VIEW 2] 사원·부서·직무 통합 View

HR 데이터를 활용하여 '사원·부서·직무 통합 View'를 제공하는 일반 View 를 생성하고 조회하십시오. |
| 조건 | 지정된 이름의 일반 View 를 생성한 뒤 결과를 조회하십시오. |

실험결과

DBaver 26.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Auto

postgres hr@skala_db

postgres day02_PostgresQL_HR_종합실습_문제.sql postgres day02_PostgresQL_HR_emp_dept_스키마.sql postgres day02_책리_임유리.sql

Filter connections by name

postgres Sample Database (SQLite)

postgres localst5432

postgres

skala_db

Schemas

Tables

courses

enrollments

majors

students

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

day02

public

hr

Tables

countries

departments

employees

job_history

jobs

locations

regions

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

Event Triggers

Extensions

Storage

System info

Roles

Administrator

System info

-- [VIEW 2] 사원·부서·직무 통합 View
-- 목적: HR 데이터를 활용하여 '사원·부서·직무 통합 View'를 제공하는 일반 View를 생성하고 조회하십시오.
-- 조건: 지정된 이름의 일반 View를 생성한 뒤 결과를 조회하십시오.
-- TODO: 아래에 SQL을 작성하십시오.
DROP VIEW IF EXISTS v_employee_department_job CASCADE;
CREATE VIEW v_employee_department_job AS
SELECT
e.employee_id,
concat_ws(' ', e.first_name, e.last_name) AS employee_name,
COALESCE(d.department_name, '부서 미배정') AS department_name,
COALESCE(j.job_title, '직무 미배정') AS job_title,
e.salary,
e.hire_date
FROM employees e
LEFT JOIN departments d ON d.department_id = e.department_id
LEFT JOIN jobs j ON j.job_id = e.job_id;
SELECT *
FROM v_employee_department_job
ORDER BY employee_id;

v_employee_department_job 1 x Statistics 1

DROP VIEW IF EXISTS v_employee_department_job CASCADE; Enter a SQL expression to filter results (use Ctrl+Space)

Grid

employee_id AZ employee_name AZ department_name AZ job_title salary hire_date

1 100 Steven King Executive President 24,000 2003-06-17

2 101 Neena Kochhar Executive Administration Vice President 17,000 2005-09-21

3 102 Lex De Haan Executive Administration Vice President 17,000 2001-01-13

4 103 Alexander Humold IT Programmer 9,000 2006-01-03

5 104 Bruce Ernst IT Programmer 6,000 2007-05-21

6 105 David Austin IT Programmer 4,800 2005-06-25

7 106 Valli Pataballa IT Programmer 4,800 2006-02-05

8 107 Diana Lorentz IT Programmer 4,200 2007-02-07

9 108 Nancy Greenberg Finance Finance Manager 12,000 2002-08-17

10 109 Daniel Faviet Finance Accountant 9,000 2002-08-16

11 110 John Chen Finance Accountant 8,200 2005-09-28

12 111 Ismael Sciarra Finance Accountant 7,700 2005-09-30

13 112 Jose Manuel Urman Finance Accountant 7,800 2006-03-07

14 113 Luis Popp Finance Accountant 6,900 2007-12-07

15 114 Den Raphaely Purchasing Purchasing Manager 11,000 2002-12-07

16 115 Alexander Khoo Purchasing Purchasing Clerk 3,100 2003-05-18

17 116 Shelli Baida Purchasing Purchasing Clerk 2,900 2005-12-24

18 117 Sigal Tobias Purchasing Purchasing Clerk 2,800 2005-07-24

19 118 Guy Himuro Purchasing Purchasing Clerk 2,600 2006-11-15

20 119 Karen Colmenares Purchasing Purchasing Clerk 2,500 2007-08-10

21 120 Matthew Weiss Shipping Stock Manager 8,000 2004-07-18

22 121 Adam Fripp Shipping Stock Manager 8,200 2005-04-10

23 122 Payam Kaufing Shipping Stock Manager 7,900 2003-05-01

24 145 John Russell Sales Sales Manager 14,000 2004-10-01

25 146 Karen Partners Sales Sales Manager 13,500 2005-01-05

26 147 Alberto Erzuriz Sales Sales Manager 12,000 2005-03-10

27 148 Gerald Cambrault Sales Sales Manager 11,000 2007-10-15

28 149 Eleni Zlotkey Sales Sales Manager 10,500 2008-01-29

29 150 Peter Tucker Sales Sales Representative 10,000 2005-01-30

30 151 David Bernstein Sales Sales Representative 9,500 2005-03-24

31 152 Peter Hall Sales Sales Representative 9,000 2005-08-20

32 153 Christopher Olsen Sales Sales Representative 8,000 2006-03-30

33 154 Nanette Cambrault Sales Sales Representative 7,500 2006-12-09

34 155 Oliver Tuvault Sales Sales Representative 7,000 2007-11-23

35 200 Jennifer Whalen Administration Administration Assistant 4,400 2003-09-17

36 201 Michael Hartstein Marketing Marketing Manager 13,000 2004-02-17

37 202 Pat Fay Marketing Marketing Representative 6,000 2005-08-17

38 203 Susan Mavris Human Resources Human Resources Representative 6,500 2002-06-07

39 204 Hermann Baer Public Relations Public Relations Representative 10,000 2002-06-07

40 205 Shelley Higgins Accounting Accounting Manager 12,000 2002-06-07

41 206 William Gietz Accounting Public Accountant 8,300 2002-06-07

42 220 Minjun Kim R&D Korea Programmer 10,500 2019-03-04

Value x

100

Refresh Save Cancel

45 row(s) fetched - 0.000s (0.001s fetch), on 2026-07-28 at 21:53:43

Export data

200

45

572 : 22 : 18443

Sel: 0 | 0

KST en Writable Smart insert

DBeaver 25.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Commit Rollback Auto

Connections postgres Script-3 <postgres> day02_PostgresQL_HR_종합실습_문제.sql <postgres> day02_PostgresQL_HR_emp_dept_샘플_스키마.sql <postgres> day02_책리_임유리.sql

Filter connections by name

- postgres localhost:5432
 - postgres
 - skala_db
 - Schemas
 - app
 - Tables

courses 24K
enrollments 24K
majors 40K
students 56K

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

Event Triggers

Extensions

Storage

System info

Roles

Administrator

System info

Files

Name

Bookmarks

Dashboards

Diagrams

General

```
-- [VIEW 2] 사원·부서·직무 통합 View
-- 목적: HR 데이터를 통합하여 '사원·부서·직무' 통합 View를 제공하는 일관 View를 생성하고 조회하십시오.
-- 조건: 지정된 이름의 통합 View를 생성한 뒤 결과를 조회하십시오.
-- TODO: 아래의 SQL을 작성하십시오.
DROP VIEW IF EXISTS v_employee_department_job CASCADE;
CREATE VIEW v_employee_department_job AS
SELECT
    e.employee_id,
    concat_ws(' ', e.first_name, e.last_name) AS employee_name,
    COALESCE(d.department_name, '부서 미배정') AS department_name,
    COALESCE(j.job_title, '직무 미배정') AS job_title,
    e.salary,
    e.hire_date
FROM employees e
LEFT JOIN departments d ON d.department_id = e.department_id
LEFT JOIN jobs j ON j.job_id = e.job_id;

SELECT *
FROM v_employee_department_job
ORDER BY employee_id;
```

v_employee_department_job 1 x Statistics 1

Grid

10 employee_id AZ employee_name AZ department_name AZ job_title 123 salary hire_date

| | | | | | | |
|----|-----|-------------------|------------------|---------------------------------|--------|------------|
| 5 | 104 | Bruce Ernst | IT | Programmer | 6,000 | 2007-05-21 |
| 6 | 105 | David Austin | IT | Programmer | 4,800 | 2005-06-25 |
| 7 | 106 | Valli Pataballa | IT | Programmer | 4,800 | 2006-02-05 |
| 8 | 107 | Diana Lorentz | IT | Programmer | 4,200 | 2007-02-07 |
| 9 | 108 | Nancy Greenberg | Finance | Finance Manager | 12,000 | 2002-08-17 |
| 10 | 109 | Daniel Faviet | Finance | Accountant | 9,000 | 2002-08-16 |
| 11 | 110 | John Chen | Finance | Accountant | 8,200 | 2005-09-28 |
| 12 | 111 | Ismael Scarra | Finance | Accountant | 7,700 | 2005-09-30 |
| 13 | 112 | José Manuel Urman | Finance | Accountant | 7,800 | 2006-03-07 |
| 14 | 113 | Luis Popp | Finance | Accountant | 6,900 | 2007-12-07 |
| 15 | 114 | Den Raphaely | Purchasing | Purchasing Manager | 11,000 | 2002-12-07 |
| 16 | 115 | Alexander Khoo | Purchasing | Purchasing Clerk | 3,100 | 2003-05-18 |
| 17 | 116 | Shelli Baida | Purchasing | Purchasing Clerk | 2,900 | 2005-12-24 |
| 18 | 117 | Sigal Tobias | Purchasing | Purchasing Clerk | 2,800 | 2005-07-24 |
| 19 | 118 | Guy Himuro | Purchasing | Purchasing Clerk | 2,600 | 2006-11-15 |
| 20 | 119 | Karen Colmenares | Purchasing | Purchasing Clerk | 2,500 | 2007-08-10 |
| 21 | 120 | Matthew Weiss | Shipping | Stock Manager | 8,000 | 2004-07-18 |
| 22 | 121 | Adam Fripp | Shipping | Stock Manager | 8,200 | 2005-04-10 |
| 23 | 122 | Payam Kaufling | Shipping | Stock Manager | 7,900 | 2003-05-01 |
| 24 | 145 | John Russell | Sales | Sales Manager | 14,000 | 2004-10-01 |
| 25 | 146 | Karen Partners | Sales | Sales Manager | 13,500 | 2005-01-05 |
| 26 | 147 | Alberto Errazuriz | Sales | Sales Manager | 12,000 | 2005-03-10 |
| 27 | 148 | Gerald Cambrault | Sales | Sales Manager | 11,000 | 2007-10-15 |
| 28 | 149 | Elvyn Zlotsev | Sales | Sales Manager | 10,500 | 2008-01-29 |
| 29 | 150 | Peter Tucker | Sales | Sales Representative | 10,000 | 2005-01-30 |
| 30 | 151 | David Bernstein | Sales | Sales Representative | 9,500 | 2005-03-24 |
| 31 | 152 | Peter Hall | Sales | Sales Representative | 9,000 | 2005-08-20 |
| 32 | 153 | Christopher Olsen | Sales | Sales Representative | 8,000 | 2006-03-30 |
| 33 | 154 | Nanette Cambrault | Sales | Sales Representative | 7,500 | 2006-12-09 |
| 34 | 155 | Oliver Tuvault | Sales | Sales Representative | 7,000 | 2007-11-23 |
| 35 | 200 | Jennifer Whalen | Administration | Administration Assistant | 4,400 | 2003-09-17 |
| 36 | 201 | Michael Hartstein | Marketing | Marketing Manager | 13,000 | 2004-02-17 |
| 37 | 202 | Pat Fay | Marketing | Marketing Representative | 6,000 | 2005-08-17 |
| 38 | 203 | Susan Mavris | Human Resources | Human Resources Representative | 6,500 | 2002-06-07 |
| 39 | 204 | Hermann Baer | Public Relations | Public Relations Representative | 10,000 | 2002-06-07 |
| 40 | 205 | Shelley Higgins | Accounting | Accounting Manager | 12,000 | 2002-06-07 |
| 41 | 206 | William Gietz | Accounting | Public Accountant | 8,300 | 2002-06-07 |
| 42 | 220 | Mihyun Kim | R&D Korea | Programmer | 10,500 | 2019-03-04 |
| 43 | 221 | Seoyeon Lee | R&D Korea | Programmer | 8,500 | 2020-07-13 |
| 44 | 222 | Jihoon Park | R&D Korea | Programmer | 7,800 | 2021-11-01 |
| 45 | 223 | Yuna Choi | 부서 미배정 | Marketing Representative | 6,200 | 2023-02-20 |

Value

206

Refresh Save Cancel Export data 200 45 45 row(s) fetched - 0.008s (0.001s fetch), on 2026-07-28 at 21:53:43

KST en Writable Smart Insert 572 : 22 : 18443 Sel: 0 | 0

E-3. 부서별 급여 통계 View

| | |
|------------|---|
| 문제 번호(A-H) | E-3 |
| 분류 | VIEW 종합실습 |
| 문제 | [VIEW 3] 부서별 급여 통계 View
HR 데이터를 활용하여 '부서별 급여 통계 View'를 제공하는 일반 View 를 생성하고 조회하십시오. |
| 조건 | 지정된 이름의 일반 View 를 생성한 뒤 결과를 조회하십시오. |

실험결과

DBBeaver 26.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Auto

postgres hr@skala.db

Connections

Filter connections by name

postgres localhost:5432

postgres

skala.db

Schemas

Tables

courses

enrollments 24K

majors 40K

students 56K

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

day02

public

hr

countries 40K

departments 72K

employees 88K

job_history 56K

jobs 24K

locations 40K

regions 40K

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

Event Triggers

Extensions

Storage

System info

Roles

Administer

System info

Files

Name

Bookmarks

Dashboards

Diagrams

General

-- [VIEW 3] 부서별 급여 통계 View
-- 목적: HR 데이터를 활용하여 '부서별 급여 통계 View'를 제공하는 일반 View를 생성하고 조회하십시오.
-- 조건: 지정된 이름의 일반 View를 생성한 뒤 결과를 조회하십시오.
-- TODO: 아래에 SQL을 작성하십시오.

DROP VIEW IF EXISTS v_department_salary_stats CASCADE;
CREATE VIEW v_department_salary_stats AS
SELECT
COALESCE(e.department_id, 0) AS department_key,
COALESCE(d.department_name, '부서 미배정') AS department_name,
COUNT(*) AS employee_count,
SUM(e.salary) AS total_salary,
ROUND(AVG(e.salary), 2) AS avg_salary,
MIN(e.salary) AS min_salary,
MAX(e.salary) AS max_salary
FROM employees e
LEFT JOIN departments d ON d.department_id = e.department_id
GROUP BY e.department_id, d.department_name;

SELECT *
FROM v_department_salary_stats
ORDER BY department_key;

v_department_salary_stats 1

Statistics 1

DROP VIEW IF EXISTS v_department_salary_stats CASCADE; CR

Enter a SQL expression to filter results (use Ctrl+Space)

department_key

A-Z department_name

123 employee_count

123 total_salary

123 avg_salary

123 min_salary

123 max_salary

Value

1 0 부서 미배정 1 6,200 6,200 6,200

2 10 Administration 1 4,400 4,400 4,400

3 20 Marketing 2 19,000 9,500 6,000

4 30 Purchasing 6 24,900 4,150 2,500

5 40 Human Resources 1 6,500 6,500 6,500

6 50 Shipping 3 24,100 8,033.33 7,900

7 60 IT 5 28,800 5,760 4,200

8 70 Public Relations 1 10,000 10,000 10,000

9 80 Sales 11 112,000 10,181.82 7,000

10 90 Executive 3 58,000 19,333.33 17,000

11 100 Finance 6 51,600 8,600 6,900

12 110 Accounting 2 20,300 10,150 8,300

13 120 R&D Korea 3 26,800 8,933.33 7,800

Refresh

Save

Cancel

Export data

200

13

13 row(s) fetched - 0.002s on 2026-07-28 at 21:54:10

KST en Writable

Smart insert

596 : 1 : 19187

Sel: 0 | 0

E-4. 관리자별 부하 직원 View

| | |
|------------|---|
| 문제 번호(A-H) | E-4 |
| 분류 | VIEW 종합실습 |
| 문제 | [VIEW 4] 관리자별 부하 직원 View

HR 데이터를 활용하여 '관리자별 부하 직원 View'를 제공하는 일반 View 를 생성하고 조회하십시오. |
| 조건 | 지정된 이름의 일반 View 를 생성한 뒤 결과를 조회하십시오. |

실험결과

DBBeaver 26.1.3 - <postgres> day02_관리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Auto

postgres hr@skala_db

Connections

Filter connections by name

DBBeaver Sample Database (SQLite)

postgres localhost:5432

postgres

skala_db

Schemas

app

Tables

courses

enrollments

majors

students

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

day02

public

Tables

countries

departments

employees

job_history

jobs

locations

regions

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

Event Triggers

Extensions

Storage

System info

Roles

Administer

System info

Files

Bookmarks

Dashboards

Diagrams

-- [VIEW 4] 관리자별 부하 직원 View
-- 문제: HR 데이터를 활용하여 '관리자별 부하 직원 View'를 제공하는 일반 View를 생성하고 조회하십시오.
-- 조건: 지정된 이름의 일반 View를 생성한 뒤 결과를 조회하십시오.
-- TODO: 아래에 SQL을 작성하십시오.

DROP VIEW IF EXISTS v_manager_subordinates CASCADE;
CREATE VIEW v_manager_subordinates AS
SELECT
 m.employee_id AS manager_id,
 concat_ws(' ', m.first_name, m.last_name) AS manager_name,
 e.employee_id AS employee_id,
 concat_ws(' ', e.first_name, e.last_name) AS employee_name,
 COALESCE(d.department_name, '부서 미배정') AS department_name
FROM employees e
INNER JOIN employees m ON m.employee_id = e.manager_id
LEFT JOIN departments d ON d.department_id = e.department_id;

SELECT *
FROM v_manager_subordinates
ORDER BY manager_id, employee_id;

v_manager_subordinates 1 Statistics 1

DROP VIEW IF EXISTS v_manager_subordinates CASCADE; CREATE

| | manager_id | AZ manager_name | employee_id | AZ employee_name | AZ department_name |
|----|------------|-------------------|-------------|-------------------|--------------------|
| 1 | 100 | Steven King | 101 | Neena Kochhar | Executive |
| 2 | 100 | Steven King | 102 | Lex De Haan | Executive |
| 3 | 100 | Steven King | 114 | Den Raphaely | Purchasing |
| 4 | 100 | Steven King | 120 | Matthew Weiss | Shipping |
| 5 | 100 | Steven King | 121 | Adam Fripp | Shipping |
| 6 | 100 | Steven King | 122 | Payam Kaufing | Shipping |
| 7 | 100 | Steven King | 145 | John Russell | Sales |
| 8 | 100 | Steven King | 146 | Karen Partners | Sales |
| 9 | 100 | Steven King | 147 | Alberto Erazuiz | Sales |
| 10 | 100 | Steven King | 148 | Gerald Cambrault | Sales |
| 11 | 100 | Steven King | 149 | Eleni Zlotkey | Sales |
| 12 | 100 | Steven King | 201 | Michael Hartstein | Marketing |
| 13 | 101 | Neena Kochhar | 108 | Nancy Greenberg | Finance |
| 14 | 101 | Neena Kochhar | 200 | Jennifer Whalen | Administration |
| 15 | 101 | Neena Kochhar | 203 | Susan Mavris | Human Resources |
| 16 | 101 | Neena Kochhar | 204 | Hermann Baer | Public Relations |
| 17 | 101 | Neena Kochhar | 205 | Shelley Higgins | Accounting |
| 18 | 102 | Lex De Haan | 103 | Alexander Hunold | IT |
| 19 | 102 | Lex De Haan | 220 | Minjun Kim | R&D Korea |
| 20 | 103 | Alexander Hunold | 104 | Bruce Ernst | IT |
| 21 | 103 | Alexander Hunold | 105 | David Austin | IT |
| 22 | 103 | Alexander Hunold | 106 | Valli Pataballa | IT |
| 23 | 103 | Alexander Hunold | 107 | Diana Lorentz | IT |
| 24 | 108 | Nancy Greenberg | 109 | Daniel Faviet | Finance |
| 25 | 108 | Nancy Greenberg | 110 | John Chen | Finance |
| 26 | 108 | Nancy Greenberg | 111 | Ismael Sciarra | Finance |
| 27 | 108 | Nancy Greenberg | 112 | Jose Manuel Urman | Finance |
| 28 | 108 | Nancy Greenberg | 113 | Luis Popp | Finance |
| 29 | 114 | Den Raphaely | 115 | Alexander Khoo | Purchasing |
| 30 | 114 | Den Raphaely | 116 | Shelli Baida | Purchasing |
| 31 | 114 | Den Raphaely | 117 | Sigal Tobias | Purchasing |
| 32 | 114 | Den Raphaely | 118 | Guy Himuro | Purchasing |
| 33 | 114 | Den Raphaely | 119 | Karen Colmenares | Purchasing |
| 34 | 145 | John Russell | 150 | Peter Tucker | Sales |
| 35 | 145 | John Russell | 151 | David Bernstein | Sales |
| 36 | 145 | John Russell | 152 | Peter Hall | Sales |
| 37 | 145 | John Russell | 153 | Christopher Olsen | Sales |
| 38 | 145 | John Russell | 154 | Nanette Cambrault | Sales |
| 39 | 145 | John Russell | 155 | Oliver Tuvault | Sales |
| 40 | 201 | Michael Hartstein | 202 | Pat Fay | Marketing |
| 41 | 201 | Michael Hartstein | 223 | Yuna Choi | 부서 미배정 |

Refresh Save Cancel Export data 200 44 44 row(s) fetched - 0.005s, on 2026-07-28 at 21:54:45

General KST en Writable Smart Insert 617 : 1 : 19909 Sel: 0 | 0

The screenshot shows the DBeaver SQL editor interface. The left sidebar displays the database schema for 'postgres' on 'localhost:5432', including tables like 'enrollments', 'majors', 'students', and 'courses'. The main editor area contains a SQL script that creates a view named 'v_manager_subordinates'. The script includes comments in Korean and SQL code for dropping the view if it exists, creating it, and selecting data from the 'employees' table joined with 'departments'.

The SQL script is as follows:

```
-- [VIEW 4] 관리자별 부하 직원 View
-- 목적: HR 데이터를 활용하여 '관리자별 부하 직원 View'를 제공하는 일반 View를 생성하고 조회하십시오.
-- 조건: 직장을 이월한 일반 View를 생성한 뒤 결과를 조회하십시오.
-- TODO: 약간의 SQL을 작성하십시오.

DROP VIEW IF EXISTS v_manager_subordinates CASCADE;
CREATE VIEW v_manager_subordinates AS
SELECT
    e.employee_id AS manager_id,
    concat_ws(' ', m.first_name, m.last_name) AS manager_name,
    e.employee_id AS employee_id,
    concat_ws(' ', e.first_name, e.last_name) AS employee_name,
    COALESCE(d.department_name, '부서 미매칭') AS department_name
FROM employees e
INNER JOIN employees m ON m.employee_id = e.manager_id
LEFT JOIN departments d ON d.department_id = e.department_id;

SELECT *
FROM v_manager_subordinates
ORDER BY manager_id, employee_id;
```

The results pane shows the data for the 'v_manager_subordinates' view. The table has 44 rows. The columns are: manager_id, manager_name, employee_id, employee_name, and department_name. The data is sorted by manager_id and then employee_id.

| manager_id | manager_name | employee_id | employee_name | department_name |
|------------|-----------------------|-----------------------|------------------|-----------------|
| 5 | 100 Steven King | 121 Adam Fripp | Shipping | |
| 6 | 100 Steven King | 122 Payam Kaufing | Shipping | |
| 7 | 100 Steven King | 145 John Russell | Sales | |
| 8 | 100 Steven King | 146 Karen Partners | Sales | |
| 9 | 100 Steven King | 147 Alberto Erazuiz | Sales | |
| 10 | 100 Steven King | 148 Gerald Cambrault | Sales | |
| 11 | 100 Steven King | 149 Eleni Zlotkey | Sales | |
| 12 | 100 Steven King | 201 Michael Hartstein | Marketing | |
| 13 | 101 Neena Kochhar | 108 Nancy Greenberg | Finance | |
| 14 | 101 Neena Kochhar | 200 Jennifer Whalen | Administration | |
| 15 | 101 Neena Kochhar | 203 Susan Mavris | Human Resources | |
| 16 | 101 Neena Kochhar | 204 Hermann Baer | Public Relations | |
| 17 | 101 Neena Kochhar | 205 Shelley Higgins | Accounting | |
| 18 | 102 Lex De Haan | 103 Alexander Hunold | IT | |
| 19 | 102 Lex De Haan | 220 Minjun Kim | R&D Korea | |
| 20 | 103 Alexander Hunold | 104 Bruce Ernst | IT | |
| 21 | 103 Alexander Hunold | 105 David Austin | IT | |
| 22 | 103 Alexander Hunold | 106 Valli Pataballa | IT | |
| 23 | 103 Alexander Hunold | 107 Diana Lorentz | IT | |
| 24 | 108 Nancy Greenberg | 109 Daniel Fariet | Finance | |
| 25 | 108 Nancy Greenberg | 110 John Chen | Finance | |
| 26 | 108 Nancy Greenberg | 111 Ismael Scarra | Finance | |
| 27 | 108 Nancy Greenberg | 112 Jose Manuel Urman | Finance | |
| 28 | 108 Nancy Greenberg | 113 Luis Popp | Finance | |
| 29 | 114 Den Raphaely | 115 Alexander Khoo | Purchasing | |
| 30 | 114 Den Raphaely | 116 Shelli Baida | Purchasing | |
| 31 | 114 Den Raphaely | 117 Sigal Tobias | Purchasing | |
| 32 | 114 Den Raphaely | 118 Guy Himuro | Purchasing | |
| 33 | 114 Den Raphaely | 119 Karen Colmenares | Purchasing | |
| 34 | 145 John Russell | 150 Peter Tucker | Sales | |
| 35 | 145 John Russell | 151 David Bernstein | Sales | |
| 36 | 145 John Russell | 152 Peter Hall | Sales | |
| 37 | 145 John Russell | 153 Christopher Olsen | Sales | |
| 38 | 145 John Russell | 154 Nanette Cambrault | Sales | |
| 39 | 145 John Russell | 155 Oliver Tuvault | Sales | |
| 40 | 201 Michael Hartstein | 202 Pat Fay | Marketing | |
| 41 | 201 Michael Hartstein | 223 Yuna Choi | 부서 미매칭 | |
| 42 | 205 Shelley Higgins | 206 William Gietz | Accounting | |
| 43 | 220 Minjun Kim | 221 Seoyeon Lee | R&D Korea | |
| 44 | 220 Minjun Kim | 222 Jihoon Park | R&D Korea | |

E-5. 직무 이력 상세 View

| | |
|------------|---|
| 문제 번호(A-H) | E-5 |
| 분류 | VIEW 종합실습 |
| 문제 | [VIEW 5] 직무 이력 상세 View
HR 데이터를 활용하여 '직무 이력 상세 View'를 제공하는 일반 View 를 생성하고 조회하십시오. |
| 조건 | 지정된 이름의 일반 View 를 생성한 뒤 결과를 조회하십시오. |

실험결과

DBaver 26.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Filter connections by name

postgres Sample Database (SQLite)

postgres localhost:5432

postgres

skala_db

Schemas

app

courses

enrollments

majors

students

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

day02

public

hr

countries

departments

employees

job_history

jobs

locations

regions

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

Event Triggers

Extensions

Storage

System Info

Roles

Administer

System info

<postgres> Script-3

<postgres> day02_PostgresQL_HR_종합실습-문제.sql

<postgres> day02_PostgresQL_HR_emp_dept_삼출스키마.sql

<postgres> day02_책리_임유리.sql

-- [VIEW 5] 직무 이력 상세 View

-- 목적: HR 데이터를 활용하여 '직무 이력 상세 View'를 제공하는 일반 View를 생성하고 조회하십시오.

-- 조건: 지정된 이름의 일반 View를 생성한 뒤 결과를 조회하십시오.

-- TODO: 아래에 SQL을 작성하십시오.

DROP VIEW IF EXISTS v_job_history_detail CASCADE;

CREATE VIEW v_job_history_detail AS

SELECT

jh.employee_id,

concat_ws(' ', e.first_name, e.last_name) AS employee_name,

jh.start_date,

jh.end_date,

jh.end_date - jh.start_date AS worked_days,

jh.job_id,

j.job_title,

COALESCE(d.department_name, '부서 미매칭') AS department_name

FROM job_history jh

INNER JOIN employees e ON e.employee_id = jh.employee_id

INNER JOIN jobs j ON j.job_id = jh.job_id

LEFT JOIN departments d ON d.department_id = jh.department_id;

SELECT *

FROM v_job_history_detail

ORDER BY employee_id, start_date;

v_job_history_detail 1

Statistics 1

DROP VIEW IF EXISTS v_job_history_detail CASCADE; CREATE |

Enter a SQL expression to filter results (use Ctrl+Space)

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F-1. 부서별 급여 요약

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| 문제 번호(A-H) | F-1 |
| 분류 | MATERIALIZED VIEW 종합실습 |
| 문제 | [Materialized View 1] 부서별 급여 요약
HR 데이터를 활용하여 '부서별 급여 요약' Materialized View 를 생성하고 조회하십시오. |
| 조건 | Materialized View 와 UNIQUE INDEX 를 생성한 뒤 결과를 조회하십시오. |

실험결과

The screenshot shows the DBeaver 26.1.3 interface with a PostgreSQL database named 'hr' connected. The left sidebar displays the database structure, including schemas, tables, and indexes. The main window shows a SQL script for creating a materialized view and an index. The bottom pane displays the execution results of the SQL script, showing a table with columns: department_key, department_name, employee_count, total_salary, and avg_salary. The table contains 13 rows of data for various departments.

| department_key | department_name | employee_count | total_salary | avg_salary |
|----------------|---------------------|----------------|--------------|------------|
| 0 | 부서 미배정 | 1 | 6,200 | 6,200 |
| 2 | 10 Administration | 4 | 4,400 | 4,400 |
| 3 | 20 Marketing | 2 | 19,000 | 9,500 |
| 4 | 30 Purchasing | 6 | 24,900 | 4,150 |
| 5 | 40 Human Resources | 1 | 6,500 | 6,500 |
| 6 | 50 Shipping | 3 | 24,100 | 8,033.33 |
| 7 | 60 IT | 5 | 28,800 | 5,760 |
| 8 | 70 Public Relations | 1 | 10,000 | 10,000 |
| 9 | 80 Sales | 11 | 112,000 | 10,181.82 |
| 10 | 90 Executive | 3 | 58,000 | 19,333.33 |
| 11 | 100 Finance | 6 | 51,600 | 8,600 |
| 12 | 110 Accounting | 2 | 20,300 | 10,150 |
| 13 | 120 R&D Korea | 3 | 26,800 | 8,933.33 |

F-2. 국가별 사원 현황

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|------------|--|
| 문제 번호(A-H) | F-2 |
| 분류 | MATERIALIZED VIEW 종합실습 |
| 문제 | [Materialized View 2] 국가별 사원 현황
HR 데이터를 활용하여 '국가별 사원 현황' Materialized View 를 생성하고 조회하십시오. |
| 조건 | Materialized View 와 UNIQUE INDEX 를 생성한 뒤 결과를 조회하십시오. |

실험결과

DBaver 26.13 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Filter connections by name

postgres localhost:5432

postgres

skala_db

Schemas

Tables

countries

enrollments

majors

students

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

day02

public

hr

Tables

countries

departments

employees

job_history

jobs

locations

regions

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

Event Triggers

Extensions

Storage

System info

Roles

Administrator

System info

Script-3

postgres> day02_PostgreSQL_HR_종합실습_문제.sql

postgres> day02_PostgreSQL_HR_emp_dept_삼키마.sql

postgres> day02_책리_임유리.sql

-- [Materialized View 2] 국가별 사원 현황
-- 문제: HR 데이터를 활용하여 '국가별 사원 현황' Materialized View를 생성하고 조회하십시오.
-- 조건: Materialized View의 UNIQUE INDEX를 생성한 뒤 결과를 조회하십시오.
-- TODO: 아래의 SQL을 작성하십시오.
DROP MATERIALIZED VIEW IF EXISTS mv_country_employee_summary CASCADE;
CREATE MATERIALIZED VIEW mv_country_employee_summary AS
SELECT
 c.country_id,
 c.country_name,
 COUNT(e.employee_id) AS employee_count,
 COALESCE(SUM(e.salary), 0) AS total_salary,
 ROUND(AVG(e.salary), 2) AS avg_salary
FROM countries c
LEFT JOIN locations l ON l.country_id = c.country_id
LEFT JOIN departments d ON d.location_id = l.location_id
LEFT JOIN employees e ON e.department_id = d.department_id
GROUP BY c.country_id, c.country_name
WITH DATA;

CREATE UNIQUE INDEX ux_mv_country_employee_summary_country_id
ON mv_country_employee_summary(country_id);

SELECT *
FROM mv_country_employee_summary
ORDER BY country_id;

REFRESH MATERIALIZED VIEW CONCURRENTLY mv_country_employee_summary;

mv_country_employee_summary 1 X Statistics 1

Enter a SQL expression to filter results (use Ctrl+Space)

country_id country_name employee_count total_salary avg_salary

1 AU Australia 0 0 [NULL]

2 BR Brazil 0 0 [NULL]

3 CA Canada 2 19,000 9,500

4 DE Germany 1 10,000 10,000

5 EG Egypt 0 0 [NULL]

6 FR France 0 0 [NULL]

7 IN India 0 0 [NULL]

8 IT Italy 0 0 [NULL]

9 JP Japan 0 0 [NULL]

10 KR South Korea 3 26,800 8,933.33

11 SG Singapore 0 0 [NULL]

12 UK United Kingdom 12 118,500 9,875

13 US United States 26 212,100 8,157.69

14 ZA South Africa 0 0 [NULL]

Value X

AU

Refresh Save Cancel

Export data

200

14

14 row(s) fetched - 0.028s, on 2026-07-28 at 21:56:25

KST en Writable

Smart insert

703 : 1 : 22766

Set: 0 | 0

G-1. 부서와 직무별 급여 합계

| | |
|------------|---|
| 문제 번호(A-H) | G-1 |
| 분류 | CUBE 종합실습 |
| 문제 | [CUBE 1] 부서와 직무별 급여 합계

HR 데이터를 활용하여 '부서와 직무별 급여 합계' 결과를 조회하는 SQL 을 작성하십시오. |
| 조건 | GROUP BY CUBE 와 GROUPING 함수를 사용해 상세·소계·전체 합계를 구하십시오. |

실험결과

DBaver 26.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Filter connections by name

postgres local host 5432

postgres

skala_db

Schemas

Tables

courses

enrollments 24K

majors 40K

students 56K

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

day02

public

hr

Tables

countries 40K

departments 72K

employees 88K

job_history 56K

jobs 24K

locations 40K

regions 40K

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

Event Triggers

Extensions

Storage

System Info

Roles

Administer

System Info

Script-3

postgres day02_PostgresQL_HR_종합실습_문제.sql

postgres day02_PostgresQL_HR_emp_dept_상용_스키마.sql

postgres day02_책리_임유리.sql

-- G. CUBE 종합실습 (2문제)

-- [CUBE 1] 부서와 직무별 급여 합계

-- 용지: HR 데이터를 활용하여 '부서와 직무별 급여 합계' 결과를 조회하는 SQL 을 작성하십시오.

-- 조건: GROUP BY CUBE 와 GROUPING 함수를 사용해 상세·소계·전체 합계를 구하십시오.

-- TODO: 아래에 SQL 을 작성하십시오.

SELECT

CASE

WHEN GROUPING(d.department_name) = 1 THEN '모든 부서'

ELSE COALESCE(d.department_name, '부서 미배정')

END AS "부서명",

CASE

WHEN GROUPING(j.job_title) = 1 THEN '모든 직무'

ELSE COALESCE(j.job_title, '직무 미배정')

END AS "직무명",

COUNT(*) AS "직원 수",

SUM(e.salary) AS "급여 합계"

FROM employees e

LEFT JOIN departments d ON d.department_id = e.department_id

LEFT JOIN jobs j ON j.job_id = e.job_id

GROUP BY CUBE(d.department_name, j.job_title)

ORDER BY

GROUPING(d.department_name), d.department_name NULLS LAST,

GROUPING(j.job_title), j.job_title NULLS LAST;

employees(*) 1 X

SELECT CASE WHEN GROUPING(d.department_name) = 1 TH

Enter a SQL expression to filter results (use Ctrl+Space)

1 Accounting Accounting Manager 1 12,000

2 Accounting Public Accountant 1 8,300

3 Accounting 모든 직무 2 20,300

4 Administration Administration Assistant 1 4,400

5 Administration 모든 직무 1 4,400

6 Executive Administration Vice President 2 34,000

7 Executive President 1 24,000

8 Executive 모든 직무 3 58,000

9 Finance Accountant 5 39,600

10 Finance Finance Manager 1 12,000

11 Finance 모든 직무 6 51,600

12 Human Resources Human Resources Representative 1 6,500

13 Human Resources 모든 직무 1 6,500

14 IT Programmer 5 28,800

15 IT 모든 직무 5 28,800

16 Marketing Marketing Manager 1 13,000

17 Marketing Marketing Representative 1 6,000

18 Marketing 모든 직무 2 19,000

19 Public Relations Public Relations Representative 1 10,000

20 Public Relations 모든 직무 1 10,000

21 Purchasing Purchasing Clerk 5 13,900

22 Purchasing Purchasing Manager 1 11,000

23 Purchasing 모든 직무 6 24,900

24 R&D Korea Programmer 3 26,800

25 R&D Korea 모든 직무 5 61,000

26 Sales Sales Representative 6 51,000

27 Sales 모든 직무 11 112,000

28 Shipping Stock Manager 3 24,100

29 Shipping 모든 직무 3 24,100

30 부서 미배정 Marketing Representative 1 6,200

31 부서 미배정 모든 직무 1 6,200

32 모든 부서 Accountant 5 39,600

33 모든 부서 Accounting Manager 1 12,000

34 모든 부서 Administration Assistant 1 4,400

35 모든 부서 Administration Vice President 2 34,000

36 모든 부서 Finance Manager 1 12,000

37 모든 부서 Human Resources Representative 1 6,500

38

Value X

Text

Accounting

Refresh

Save

Cancel

Export data

200

50

500 row(s) fetched - 0.002s, on 2026-07-28 at 21:56:53

KST en Writable Smart insert 729 : 51 : 23725 Sel: 0 | 0

DBeaver 26.1.3 - <postgres> day02_쿼리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Rollback Commit SQL postgres hr@skala_db

Connections postgres localhost:5432 postgres skala_db Schemas app Tables courses 24K enrollments 40K majors 56K students 56K Foreign Tables Views Materialized Views Indexes Functions Sequences Data types Aggregate functions Event Triggers Extensions Storage System Info Roles Administrator System info

Filter connections by name

day02 public hr Tables countries 40K departments 72K employees 80K job_history 56K jobs 24K locations 40K regions 40K Foreign Tables Views Materialized Views Indexes Functions Sequences Data types Aggregate functions Event Triggers Extensions Storage System Info Roles Administrator System info

Files Name Bookmarks Dashboards Diagrams General

Script-3 <postgres> day02_PostgreSQL_HR_종합실습_문제.sql <postgres> day02_PostgreSQL_HR_emp_dept_샘플_스키마.sql <postgres> day02_쿼리_임유리.sql

-- G. CUBE 종합실습 (2분짜리)

-- [CUBE 1] 부서와 직무별 급여 집계
-- 문제: HR 데이터를 활용하여 '부서'와 '직무'별 급여 집계 결과를 조회하는 SQL을 작성하십시오.
-- 조건: GROUP BY CUBE와 GROUPING 함수를 사용해 상세·요약 집계 결과를 구하십시오.
-- TODO: 아래에 SQL을 작성하십시오.

```
SELECT
  CASE
    WHEN GROUPING(d.department_name) = 1 THEN '모든 부서'
    ELSE COALESCE(d.department_name, '부서 미배정')
  END AS "부서명",
  CASE
    WHEN GROUPING(j.job_title) = 1 THEN '모든 직무'
    ELSE COALESCE(j.job_title, '직무 미배정')
  END AS "직무명",
  COUNT(*) AS "직원 수",
  SUM(e.salary) AS "급여 합계"
FROM employees e
LEFT JOIN departments d ON d.department_id = e.department_id
LEFT JOIN jobs j ON j.job_id = e.job_id
GROUP BY CUBE(d.department_name, j.job_title)
ORDER BY
  GROUPING(d.department_name), d.department_name NULLS LAST,
  GROUPING(j.job_title), j.job_title NULLS LAST;
```

employees(←) 1 ×

SELECT CASE WHEN GROUPING(d.department_name) = 1 THEN '모든 부서' ELSE COALESCE(d.department_name, '부서 미배정') END AS "부서명", CASE WHEN GROUPING(j.job_title) = 1 THEN '모든 직무' ELSE COALESCE(j.job_title, '직무 미배정') END AS "직무명", COUNT(*) AS "직원 수", SUM(e.salary) AS "급여 합계" FROM employees e LEFT JOIN departments d ON d.department_id = e.department_id LEFT JOIN jobs j ON j.job_id = e.job_id GROUP BY CUBE(d.department_name, j.job_title) ORDER BY GROUPING(d.department_name), d.department_name NULLS LAST, GROUPING(j.job_title), j.job_title NULLS LAST;

| 부서명 | 직무명 | 직원 수 | 급여 합계 |
|------------------|---------------------------------|------|---------|
| 모든 부서 | 모든 직무 | 123 | 28,800 |
| IT | Programmer | 5 | 28,800 |
| IT | 모든 직무 | 5 | 28,800 |
| Marketing | Marketing Manager | 1 | 13,000 |
| Marketing | Marketing Representative | 1 | 6,000 |
| Marketing | 모든 직무 | 2 | 19,000 |
| Public Relations | Public Relations Representative | 1 | 10,000 |
| Public Relations | 모든 직무 | 1 | 10,000 |
| Purchasing | Purchasing Clerk | 5 | 13,900 |
| Purchasing | Purchasing Manager | 1 | 11,000 |
| Purchasing | 모든 직무 | 6 | 24,900 |
| R&D Korea | Programmer | 3 | 26,800 |
| R&D Korea | 모든 직무 | 3 | 26,800 |
| Sales | Sales Manager | 5 | 61,000 |
| Sales | Sales Representative | 6 | 51,000 |
| Sales | 모든 직무 | 11 | 112,000 |
| Shipping | Stock Manager | 3 | 24,100 |
| Shipping | 모든 직무 | 3 | 24,100 |
| 부서 미배정 | Marketing Representative | 1 | 6,200 |
| 부서 미배정 | 모든 직무 | 1 | 6,200 |
| 모든 부서 | Accountant | 5 | 39,600 |
| 모든 부서 | Accounting Manager | 1 | 12,000 |
| 모든 부서 | Administration Assistant | 1 | 4,400 |
| 모든 부서 | Administration Vice President | 2 | 34,000 |
| 모든 부서 | Finance Manager | 1 | 12,000 |
| 모든 부서 | Human Resources Representative | 1 | 6,500 |
| 모든 부서 | Marketing Manager | 1 | 13,000 |
| 모든 부서 | Marketing Representative | 2 | 12,200 |
| 모든 부서 | President | 1 | 24,000 |
| 모든 부서 | Programmer | 8 | 55,600 |
| 모든 부서 | Public Accountant | 1 | 8,300 |
| 모든 부서 | Public Relations Representative | 1 | 10,000 |
| 모든 부서 | Purchasing Clerk | 5 | 13,900 |
| 모든 부서 | Purchasing Manager | 1 | 11,000 |
| 모든 부서 | Sales Manager | 5 | 61,000 |
| 모든 부서 | Sales Representative | 6 | 51,000 |
| 모든 부서 | Stock Manager | 3 | 24,100 |
| 모든 부서 | 모든 직무 | 45 | 392,600 |

Refresh Save Cancel Export data 200 50 50 row(s) fetched - 0.002s, on 2026-07-28 at 21:56:53

KST en Writable Smart Insert 729 : 51 : 23725 Sel: 0 | 0

G-2. 입사 연도와 커미션 수령 여부별 인원수

| | |
|------------|---|
| 문제 번호(A-H) | G-2 |
| 분류 | CUBE 종합실습 |
| 문제 | [CUBE 2] 입사 연도와 커미션 수령 여부별 인원수
HR 데이터를 활용하여 '입사 연도와 커미션 수령 여부별 인원수' 결과를 조회하는 SQL 을 작성하십시오. |
| 조건 | GROUP BY CUBE 와 GROUPING 함수를 사용해 상세·소계·전체 합계를 구하십시오. |

실험결과

DBBeaver 26.1.3 - <postgres> day02_책리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Rollback Auto postgres hr@skala_db

Connections postgres localhost:5432

postgres

skala_db

Schemas

app

Tables

Courses

enrollments 24K

majors 40K

students 56K

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

day02

public

hr

Tables

countries 40K

departments 72K

employees 88K

job_history 56K

jobs 24K

locations 40K

regions 40K

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

Event Triggers

Extensions

Storage

System info

Roles

Administer

System info

Script-3

<postgres> day02_PostgreSQL_HR_종합실습_문제.sql

<postgres> day02_PostgreSQL_HR_emp_dept_샘플_스키마.sql

<postgres> day02_책리_임유리.sql X

-- [CUBE 2] 입사 연도와 커미션 수령 여부별 인원수
-- 목적: HR 데이터를 활용하여 '입사 연도와 커미션 수령 여부별 인원수' 결과를 조회하는 SQL 을 작성하십시오.
-- 조건: GROUP BY, CUBE와 GROUPING 함수를 사용해 상세·소계·전체 합계를 구하십시오.
-- TODO: 아래에 SQL 을 작성하십시오.
SELECT
CASE
WHEN GROUPING(EXTRACT(YEAR FROM hire_date)) = 1 THEN '모든 연도'
ELSE EXTRACT(YEAR FROM hire_date)::integer::text
END AS "입사 연도",
CASE
WHEN GROUPING(
CASE WHEN commission_pct IS NOT NULL
THEN "커미션 수령" ELSE "커미션 미수령" END
) = 1 THEN '전체'
ELSE CASE WHEN commission_pct IS NOT NULL
THEN "커미션 수령" ELSE "커미션 미수령" END
END AS "커미션 수령 여부",
COUNT(*) AS "사원 수"
FROM employees
GROUP BY CUBE(
EXTRACT(YEAR FROM hire_date),
CASE WHEN commission_pct IS NOT NULL
THEN "커미션 수령" ELSE "커미션 미수령" END
)
ORDER BY
GROUPING(EXTRACT(YEAR FROM hire_date)),
EXTRACT(YEAR FROM hire_date) NULLS LAST,
GROUPING(
CASE WHEN commission_pct IS NOT NULL
THEN "커미션 수령" ELSE "커미션 미수령" END
)
CASE WHEN commission_pct IS NOT NULL
THEN "커미션 수령" ELSE "커미션 미수령" END NULLS LAST;

Results 1 X

SELECT CASE WHEN GROUPING(EXTRACT(YEAR FROM hire_d... Enter a SQL expression to filter results (use Ctrl+Space)

1 2001 커미션 미수령 123 사원 수

2 2001 전체 1

3 2002 커미션 미수령 7

4 2002 전체 7

5 2003 커미션 미수령 4

6 2003 전체 4

7 2004 커미션 미수령 2

8 2004 커미션 수령 1

9 2004 전체 3

10 2005 커미션 미수령 8

11 2005 커미션 수령 5

12 2005 전체 13

13 2006 커미션 미수령 4

14 2006 커미션 수령 2

15 2006 전체 6

16 2007 커미션 미수령 4

17 2007 커미션 수령 2

18 2007 전체 6

19 2008 커미션 수령 1

20 2008 전체 1

21 2009 커미션 미수령 1

22 2009 전체 1

23 2020 커미션 미수령 1

24 2020 전체 1

25 2021 커미션 미수령 1

26 2021 전체 1

27 2023 커미션 미수령 1

28 2023 전체 1

29 모든 연도 커미션 미수령 34

30 모든 연도 커미션 수령 11

31 모든 연도 전체 45

Value X

Text

2001

Refresh Save Cancel Export data 200 31 31 row(s) fetched - 0.001s, on 2026-07-28 at 21:57:28

KST en Writable Smart insert 764 : 54 : 24831 Sel: 0 | 0

H-1. 국가 → 도시 → 부서 계층별 급여 합계

| | |
|------------|---|
| 문제 번호(A-H) | H-1 |
| 분류 | ROLLUP 종합실습 |
| 문제 | [ROLLUP 1] 국가 → 도시 → 부서 계층별 급여 합계
HR 데이터를 활용하여 '국가 → 도시 → 부서 계층별 급여 합계' 결과를 조회하는 SQL 을 작성하십시오. |
| 조건 | GROUP BY ROLLUP 과 GROUPING 함수를 사용해 계층별 상세·소계·전체 합계를 구하십시오. |

실험결과

DBEaver 26.1.3 - <postgres> day02_쿼리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Commit Rollback Auto

postgres hr@skala_db

Connections postgres local host 5432

Databases

postgres

skala_db

Schemas

app

Tables

courses

enrollments 24K

majors 40K

students 56K

Foreign Tables

Views

Materialized Views

Indexes

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Sequences

Data types

Aggregate functions

day02

public

hr

Tables

countries 40K

departments 72K

employees 88K

job_history 56K

jobs 24K

locations 40K

regions 40K

Foreign Tables

Views

Materialized Views

Indexes

Functions

Sequences

Data types

Aggregate functions

Event Triggers

Extensions

Storage

System info

Roles

Administer

System info

postgres> Script-3

postgres> day02_PostgresQL_HR_종합실습_문제.sql

postgres> day02_PostgresQL_HR_emp_dept_상세_스키마.sql

postgres> day02_쿼리_임유리.sql

Filter connections by name

postgres> day02_쿼리_임유리.sql

SQL Editor

-- H. ROLLUP 종합실습 (2문제)

-- [ROLLUP 1] 국가 → 도시 → 부서 계층별 급여 합계
-- 문제: HR 데이터를 활용하여 '국가 → 도시 → 부서 계층별 급여 합계' 결과를 조회하는 SQL 을 작성하십시오.
-- TODO: GROUP BY ROLLUP과 GROUPING 함수를 사용해 계층별 상세·소계·전체 합계를 구하십시오.

SELECT
CASE
WHEN GROUPING(c.country_name) = 1 THEN '전체 합계'
ELSE COALESCE(c.country_name, '국가 미배정')
END AS "국가명",
CASE
WHEN GROUPING(l.city) = 1 THEN '도시 소계'
ELSE COALESCE(l.city, '도시 미배정')
END AS "도시",
CASE
WHEN GROUPING(d.department_name) = 1 THEN
CASE WHEN GROUPING(l.city) = 0 THEN '도시 소계' ELSE '-' END
ELSE COALESCE(d.department_name, '부서 미배정')
END AS "부서명",
COUNT(e.employee_id) AS "직원 수",
COALESCE(SUM(e.salary), 0) AS "급여 합계"
FROM employees e
LEFT JOIN departments d ON d.department_id = e.department_id
LEFT JOIN locations l ON l.location_id = d.location_id
LEFT JOIN countries c ON c.country_id = l.country_id
GROUP BY ROLLUP(c.country_name, l.city, d.department_name)
ORDER BY
GROUPING(c.country_name), c.country_name NULLS LAST,
GROUPING(l.city), l.city NULLS LAST,
GROUPING(d.department_name), d.department_name NULLS LAST;

employees(+) 1 X

SELECT CASE WHEN GROUPING(c.country_name) = 1 THEN '전체 합계'

Enter a SQL expression to filter results (use Ctrl+Space)

Grid

1 Canada Toronto Marketing 2 19,000

2 Canada Toronto 도시 소계 2 19,000

3 Canada 국가 소계 - 2 19,000

4 Germany Munich Public Relations 1 10,000

5 Germany Munich 도시 소계 1 10,000

6 Germany 국가 소계 - 1 10,000

7 South Korea Seoul R&D Korea 3 26,800

8 South Korea Seoul 도시 소계 3 26,800

9 South Korea 국가 소계 - 3 26,800

10 United Kingdom London Human Resources 1 6,500

11 United Kingdom London 도시 소계 1 6,500

12 United Kingdom Oxford Sales 11 112,000

13 United Kingdom Oxford 도시 소계 11 112,000

14 United Kingdom 국가 소계 - 12 118,500

15 United States Seattle Accounting 2 20,300

16 United States Seattle Administration 1 4,400

17 United States Seattle Executive 3 58,000

18 United States Seattle Finance 6 51,600

19 United States Seattle Purchasing 6 24,900

20 United States Seattle 도시 소계 18 159,200

21 United States South San Francisco Shipping 3 24,100

22 United States South San Francisco 도시 소계 3 24,100

23 United States Southlake IT 5 28,800

24 United States Southlake 도시 소계 5 28,800

25 United States 국가 소계 26 212,100

26 국가 미배정 도시 미배정 부서 미배정 1 6,200

27 국가 미배정 도시 미배정 도시 소계 1 6,200

28 국가 미배정 국가 소계 - 1 6,200

29 전체 합계 국가 소계 - 45 392,600

Record

Refresh Save Cancel Export data 200 29 29 row(s) fetched - 0.001s, on 2026-07-28 at 21:57:47

General

H-2. 입사 연도 → 부서별 인원 및 급여 합계

| | |
|------------|---|
| 문제 번호(A-H) | H-2 |
| 분류 | ROLLUP 종합실습 |
| 문제 | [ROLLUP 2] 입사 연도 → 부서별 인원 및 급여 합계
HR 데이터를 활용하여 '입사 연도 → 부서별 인원 및 급여 합계' 결과를 조회하는 SQL 을 작성하십시오. |
| 조건 | GROUP BY ROLLUP 과 GROUPING 함수를 사용해 계층별 상세·소계·전체 합계를 구하십시오. |

실험결과

DBaver 26.1.3 - <postgres> day02_쿼리_임유리.sql

File Edit Navigate Search SQL Editor Database Window Help

Commit Rollback Auto postgres hr@skala_db postgres day02_PostgresSQL_HR_종합실습_문제.sql postgres day02_PostgresSQL_HR_emp_dept_샘플_스키마.sql postgres day02_쿼리_임유리.sql

Filter connections by name
DBeaver Sample Database (SQLite)
postgres localhost:5432
Databases
postgres
skala_db
Schemas
app
Tables
courses
enrollments 24K
majors 40K
students 56K
Foreign Tables
Views
Materialized Views
Indexes
Functions
Sequences
Data types
Aggregate functions
day02
public
hr
Tables
countries 40K
departments 72K
employees 88K
job_history 56K
jobs 24K
locations 40K
regions 40K
Foreign Tables
Views
Materialized Views
Indexes
Functions
Sequences
Data types
Aggregate functions
Event Triggers
Extensions
Storage
System info
Roles
Administer
System info

SQL Editor
-- [ROLLUP 2] 입사 연도 → 부서별 인원 및 급여 합계
-- 용도: HR 데이터를 활용하여 '입사 연도 → 부서별 인원 및 급여 합계' 결과를 조회하는 SQL 을 작성하십시오.
-- 조건: GROUP BY ROLLUP 과 GROUPING 함수를 사용해 계층별 상세·소계·전체 합계를 구하십시오.
-- TODO: 아래에 SQL 을 작성하십시오.
SELECT
CASE
WHEN GROUPING(EXTRACT(YEAR FROM e.hire_date)) = 1 THEN '전체 합계'
ELSE EXTRACT(YEAR FROM e.hire_date)::integer::text
END AS '입사 연도',
CASE
WHEN GROUPING(d.department_name) = 1 THEN
CASE
WHEN GROUPING(EXTRACT(YEAR FROM e.hire_date)) = 1
THEN '모든 부서'
ELSE '연도 소계'
END
ELSE COALESCE(d.department_name, '부서 미배정')
END AS '부서명',
COUNT(*) AS '직원 수',
SUM(e.salary) AS '급여 합계'
FROM employees e
LEFT JOIN departments d ON d.department_id = e.department_id
GROUP BY ROLLUP(EXTRACT(YEAR FROM e.hire_date),
ORDER BY
GROUPING(EXTRACT(YEAR FROM e.hire_date)),
EXTRACT(YEAR FROM e.hire_date) NULLS LAST,
GROUPING(d.department_name),
d.department_name NULLS LAST;

employees(+) 1 X
Grid
1 2001 Executive 1 17,000
2 2001 연도 소계 1 17,000
3 2002 Accounting 2 20,300
4 2002 Finance 2 21,000
5 2002 Human Resources 1 6,500
6 2002 Public Relations 1 10,000
7 2002 Purchasing 1 11,000
8 2002 연도 소계 7 68,800
9 2003 Administration 1 4,400
10 2003 Executive 1 24,000
11 2003 Purchasing 1 3,100
12 2003 Shipping 1 7,900
13 2003 연도 소계 4 39,400
14 2004 Marketing 1 13,000
15 2004 Sales 1 14,000
16 2004 Shipping 1 8,000
17 2004 연도 소계 3 35,000
18 2005 Executive 1 17,000
19 2005 Finance 2 15,900
20 2005 IT 1 4,800
21 2005 Marketing 1 6,000
22 2005 Purchasing 2 5,700
23 2005 Sales 5 54,000
24 2005 Shipping 1 8,200
25 2005 연도 소계 13 111,600
26 2006 Finance 1 7,800
27 2006 IT 2 13,800
28 2006 Purchasing 1 2,600
29 2006 Sales 2 15,500
30 2006 연도 소계 6 39,700
31 2007 Finance 1 6,900
32 2007 IT 2 10,200
Text
SELECT CASE WHEN GROUPING(EXTRACT(YEAR FROM e.hire_date)) = 1 THEN '전체 합계'
ELSE EXTRACT(YEAR FROM e.hire_date)::integer::text
END AS '입사 연도',
CASE
WHEN GROUPING(d.department_name) = 1 THEN
CASE
WHEN GROUPING(EXTRACT(YEAR FROM e.hire_date)) = 1
THEN '모든 부서'
ELSE '연도 소계'
END
ELSE COALESCE(d.department_name, '부서 미배정')
END AS '부서명',
COUNT(*) AS '직원 수',
SUM(e.salary) AS '급여 합계'
FROM employees e
LEFT JOIN departments d ON d.department_id = e.department_id
GROUP BY ROLLUP(EXTRACT(YEAR FROM e.hire_date),
ORDER BY
GROUPING(EXTRACT(YEAR FROM e.hire_date)),
EXTRACT(YEAR FROM e.hire_date) NULLS LAST,
GROUPING(d.department_name),
d.department_name NULLS LAST;

Value
Text
2001

Refresh Save Cancel Export data 200 46 46 row(s) fetched - 0.001s, on 2025-07-28 at 21:58:00

General KST en Writable Smart Insert 829 : 34 : 27172 Sel: 0 | 0

The screenshot shows the DBeaver SQL editor interface. The left sidebar displays the database schema tree for 'postgres' on 'localhost:5432'. The main editor window contains a SQL query that uses CTEs and GROUPING functions to calculate salary statistics by year and department.

```

WITH employees AS (
    SELECT * FROM hr.employees e
), departments AS (
    SELECT * FROM hr.departments d
)
SELECT CASE WHEN GROUPING(EXTRACT(YEAR FROM e.hire_date)) = 1 THEN '연도 집계'
            ELSE EXTRACT(YEAR FROM e.hire_date)::integer::text
        END AS "연도",
       CASE WHEN GROUPING(d.department_name) = 1 THEN
           CASE WHEN GROUPING(EXTRACT(YEAR FROM e.hire_date)) = 1
                THEN '모든 부서'
                ELSE '연도 소계'
           END
       ELSE COALESCE(d.department_name, '부서 미배정')
       END AS "부서명",
       COUNT(*) AS "사원 수",
       SUM(e.salary) AS "급여 합계"
FROM employees e
LEFT JOIN departments d ON d.department_id = e.department_id
GROUP BY ROLLUP(EXTRACT(YEAR FROM e.hire_date), d.department_name)
ORDER BY
    GROUPING(EXTRACT(YEAR FROM e.hire_date)),
    EXTRACT(YEAR FROM e.hire_date) NULLS LAST,
    GROUPING(d.department_name),
    d.department_name NULLS LAST;

```

The results pane shows a table with columns: 연도, 부서명, 사원 수, and 급여 합계. The data is grouped by year and department, with summary rows for each year.

| 연도 | 부서명 | 사원 수 | 급여 합계 | |
|----|-------|------------|-------|---------|
| 16 | 2004 | Shipping | 1 | 8,000 |
| 17 | 2004 | 연도 소계 | 3 | 35,000 |
| 18 | 2005 | Executive | 1 | 17,000 |
| 19 | 2005 | Finance | 2 | 15,900 |
| 20 | 2005 | IT | 1 | 4,800 |
| 21 | 2005 | Marketing | 1 | 6,000 |
| 22 | 2005 | Purchasing | 2 | 5,700 |
| 23 | 2005 | Sales | 5 | 54,000 |
| 24 | 2005 | Shipping | 1 | 8,200 |
| 25 | 2005 | 연도 소계 | 13 | 111,600 |
| 26 | 2006 | Finance | 1 | 7,800 |
| 27 | 2006 | IT | 2 | 13,800 |
| 28 | 2006 | Purchasing | 1 | 2,600 |
| 29 | 2006 | Sales | 2 | 15,500 |
| 30 | 2006 | 연도 소계 | 6 | 39,700 |
| 31 | 2007 | Finance | 1 | 6,900 |
| 32 | 2007 | IT | 2 | 10,200 |
| 33 | 2007 | Purchasing | 1 | 2,500 |
| 34 | 2007 | Sales | 2 | 18,000 |
| 35 | 2007 | 연도 소계 | 6 | 37,600 |
| 36 | 2008 | Sales | 1 | 10,500 |
| 37 | 2008 | 연도 소계 | 1 | 10,500 |
| 38 | 2019 | R&D Korea | 1 | 10,500 |
| 39 | 2019 | 연도 소계 | 1 | 10,500 |
| 40 | 2020 | R&D Korea | 1 | 8,500 |
| 41 | 2020 | 연도 소계 | 1 | 8,500 |
| 42 | 2021 | R&D Korea | 1 | 7,800 |
| 43 | 2021 | 연도 소계 | 1 | 7,800 |
| 44 | 2023 | 부서 미배정 | 1 | 6,200 |
| 45 | 2023 | 연도 소계 | 1 | 6,200 |
| 46 | 전체 집계 | 모든 부서 | 45 | 392,600 |